



Leveraging Synthetic Data from Large Language Models to Steer and Enhance Model Learning

PhD Dissertation Defense • Ajay Patel

November 24th, 2025

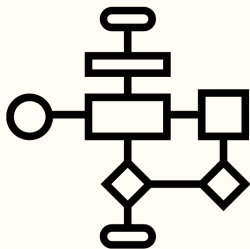
Advisor: Chris Callison-Burch

Committee: Dan Roth (chair), Lyle Ungar, Duncan Watts, Hannaneh Hajishirzi, Colin Raffel

Introduction

- Background
- Thesis Statement
- Publications Presented

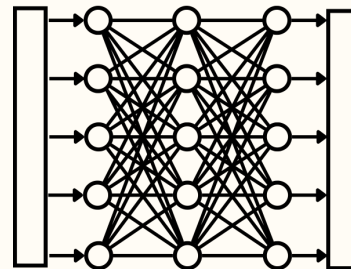
Classical AI vs. Machine Learning



Classical AI

No training data required
Hard-coded rules
Weak performance on complex tasks
Easy to interpret and control exact behavior

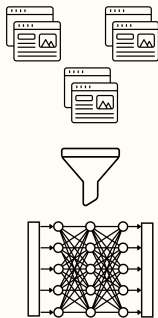
V.S.



Machine Learning

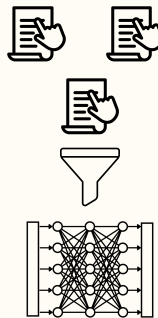
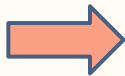
Training data required
Automatically learns rules from patterns in data
Strong performance on complex tasks
Difficult to interpret and control exact behavior

Large Language Models (LLMs)



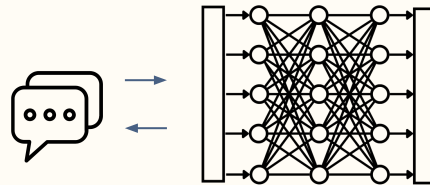
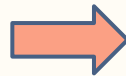
Pre-Training

Trained on large amounts
of unlabeled text data
from the internet



Fine-Tuning

Trained on a smaller
dataset of labeled
examples of a task

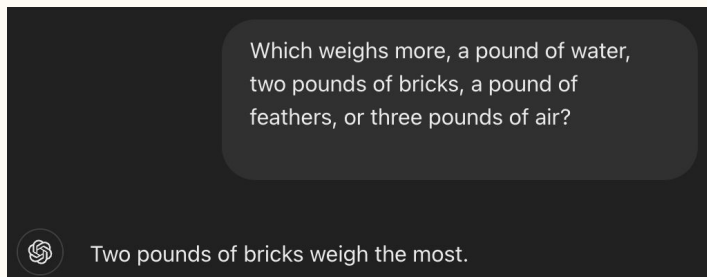


Prompting

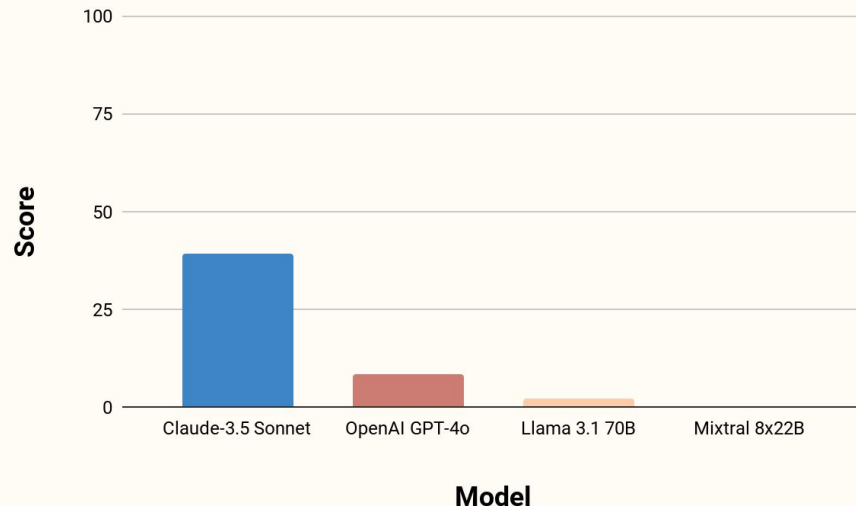
Model can perform a task
at inference-time

(Brown et al., 2020)

Large Language Models (LLMs)



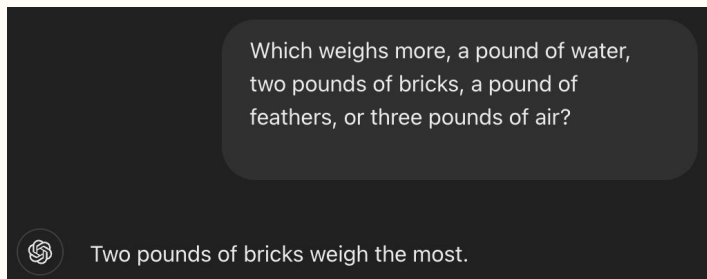
This specific type of brain teaser is likely not well-represented in the LLM's training data.



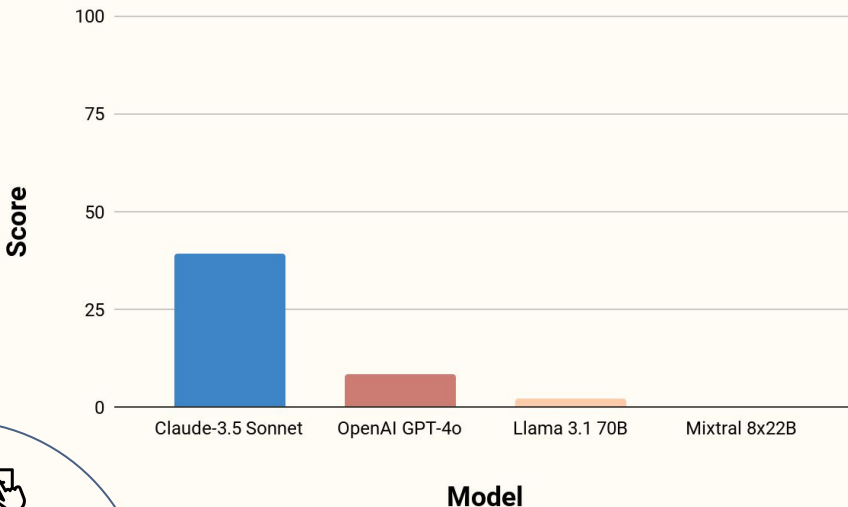
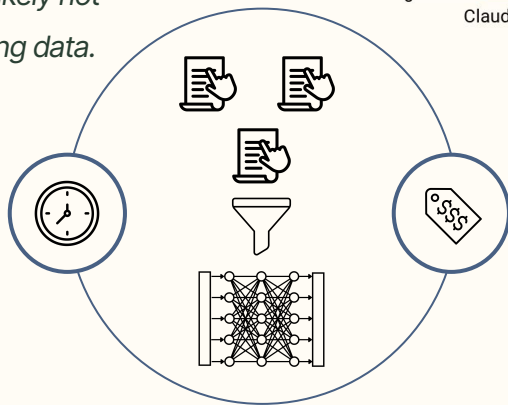
WorkArena-L2 Benchmark

(Williams and Huckle, 2024; Drouin et al., 2024)

Large Language Models (LLMs)



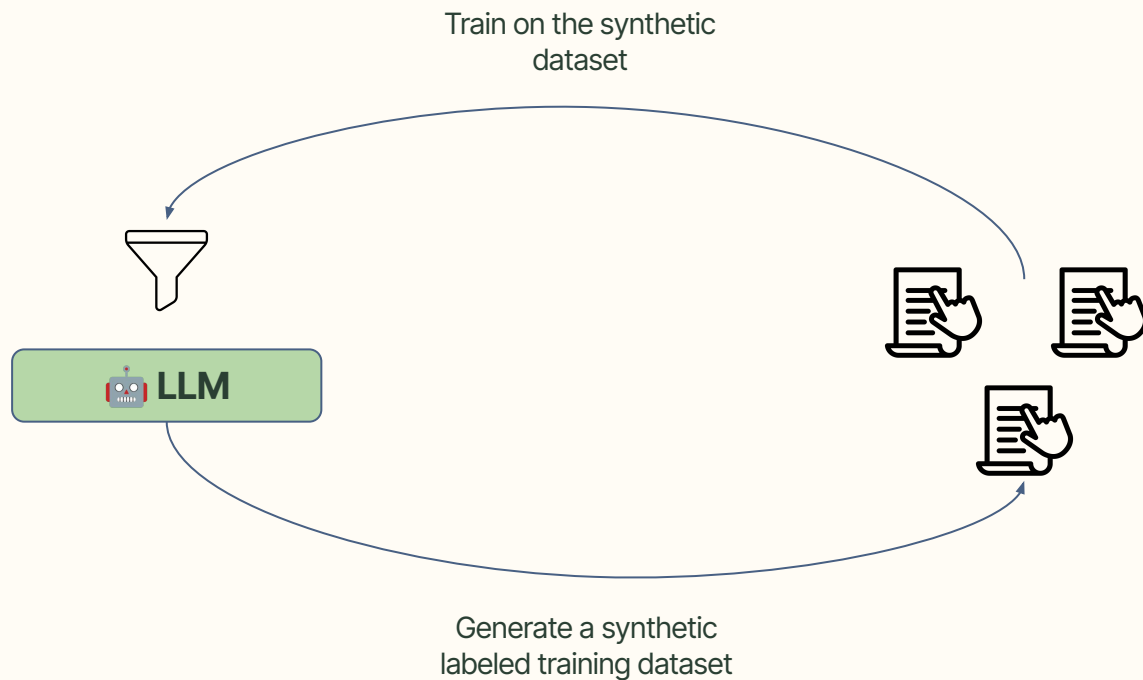
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WorkArena-L2 Benchmark

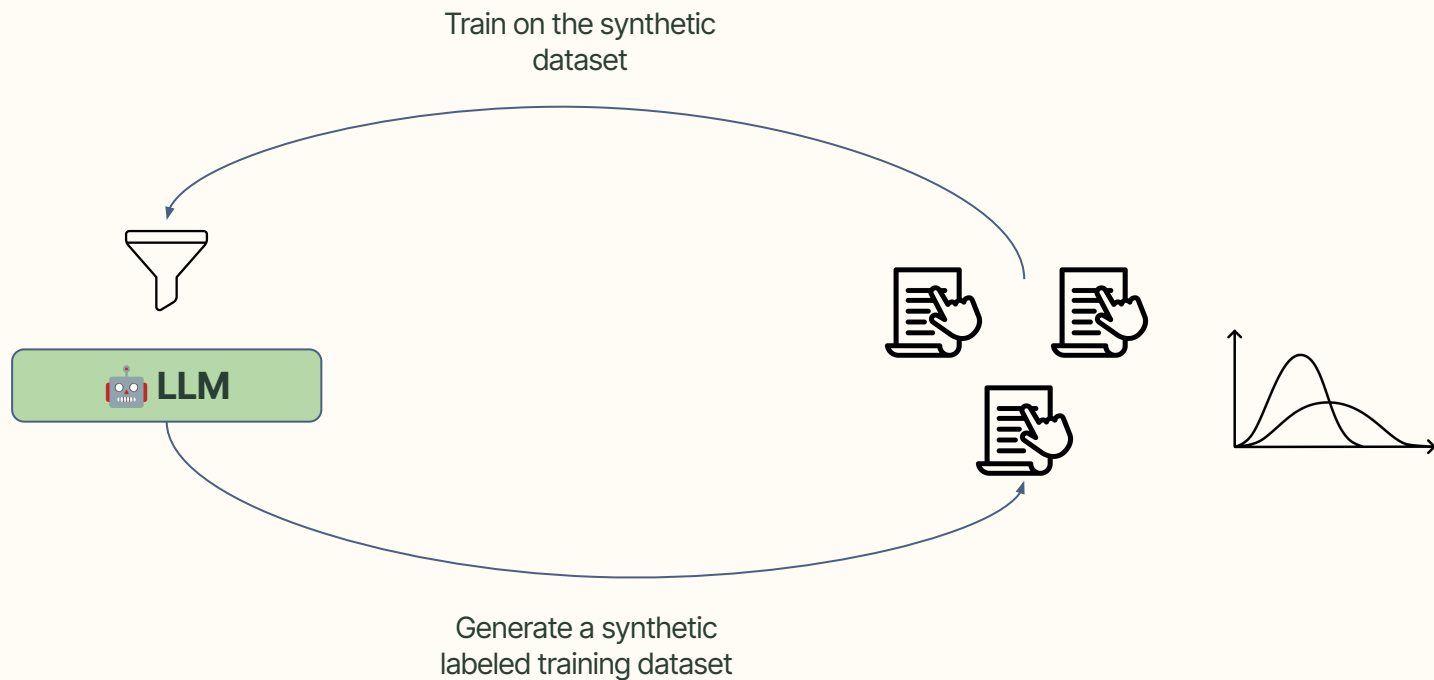
(Williams and Huckle, 2024; Drouin et al., 2024)

Synthetic Training Data



(Kumar et al., 2020a,b; Callison-Burch, 2009)

Synthetic Training Data



(Thanh-Tung and Tran, 2020)

Synthetic Training Data

The screenshot shows the Nature journal website for an article titled "AI models collapse when trained on recursively generated data". The page includes navigation links, article metadata, author information, and an abstract. Overlaid on the page are several annotations: a green box on the left, a funnel icon pointing to the breadcrumb "nature > articles > article", a circular arrow pointing from the article title to the text "Train on the synthetic dataset", and a line graph on the right with an arrow pointing to the text "Generate a synthetic labeled training dataset".

nature Train on the synthetic dataset

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[nature](#) > [articles](#) > article

Article | [Open access](#) | Published: 24 July 2024

AI models collapse when trained on recursively generated data

[Ilia Shumailov](#) ✉, [Zakhar Shumaylov](#) ✉, [Yiren Zhao](#), [Nicolas Papernot](#), [Ross Anderson](#) & [Yarin Gal](#) ✉

[Nature](#) **631**, 755–759 (2024) | [Cite this article](#)

337k Accesses | **24** Citations | **2946** Altmetric | [Metrics](#)

Abstract

Stable diffusion revolutionized image creation from descriptive text. GPT-2 (ref. ¹), GPT-3(.5) (ref. ²) and GPT-4 (ref. ³) demonstrated high performance across a variety of language tasks. ChatGPT introduced such language models to the public. It is now clear that generative artificial intelligence (AI) such as large language models (LLMs) is here to stay and will

Synthetic Training Data

Many research papers *successfully* demonstrate using LLM-generated training data to improve a model's own performance on a specific task, but the question still remains:

When and ***why*** does synthetic training data work?

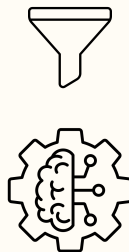
(Yu et al., 2023; Kumar et al., 2020a,b; Yoo et al., 2021; Han et al., 2021d; Ye et al., 2022; Honovich et al., 2022, *inter alia*)

Early Applications of Synthetic Data

Cat



Dog



Early Applications of Synthetic Data

Original



Dog



Flipped



Cat



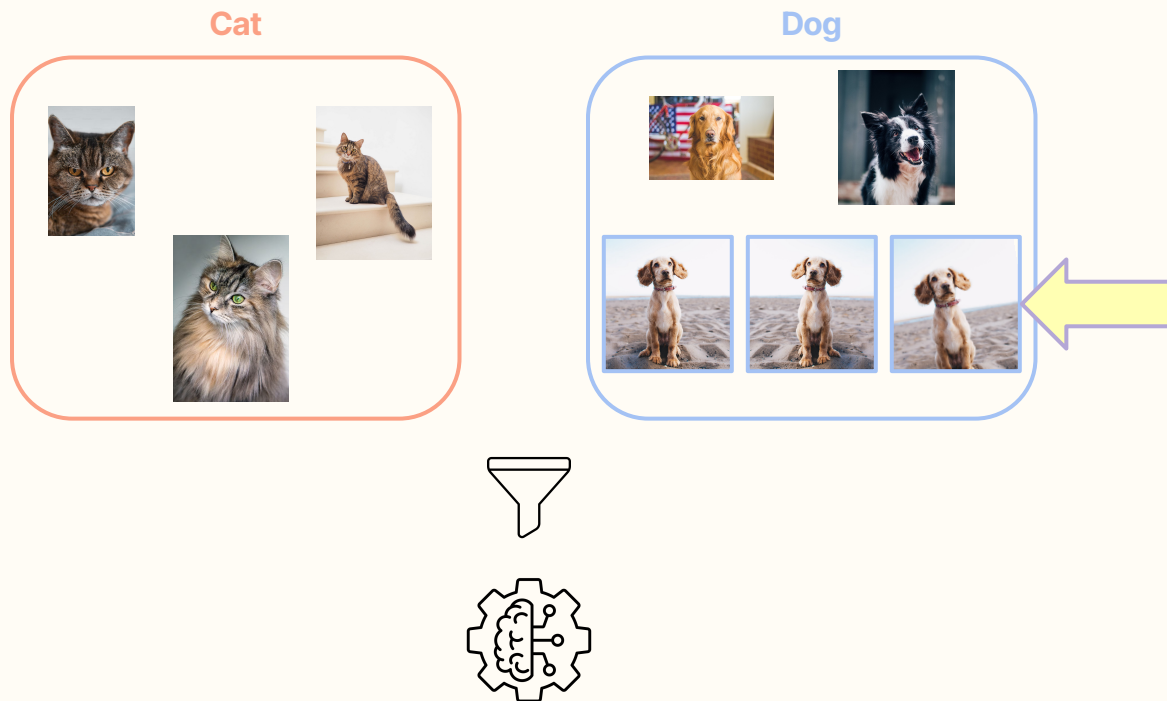
Rotated



Cat



Early Applications of Synthetic Data



(Cireřan et al., 2011)

Early Applications of Synthetic Data



Human Machine Learning Practitioner

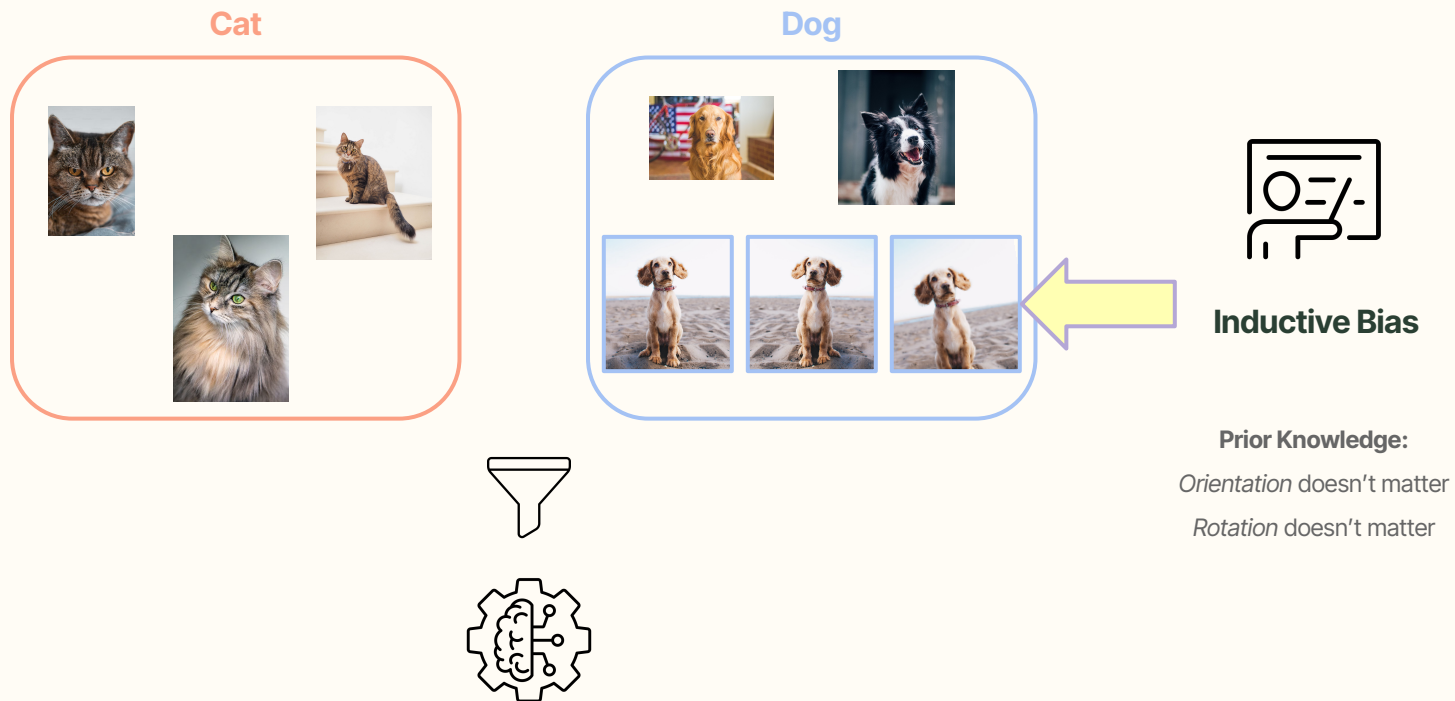
Prior Knowledge of Non-Important Features

Size of Image
Blurriness of Image
Rotation of Image
...

Prior Knowledge of Important Features

Ear Shape
Nose Shape
Fur Color
...

Inductive Bias



(Cireřan et al., 2011; Baxter, 2000)

Inductive Bias

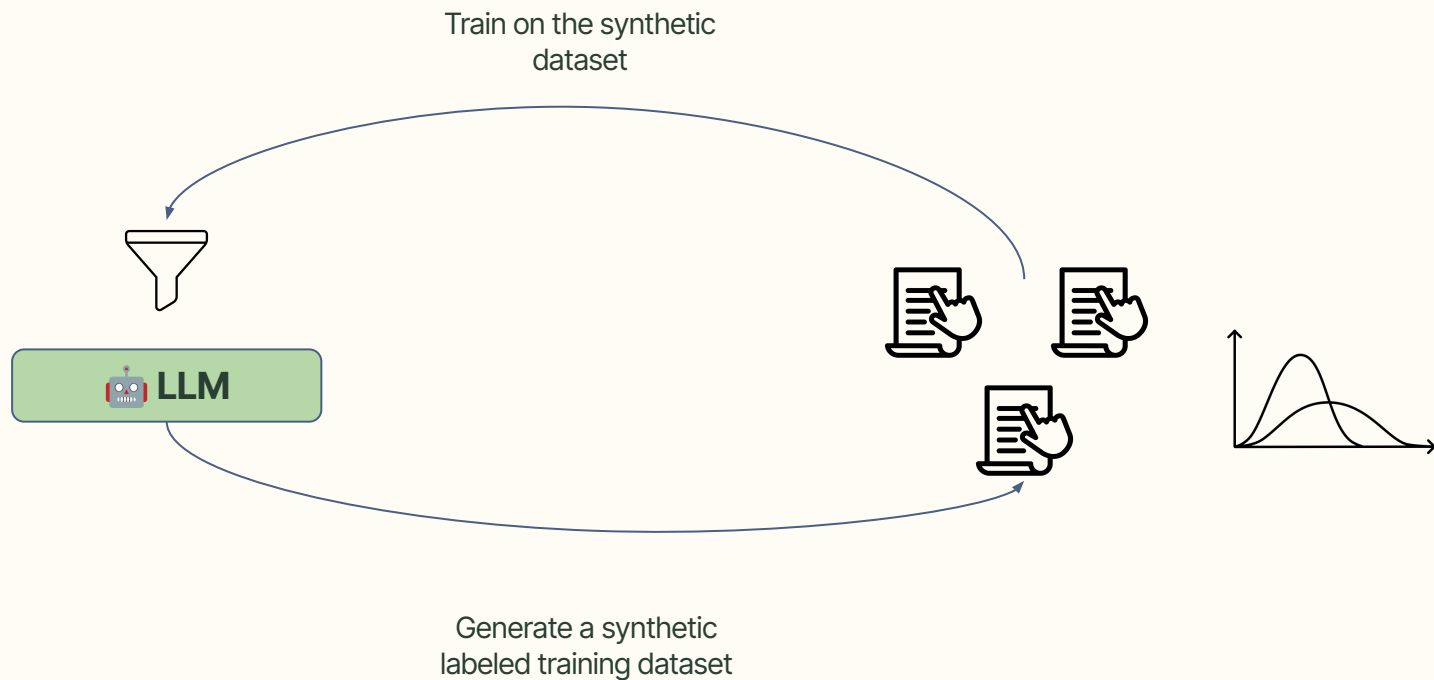
A helpful assumption the model can leverage to make better predictions and decisions that generalize.



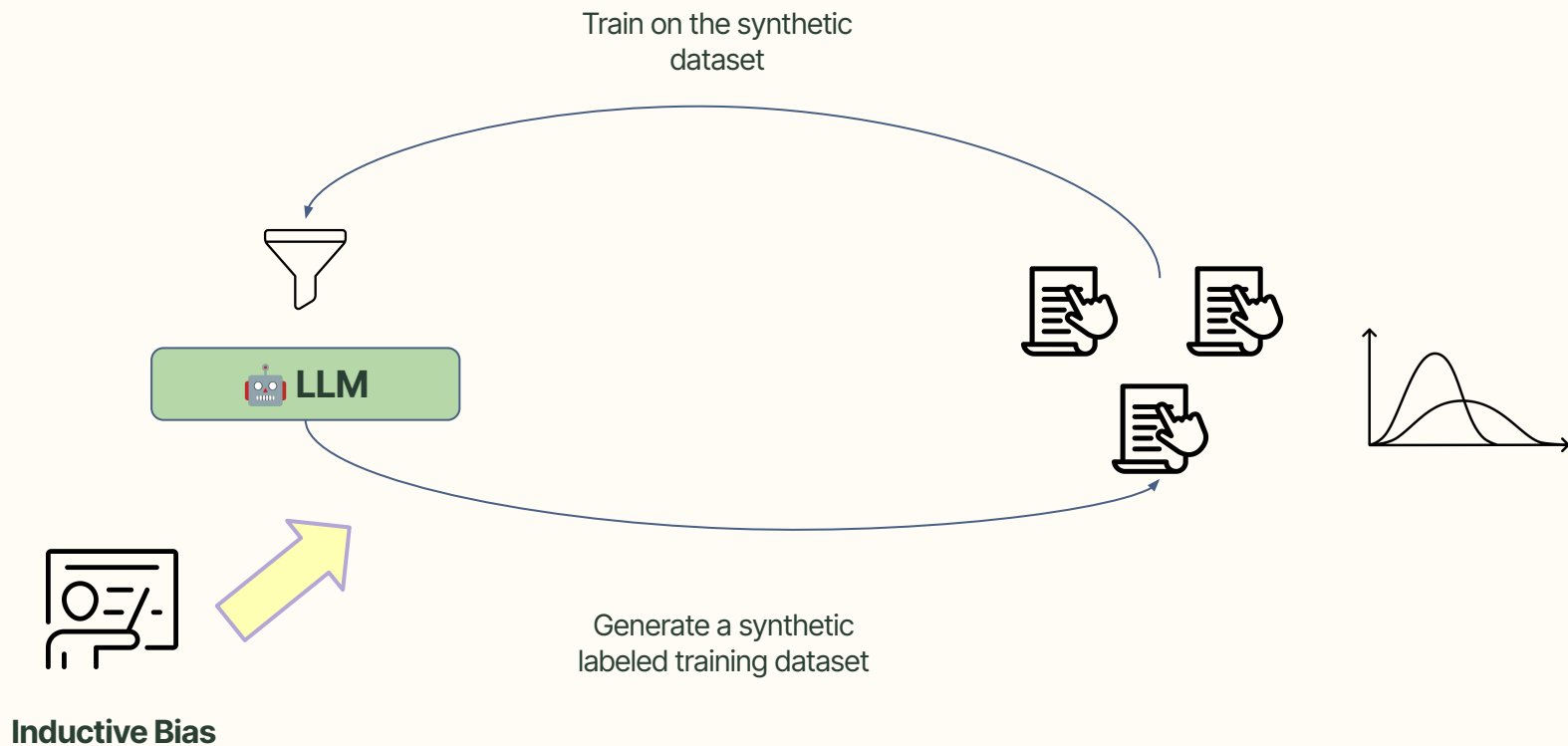
Prior Knowledge:
Orientation doesn't matter
Rotation doesn't matter

(Cireřan et al., 2011; Baxter, 2000)

Synthetic Data as an Inductive Bias



Synthetic Data as an Inductive Bias



Thesis Statement

This thesis argues **synthetic training data is most effective** at enhancing models **when it is used as an inductive bias**. I demonstrate that when synthetic training data is carefully constructed to incorporate useful learning biases relevant to a given task, model performance can **significantly improve, enabling new capabilities**. I demonstrate that this approach is **not limited to enhancing performance on a single task**; rather, applying an inductive bias through synthetic data in the pre-training phase can enrich the general knowledge and skill acquisition of LLMs as they undergo their foundational learning process.

Publications Presented

1. (Patel et al., 2023) Ajay Patel, Bryan Li, Mohammad Sadegh Rasooli, Noah Constant, Colin Raffel, and Chris Callison-Burch. Bidirectional Language Models Are Also Few-shot Learners. **In ICLR 2023.**
2. (Patel et al., 2024) Ajay Patel, Colin Raffel, and Chris Callison-Burch. DataDreamer: A Tool for Synthetic Data Generation and Reproducible LLM Workflows. **In ACL 2024.**
3. (Patel et al., 2024) Ajay Patel, Markus Hofmarcher, Claudiu Leoveanu-Condrei, Marius-Constantin Dinu, Chris Callison-Burch, and Sepp Hochreiter. Large Language Models Can Self-Improve At Web Agent Tasks. **arXiv 2024.**
4. (Patel et al., 2024) Ajay Patel*, Jiacheng Zhu*, Justin Qiu, Zachary Horvitz, Marianna Apidianaki, Kathleen McKeown, and Chris Callison-Burch. StyleDistance: Stronger Content-Independent Style Embeddings with Synthetic Parallel Examples. **In NAACL 2025.**
5. (Yang et al., 2024) Yue Yang*, Ajay Patel*, Matt Deitke, Luca Weihs, Tanmay Gupta, Ranjay Krishna, Andrew Head, Chris Callison-Burch, Mark Yatskar, Aniruddha Kembhavi, Christopher Clark. Scaling Text-Rich Image Understanding via Code-Guided Synthetic Multimodal Data Generation. **In ACL 2025.**
6. (Patel et al., 2025) Ajay Patel, Colin Raffel, Chris Callison-Burch. FineInstructions: Scaling Synthetic Pre-Training. **Working manuscript.**

* denotes co-first authorship

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§1 In-Context Learning

* denotes co-first authorship

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} § 2
**DataDreamer
Framework**

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§ 3
Fine-Tuning

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§ 4 Pre-Training

* denotes co-first authorship

Inductive Biases and Sources of Knowledge

Inductive Biases



Task Priming



Task Asymmetry



Task Rules and Heuristics



Task Structuring

Sources of Knowledge



Surfacing Latent Knowledge and Capabilities



Composing Existing Knowledge and Capabilities



Environment Feedback

Enhancing Model Performance using Synthetic Data and In-Context Learning

- Motivation
- Procedure & Experimental Setup
- Results
- Summary

Motivation

Machine Translation is a task that is *notoriously difficult* to acquire training data for.



English ⇌ Spanish
(High Resource)

She is a doctor.

⇌

Ella es doctora.

Where is the store?

⇌

¿Dónde está la tienda?

We are going home.

⇌

Vamos a casa.

...

⇌

...

Malayalam ⇌ Bulgarian
(Low Resource)



No results found.

In-Context Learning

Zero-Shot Prompting

Translate Spanish to English.

Spanish: Los árboles son importantes.

English:

Few-Shot Prompting

Translate Spanish to English.

Spanish: El clima es soleado.

English: The weather is sunny.

Spanish: Mi perro es un cachorro.

English: My dog is a puppy.

Spanish: Los árboles son importantes.

English:

} Two "in-context"
training examples

(2-shot)

Motivation

Zero-Shot Prompting

Translate Spanish to English.

Spanish: El clima es soleado.

English:

The weather is sunny.

Few-Shot Prompting

Translate Spanish to English.

Spanish: El clima es soleado.

English: *The weather is sunny.*

Spanish: Mi perro es un cachorro.

English:

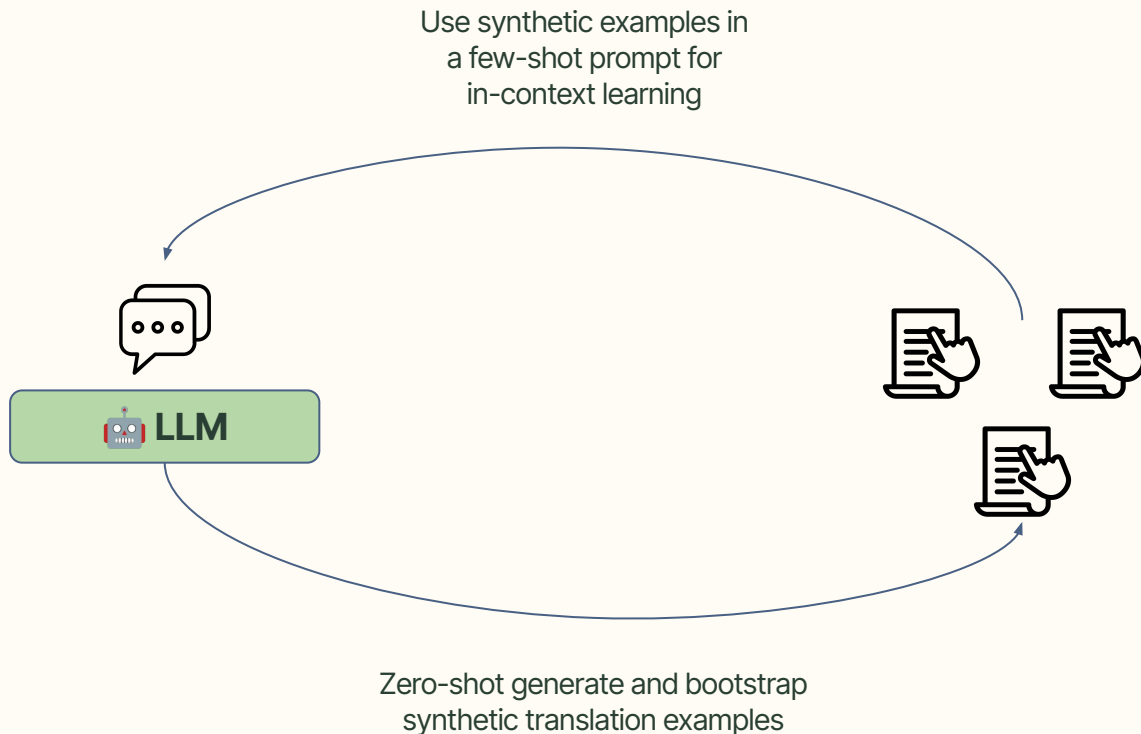
Spanish: Los árboles son importantes.

English:

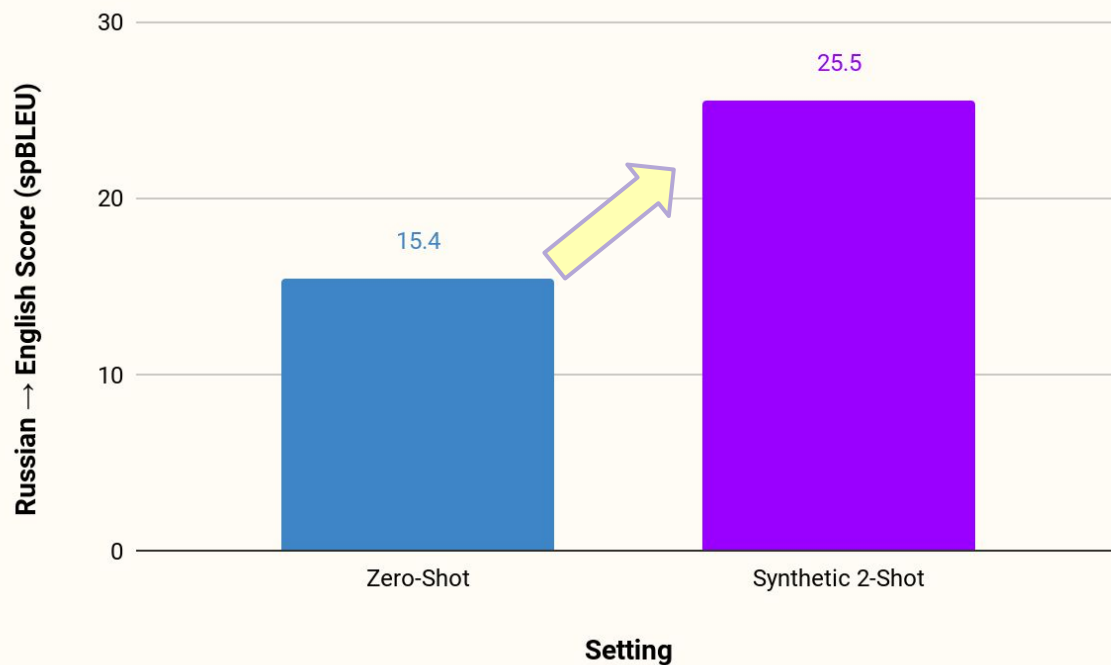
What if we replace these with "synthetic parallel examples" generated by a zero-shot prompt?

(Han et al., 2021; Patel et al., 2023)

Bootstrapping Synthetic In-Context Examples

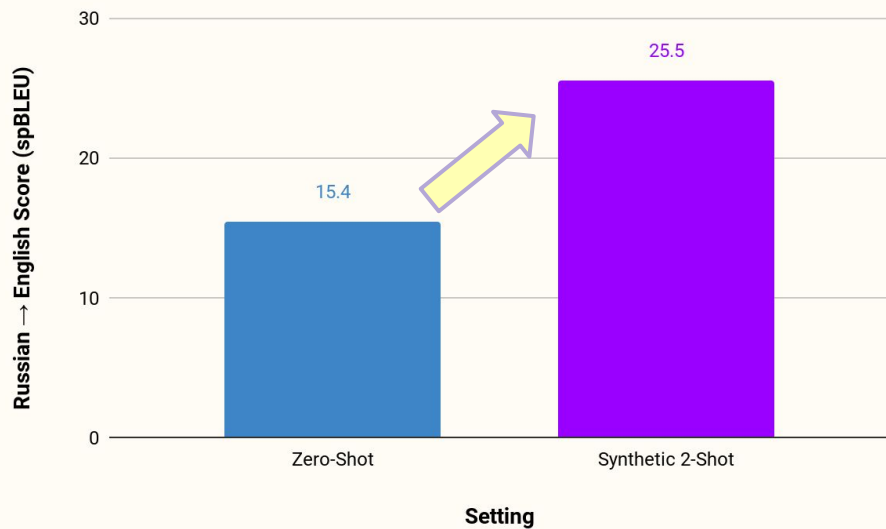


Results with Synthetic Examples



(Patel et al., 2023)

Results with Synthetic Examples



Inductive Biases



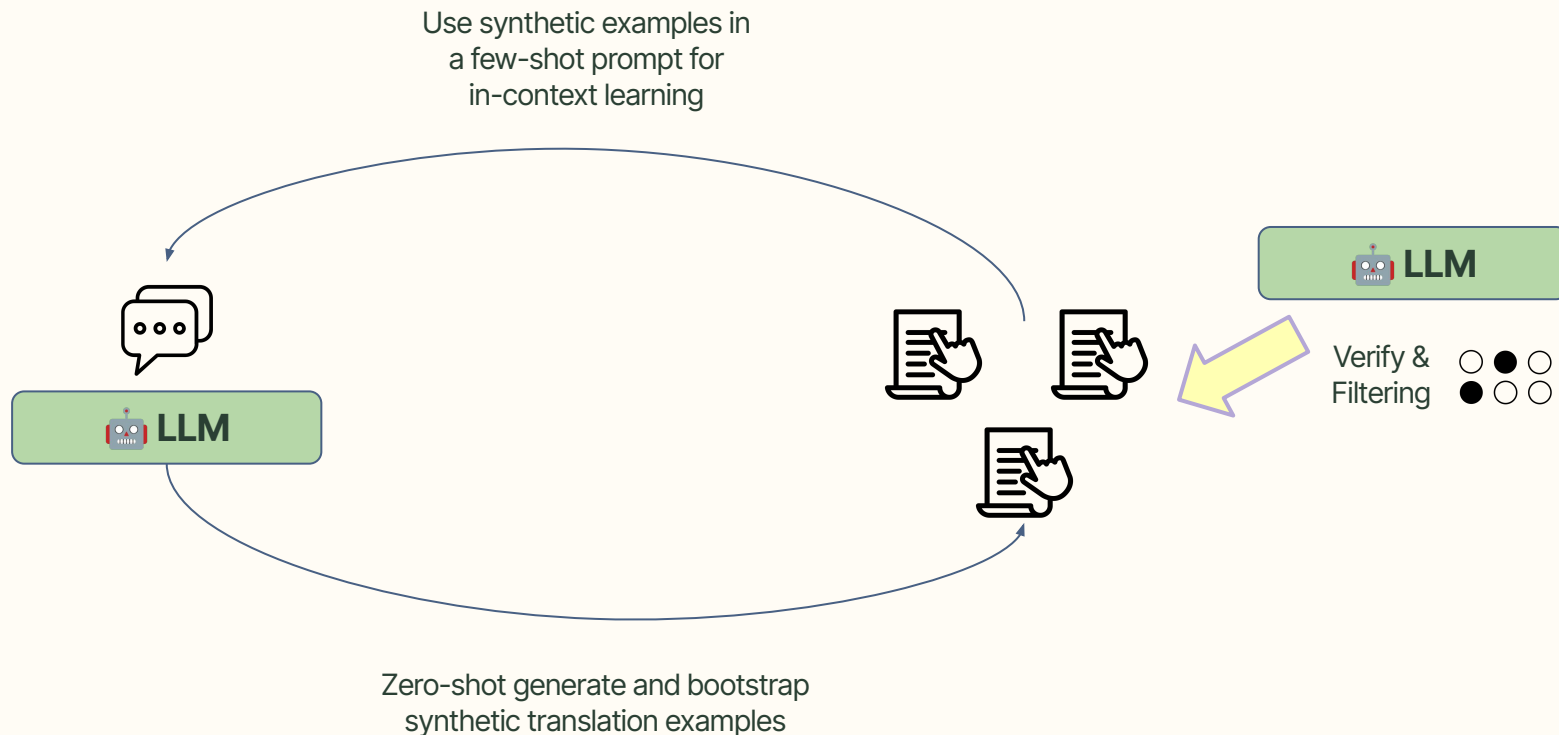
Task Priming

Sources of Knowledge

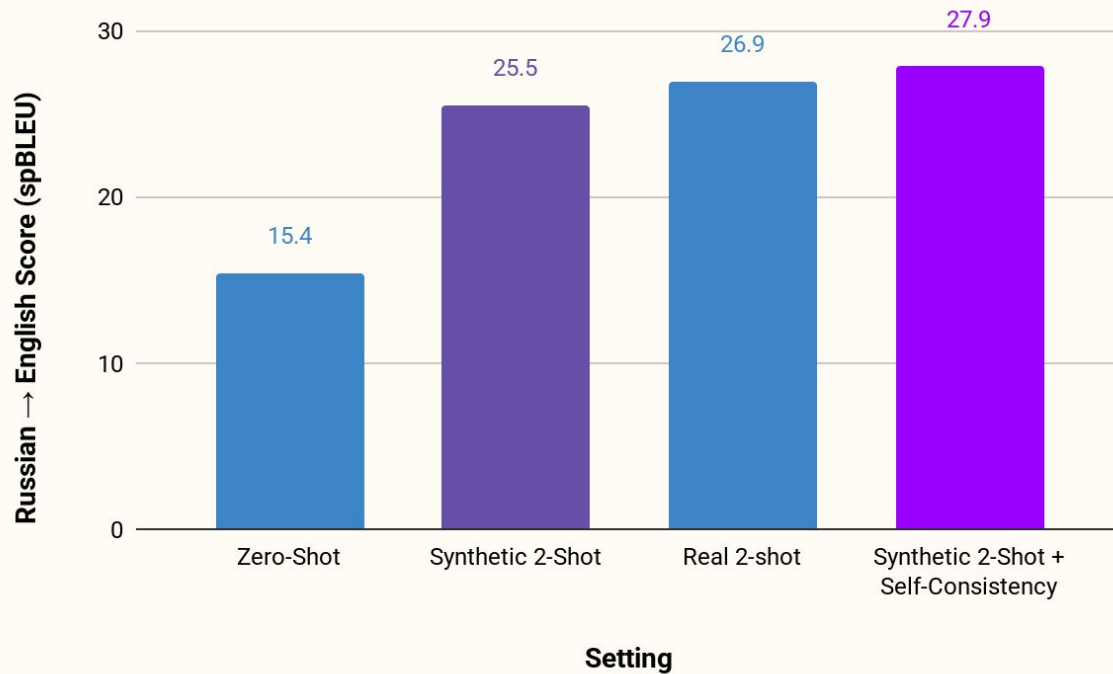


Surfacing Latent Knowledge and Capabilities

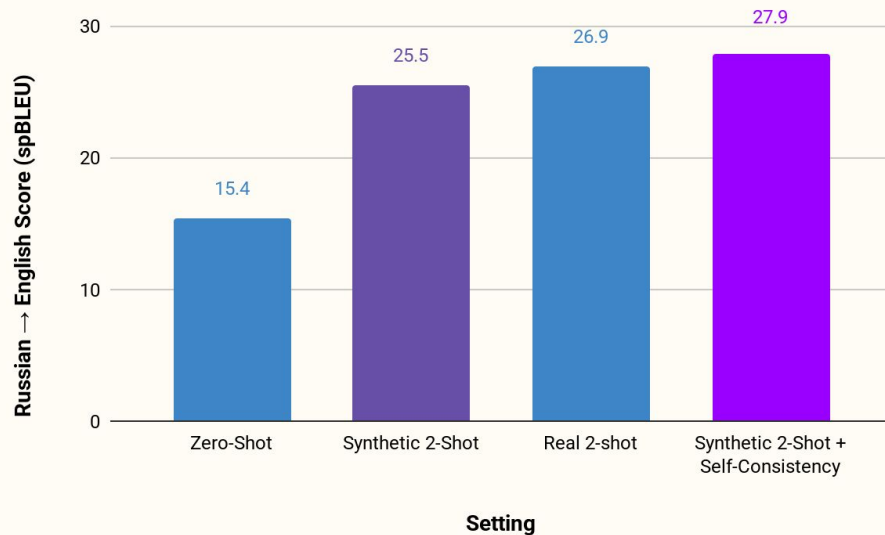
Self-Consistency



Results with Self-Consistency



Results with Self-Consistency



Inductive Biases



Task Asymmetry

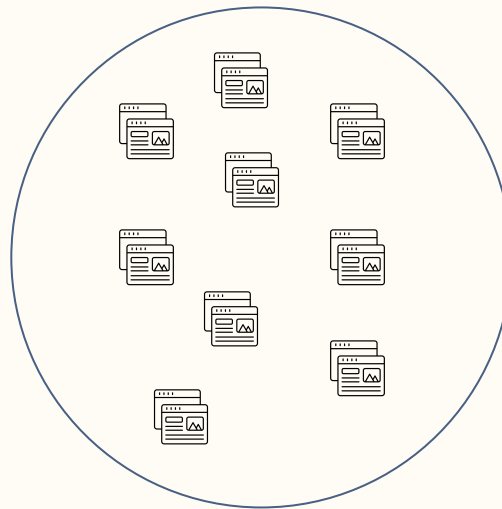
Sources of Knowledge



Composing Existing Knowledge and Capabilities

English-Centric Bootstrapping

Russian – 2%

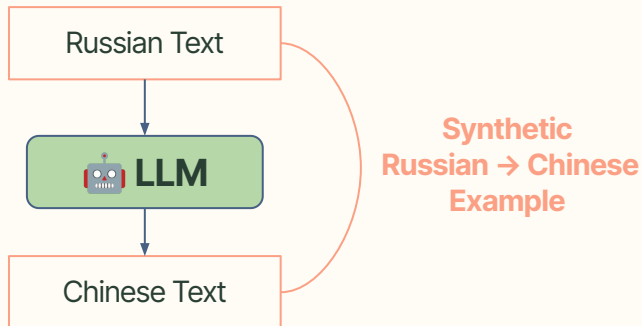


English – 76%

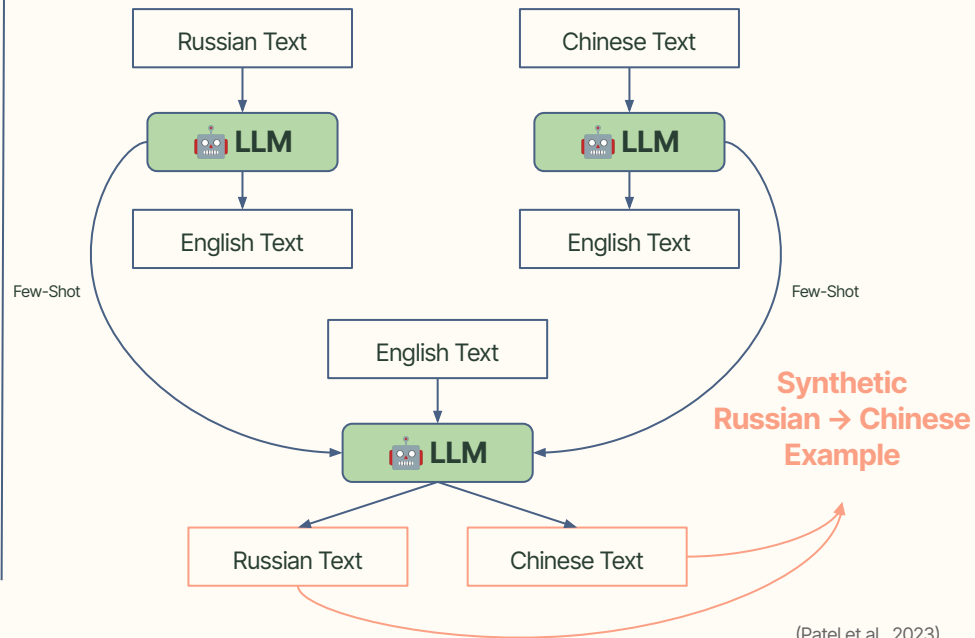
English-Centric Bootstrapping

Russian → Chinese

Standard Bootstrap

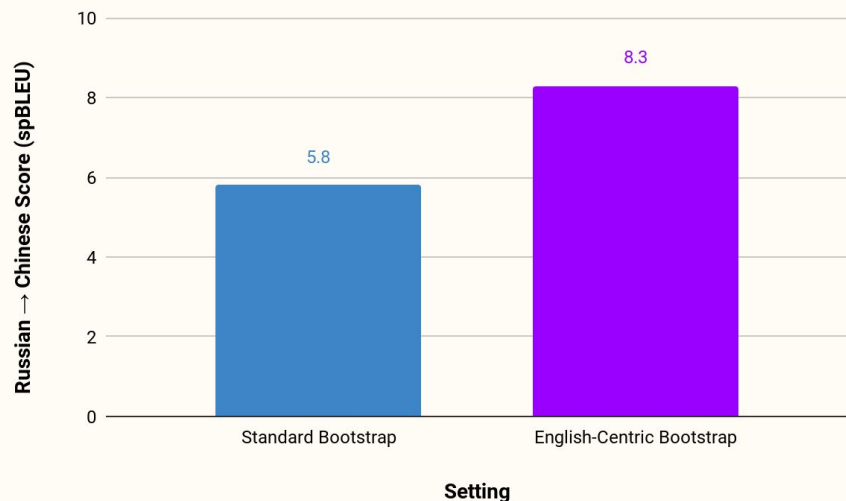


English-Centric Bootstrap



(Patel et al., 2023)

Results with English-Centric Bootstrapping



Inductive Biases



Task Asymmetry



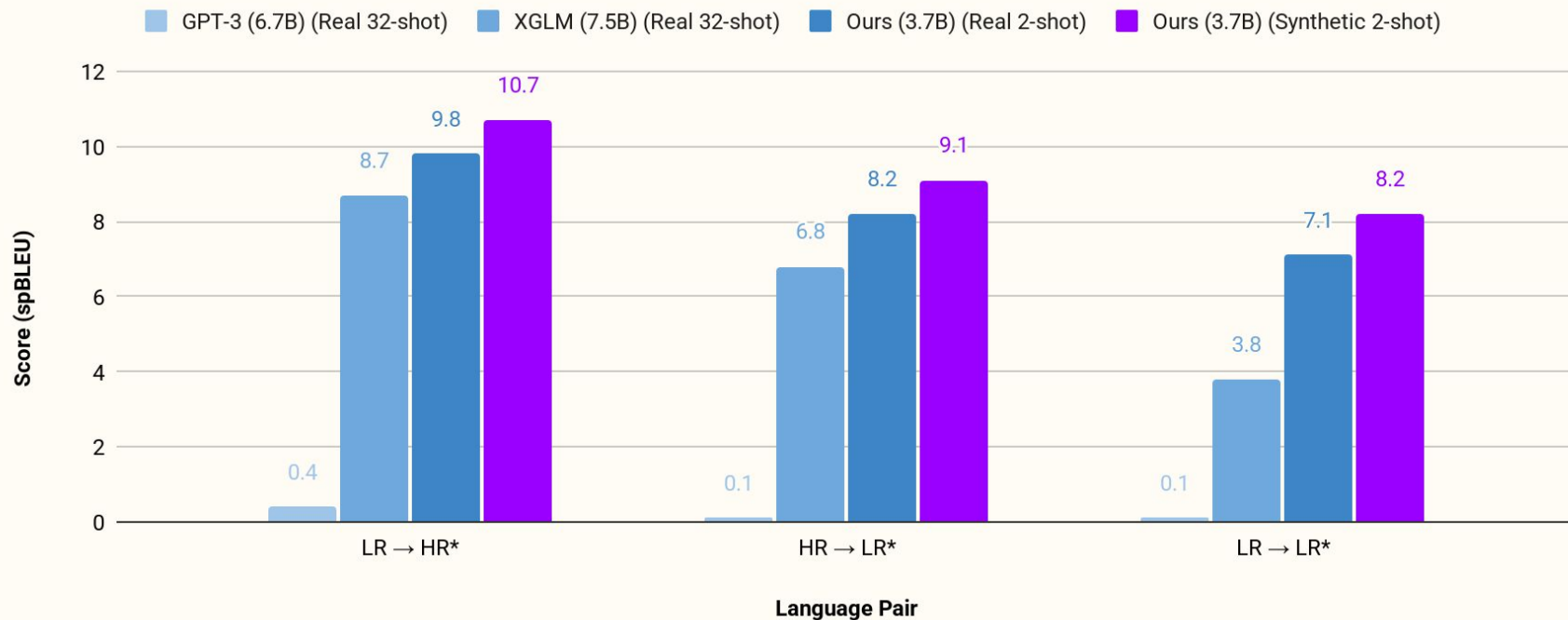
Task Structuring

Sources of Knowledge



Composing Existing Knowledge and Capabilities

Final Results



HR = High-Resource Language

LR = Low-Resource Language

* = statistically significant

(Patel et al., 2023)

Summary

- Improve zero-shot machine translation performance of an LLM with:
 - Bootstrapping of Synthetic In-Context Examples
 - Self-Consistency Filtering
 - English-Centric Bootstrapping

Inductive Biases



Task Priming



Task Asymmetry



Task Structuring

Sources of Knowledge



Surfacing Latent Knowledge and Capabilities



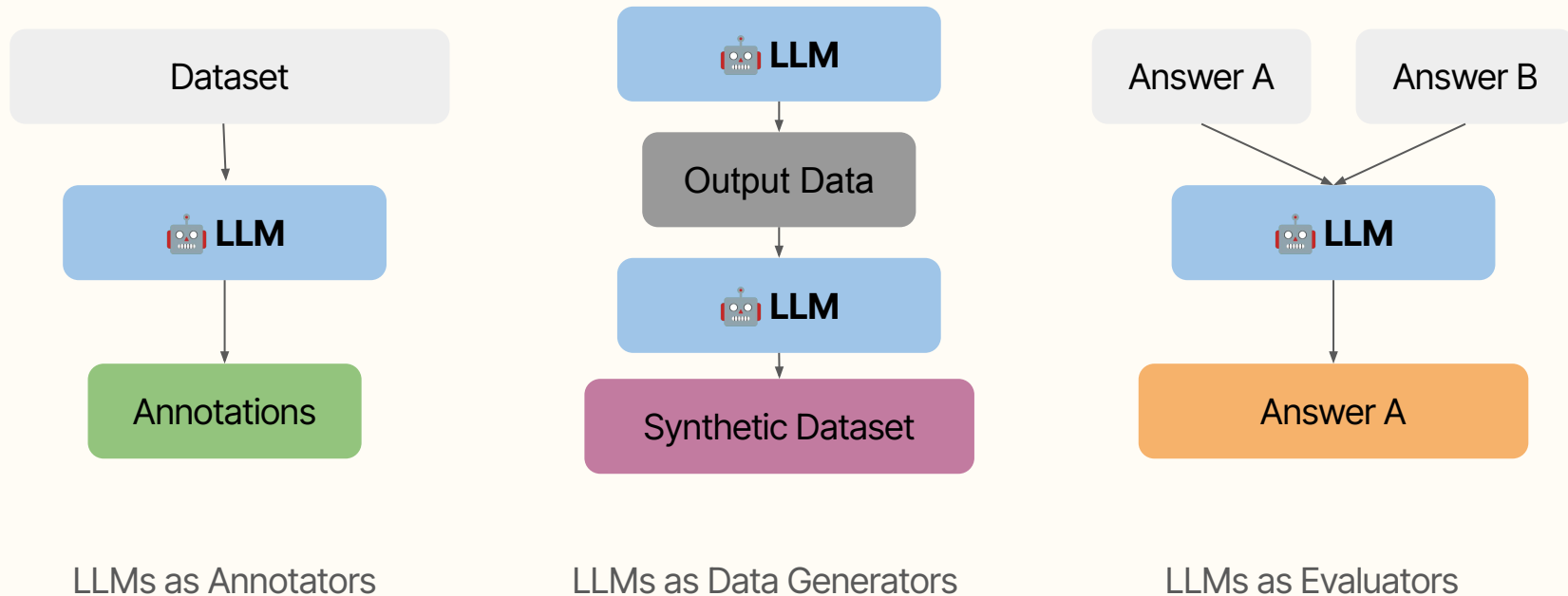
Composing Existing Knowledge and Capabilities

DataDreamer: A Framework For Synthetic Data Generation and Training Workflows

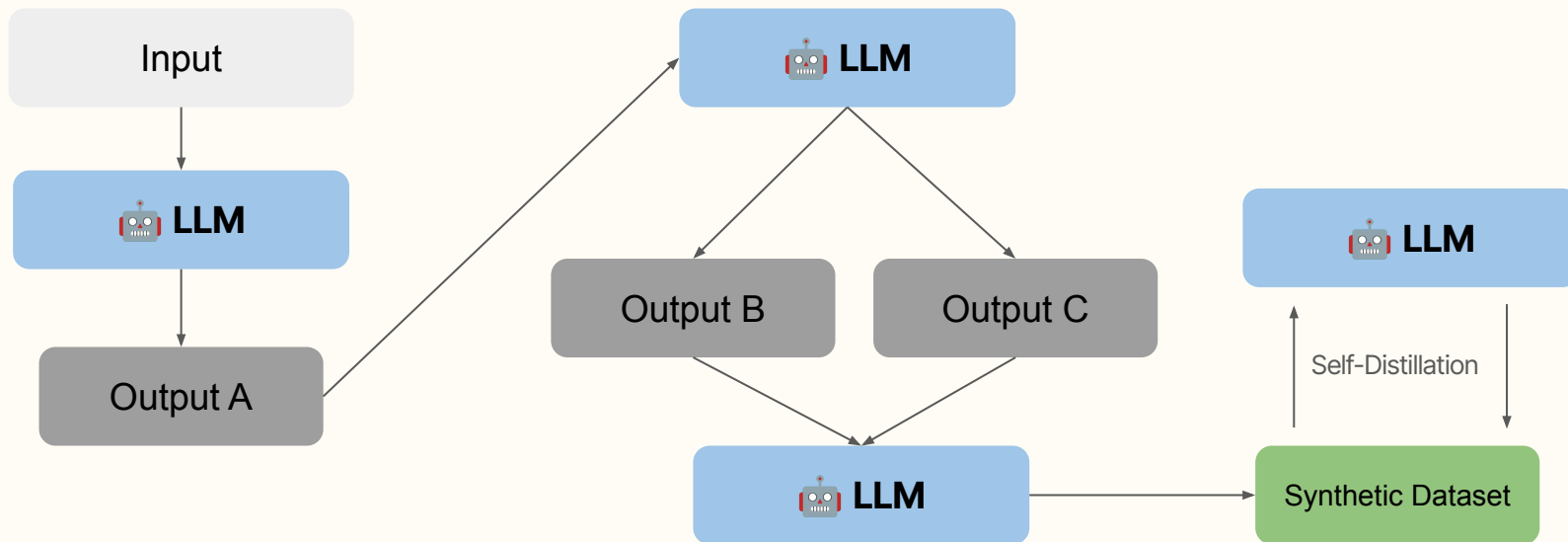


- Motivation and Current Challenges
- DataDreamer
- Impact and Case Studies
- Summary

Motivation



Motivation



Current Challenges

Multi-GPU

LLMs often require a distributed GPU setup to run or train

Slow & Expensive

Resource-intensive LLMs are slow to run and expensive to parallelize

Requires Systems Engineering

Configuring caching, quantization, parameter-efficient tuning, etc.

Increasingly Complex Workflows That Are Difficult To Reproduce / Maintain

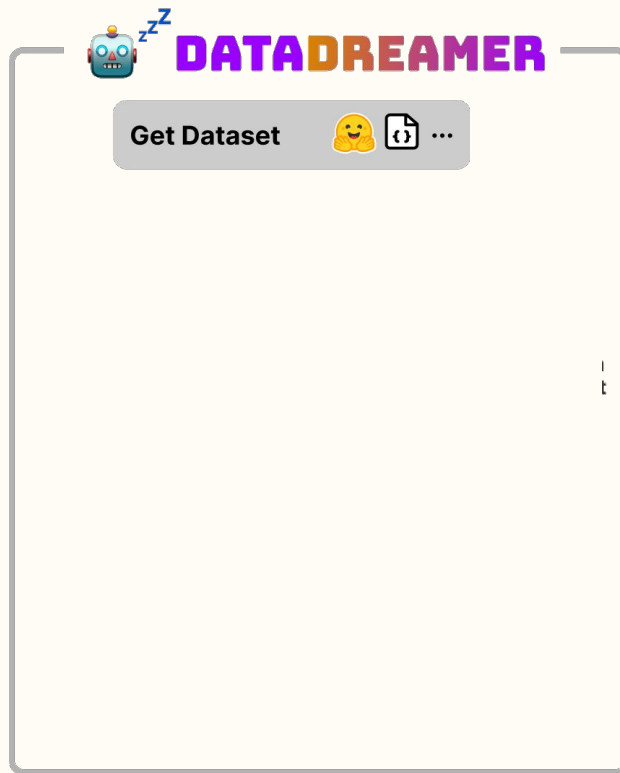
Researchers are building complex pipelines chaining multiple steps of using LLMs and training models

DataDreamer

- **A Python library that simplifies:**
 - Prompting Workflows
 - Synthetic Data Generation Workflows
 - Training & Alignment Workflows

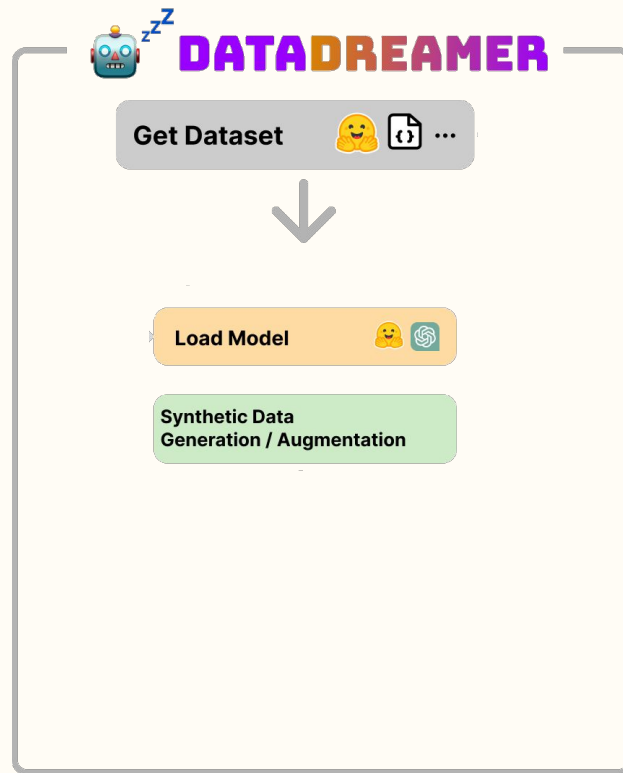
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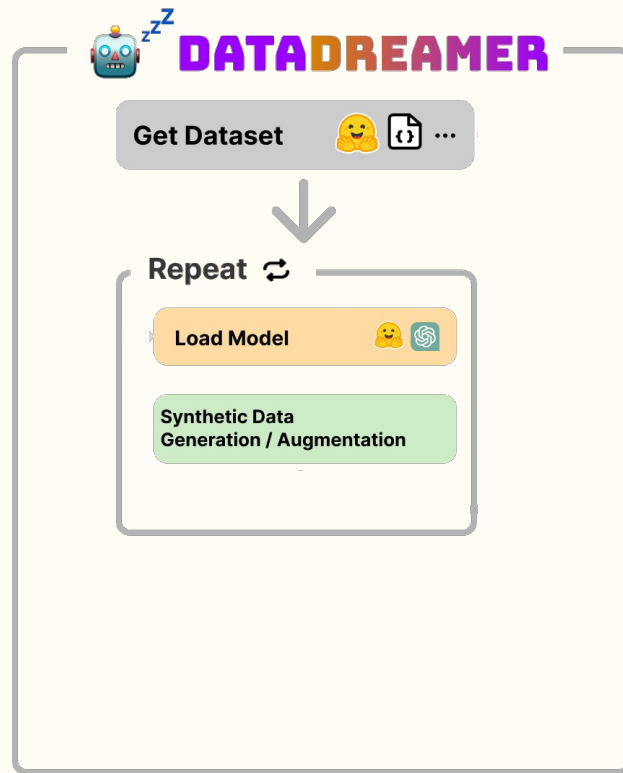
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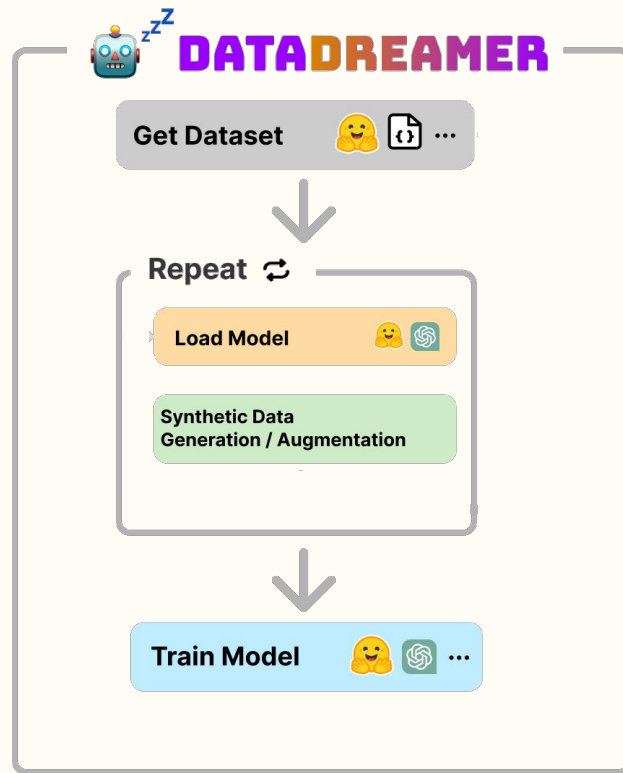
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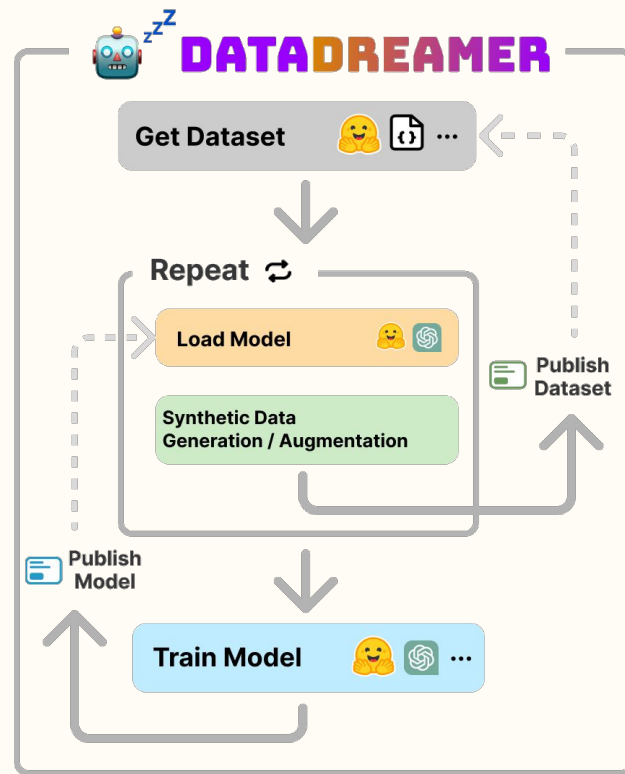
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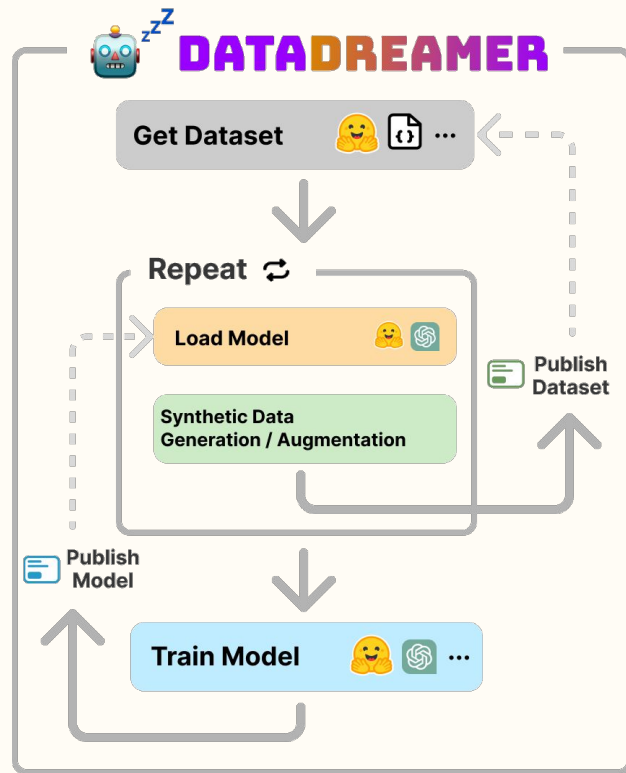
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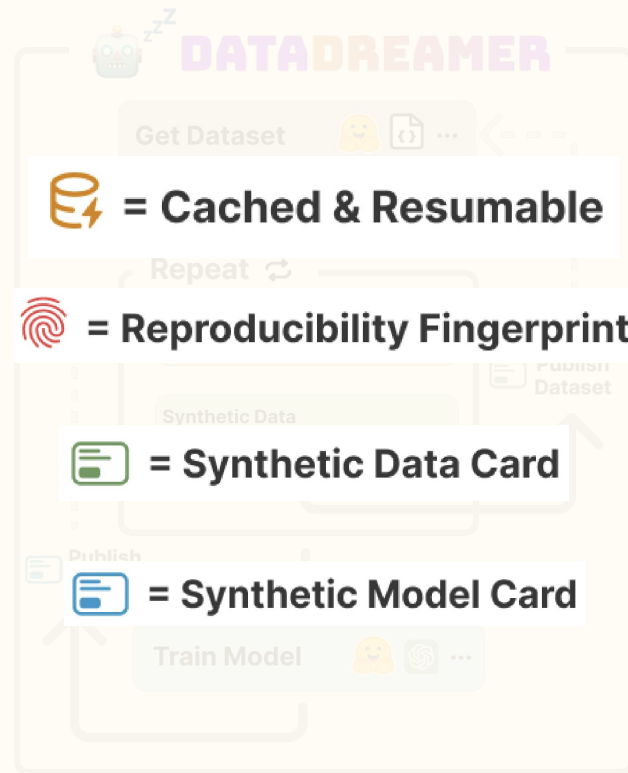
DataDreamer

- Simple code for **complex workflows** like:
 - Multi-Stage Synthetic Data Generation
 - Model Distillation
 - Instruction-Tuning
 - Alignment (RLHF & DPO)
 - Self-Improving LLMs



DataDreamer

- Simple code for **complex workflows** like:
 - Multi-Stage Synthetic Data Generation
 - Model Distillation
 - Instruction-Tuning
 - Alignment (RLHF & DPO)
 - Self-Improving LLMs



DataDreamer



ANTHROPIC

... and more

DataDreamer

Efficient Inference Techniques

Multiple GPUs, Quantization, etc...

Efficient Training Techniques

Multiple GPUs, PyTorch FSDP, QLora, etc...

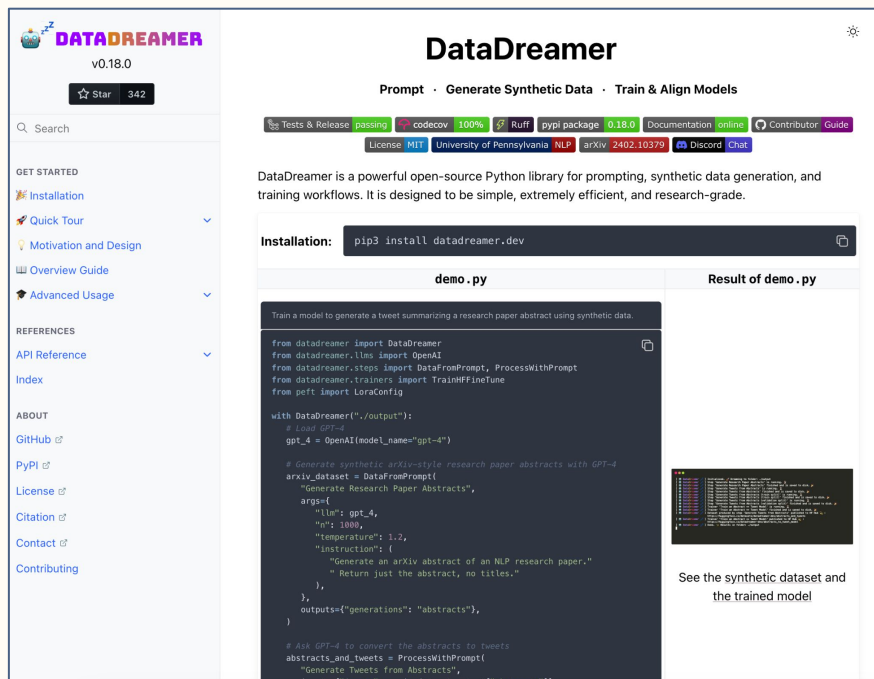
Infrastructure and Boilerplate

Caching, Resumability, Checkpointing,
Publishing, etc...

Commonly Used Workflow Steps

Few-Shot Prompting, Fine-Tuning, DPO
Alignment, etc...

Impact and Case Studies



<https://datadreamer.dev>



ACL 2024
Bangkok, Thailand

Open science, open data, and open models for reproducible NLP research



1000+
GitHub Stars

Used to generate the **PixMo-Docs** synthetic instruction-tuning data for



(Patel et al., 2024)

Summary

DataDreameer helps practitioners **implement** complex synthetic data generation and training workflows **easily**.

Inductive Biases

Task Priming

Task Asymmetry

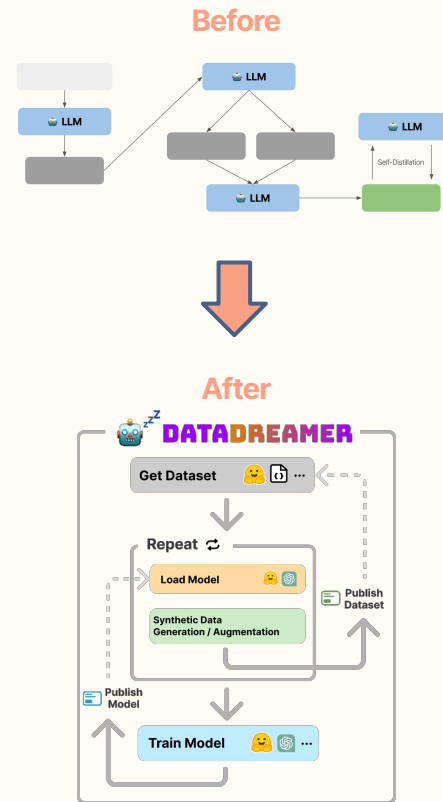
Task Rules and Heuristics

Task Structuring

Sources of Knowledge

Surfacing Latent Knowledge and Capabilities

Composing Existing Knowledge and Capabilities



(Patel et al., 2024)

Enabling New Model Capabilities by Fine-Tuning on Synthetic Data

- **Example 1:** Training Style Embeddings
- **Example 2:** Making Agent Models More Capable
- **Example 3:** Enabling Vision Language Models to Understand Text-Rich Figures

Enabling New Model Capabilities by Fine-Tuning on Synthetic Data

Example 1/3

Training Style Embeddings

Motivation

Text embeddings typically embed two texts close together if they have similar meanings.

```
cosine_similarity(  
    "I h8 that game.",  
    "I want to go too."  
) = 0.43
```

Different Content

```
cosine_similarity(  
    "I h8 that game.",  
    "I hate that game."  
) = 0.98
```

Same Content

Motivation

Style embeddings embed two texts close together if they share the **same style**, regardless of content.

```
cosine_similarity(  
    "I h8 that game.",  
    "I want 2 go too."  
) = 0.96
```

**Same Style
Different Content**

```
cosine_similarity(  
    "I h8 that game.",  
    "I want to go too."  
) = 0.43
```

**Different Style
Different Content**

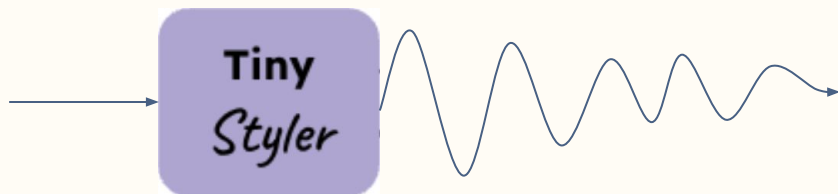
```
cosine_similarity(  
    "I h8 that game.",  
    "I hate that game."  
) = 0.38
```

**Different Style
Same Content**

Motivation

Informal Style

*"its keeping me
up at nite, i
have to know
what it is"*



Formal Style

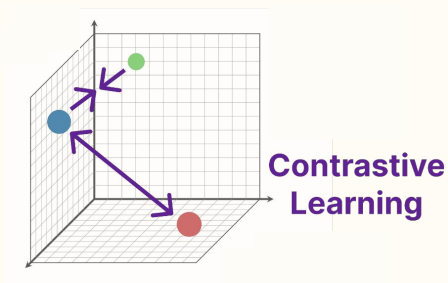
*"What is it? It is
keeping me up at
night."*

**Steered by
Style Embeddings**

(Horvitz et al., 2024; Patel et al., 2024)

Motivation

- To train embeddings, we need “triplets” as training data:
 - ● **Anchor Text**
 - ● **Positive Text** - That should be embedded **close** to the anchor text. (*same style*)
 - ● **Negative Text** - That should be embedded **far from** the anchor text. (*different style*)



- Acquiring such data for training style embeddings is challenging.
 - There are very few datasets with positive and negative parallel examples of a style feature.

Procedure

We create **StyleDistance** embeddings by using LLMs to generate **synthetic positive and negative examples** of 40 style features for precise contrastive learning.

Style Feature: Usage of Active Voice

Anchor:

● *"I adored the Dolce & Gabbana mauve leather creation with a signature lock."*

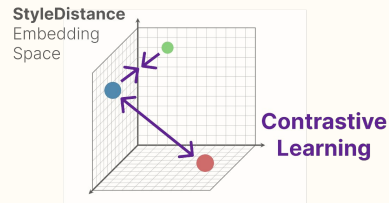
Positive & Negative Examples:

● *"I observed the impact rising temperatures had on tropical diseases."*

Same Style
Different Content

● *"The Dolce & Gabbana mauve leather creation with a signature lock was adored by me."*

Same Content
Different Style



Procedure

40 Style Features

Syntactic Features

- Usage of Conjunctions
- Usage of Articles
- Frequent Usage of Function Words
- Usage of Personal Pronouns
- Usage of Pronouns
- Usage of Active Voice
- Usage of Contractions
- Frequent Usage of Determiners
- Usage of Prepositions

Graphical and Digital Features

- Usage of Numerical Substitution
- Usage of Uppercase Letters
- Usage of Text Emojis
- Usage of Emojis
- Presence of Misspelled Words
- Usage of Only Uppercase Letters
- Usage of Only Lowercase Letters
- Usage of Numerical Digits
- Frequent Usage of Punctuation

Emotional and Cognitive Features

- Positive Sentiment Expression
- Usage of Words Indicating Affective vs. Perceptual Processes
- Usage of Words Indicating Cognitive vs. Perceptual Processes
- Usage of Words Indicating Affective vs. Cognitive Processes
- Usage of Certain Tone

Social and Interpersonal Features

- Usage of Offensive Language
- Usage of Self-Focused Language vs. You-Focused
- Usage of Self-Focused Perspective vs. Third-person Singular
- Usage of Self-Focused Language vs. Inclusive-focused
- Usage of Self-Focused Language vs. Audience-focused
- Usage of Polite Tone

Temporal and Aspectual Features

- Usage of Present-focused vs. Future-focused
- Usage of Present-focused vs. Past-focused

Stylistic and Aesthetic Features

- Usage of Metaphors
- Usage of Formal Tone
- Incorporation of Humor
- Fluency in Sentence Construction
- Complex Sentence Structure
- Usage of Sarcasm

Lexical Features

- Usage of Long Words
- Usage of Nominalizations
- Frequent Usage of Common Verbs

Procedure

40 Style Features

Syntactic Features

- Usage of Conjunctions
- Usage of Articles
- Frequent Usage of Function Words
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- Fluency in Sentence Construction
- Complex Sentence Structure
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Task Rules and Heuristics

Social and Pragmatic Features

- Usage of Informal Language
- Usage of Self-Focused Language vs. You-Focused
- Usage of Self-Focused Perspective vs. Third-person Singular
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Temporal and Aspectual Features

- Usage of Present-focused vs. Future-focused
- Usage of Present-focused vs. Past-focused

Procedure

40 style features x **100** parallel examples

Procedure

40 style features x **100** parallel examples



~**320K** unique synthetic training triplets

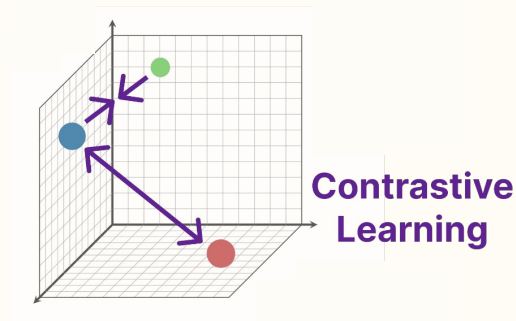
Procedure

$$L_t(\theta) = \sum_{(\text{anchor}, \text{pos}, \text{neg}) \in \mathcal{D}} [\|f_\theta(\text{anchor}) - f_\theta(\text{positive})\|_2^2 - \|f_\theta(\text{anchor}) - f_\theta(\text{negative})\|_2^2 + \text{margin}]_+$$

40 style features x **100** parallel examples

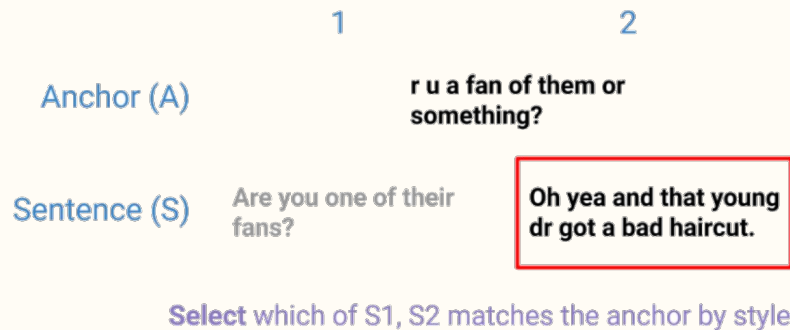


~320K unique synthetic training triplets



Evaluation

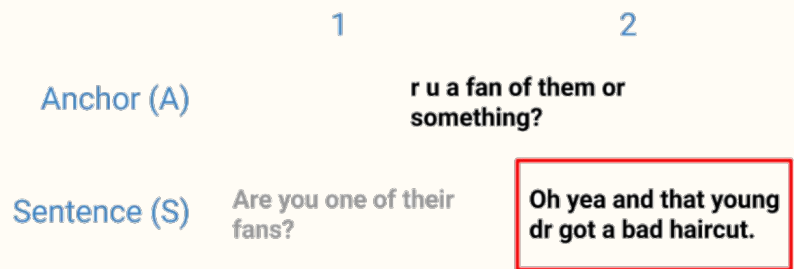
Style-or-Content Benchmark



(Wegmann and Nguyen, 2021; Wegmann et al., 2022)

Evaluation

Style-or-Content Benchmark

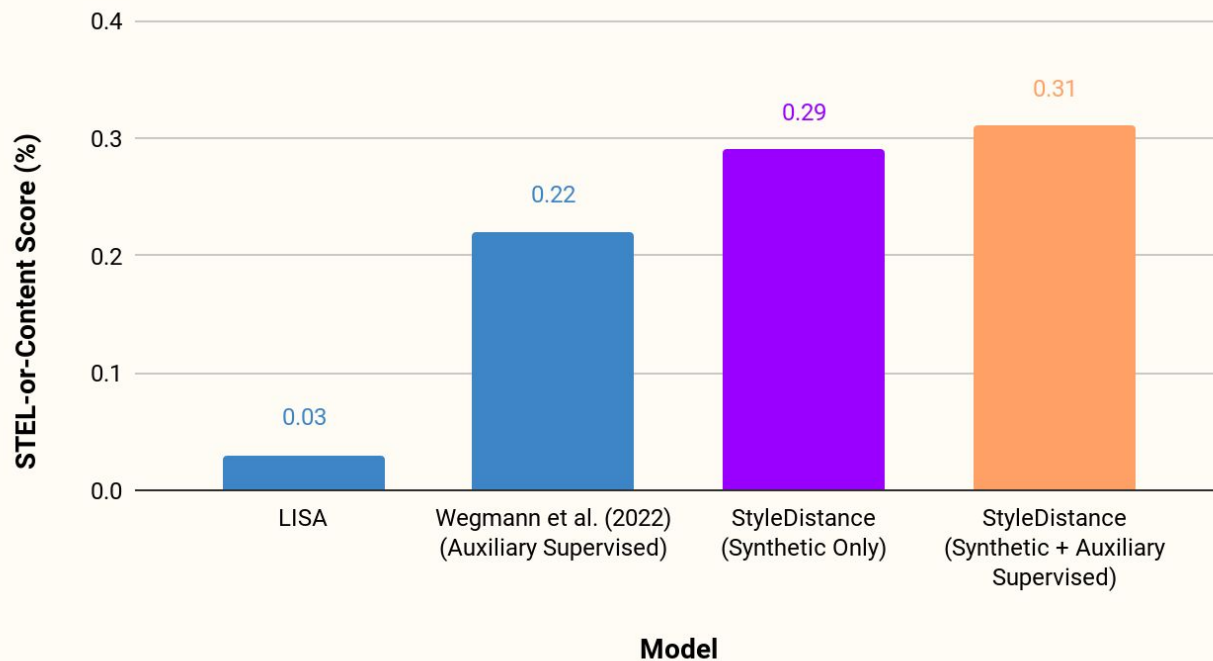


Select which of S1, S2 matches the anchor by style



In order to succeed, the model must represent **style features** *stronger* than **content features**.

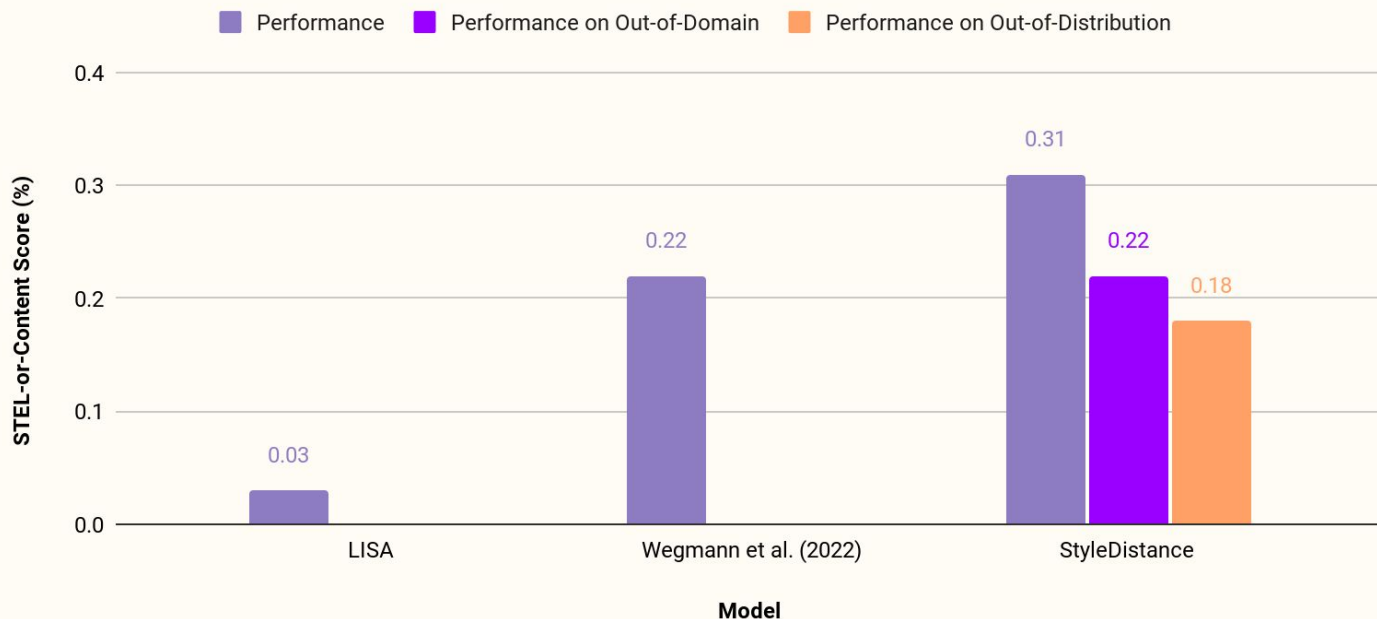
Results



(Patel et al., 2024)

Generalization Results

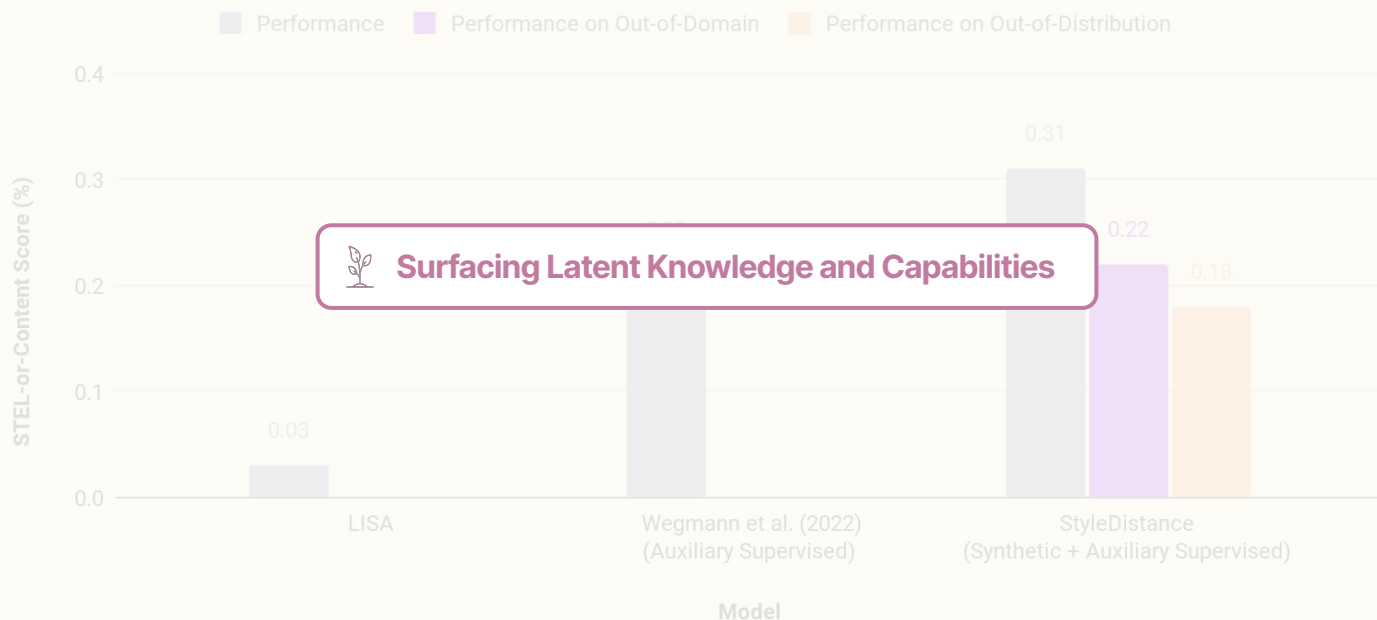
We also evaluate for **generalization** and test *out-of-domain* and *out-of-distribution* performance.



(Patel et al., 2024)

Generalization Results

We also evaluate for **generalization** and test *out-of-domain* and *out-of-distribution* performance.



Surfacing Latent Knowledge and Capabilities

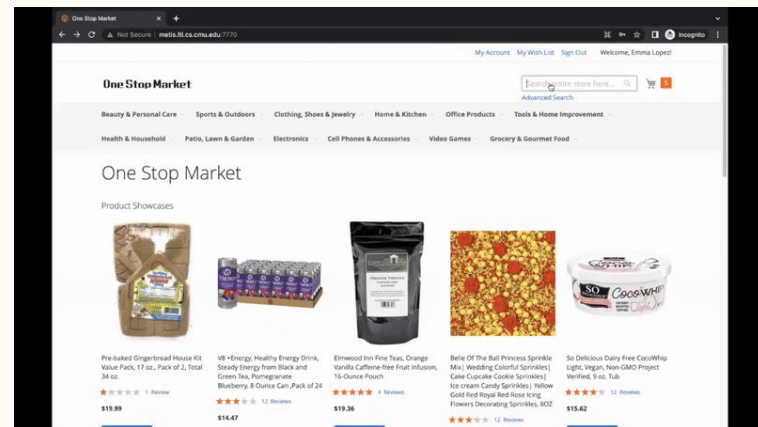
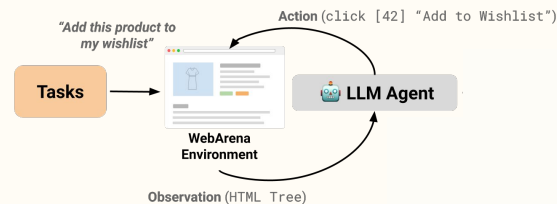
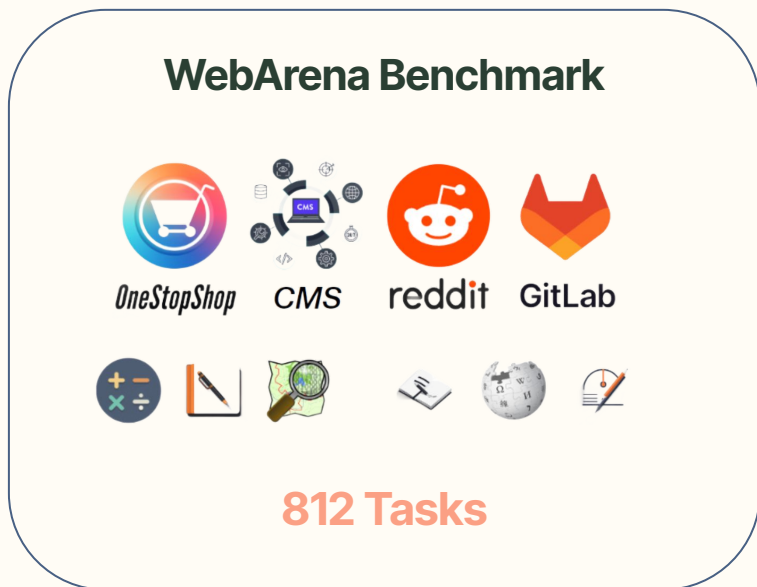
Enabling New Model Capabilities by Fine-Tuning on Synthetic Data

Example 2/3

Making Agent Models More Capable

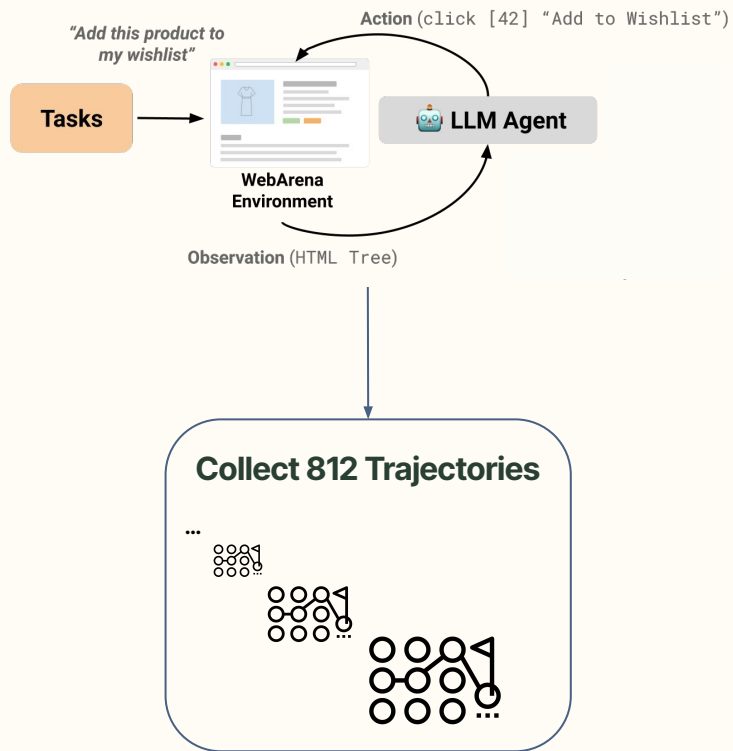
Motivation

LLMs struggle to perform **long-horizon tasks** as agents in an environment.

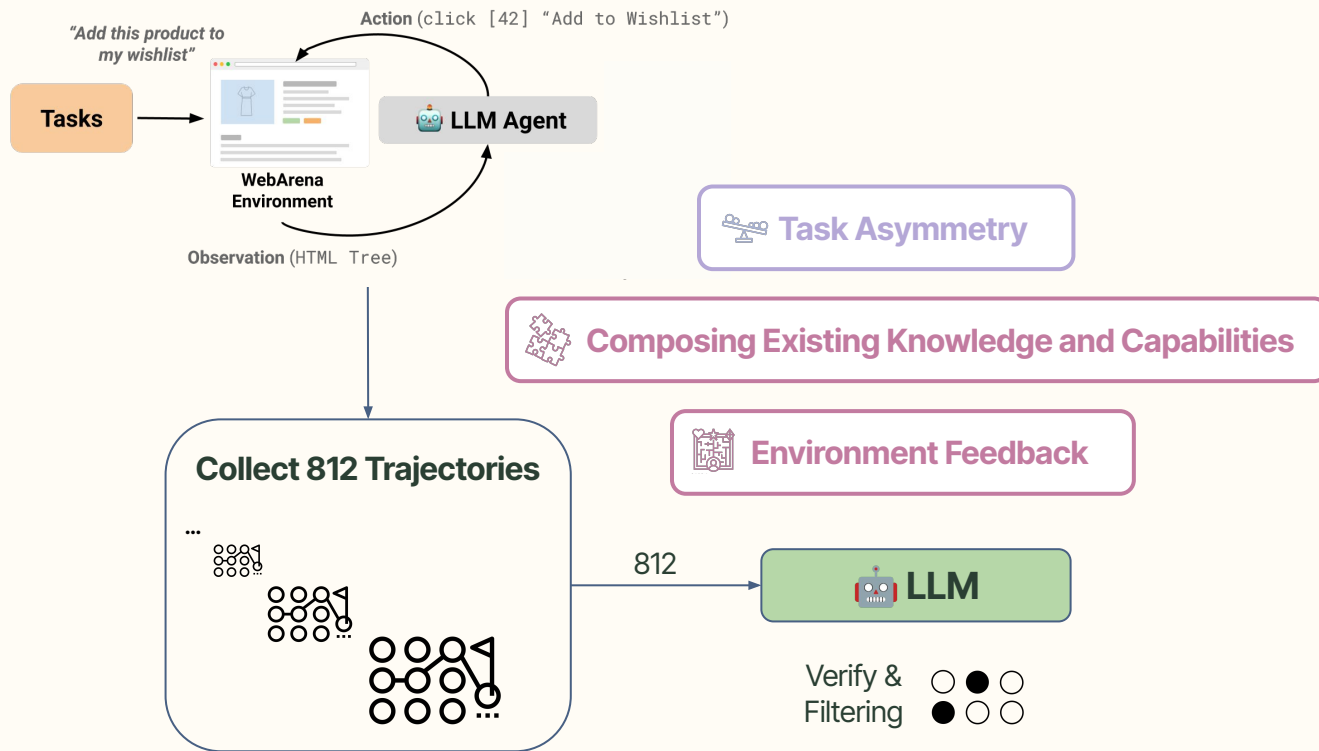


(Zhou et al., 2024)

Procedure

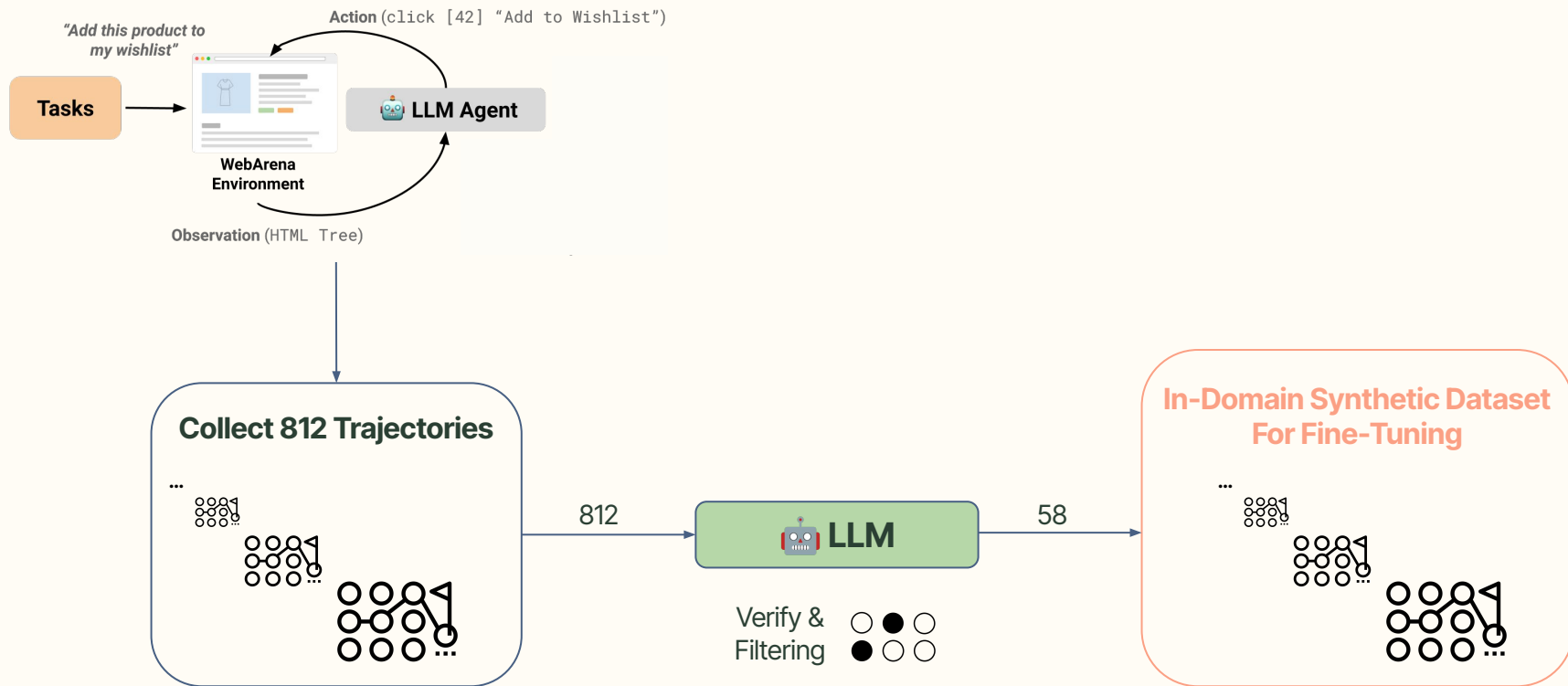


Procedure



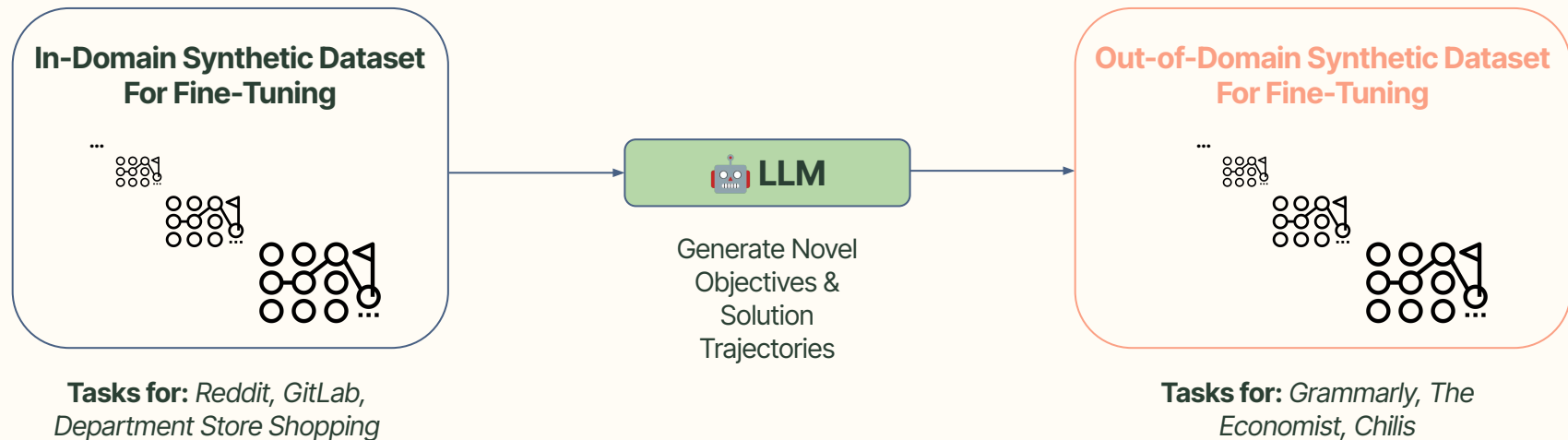
(Patel et al., 2024)

Procedure

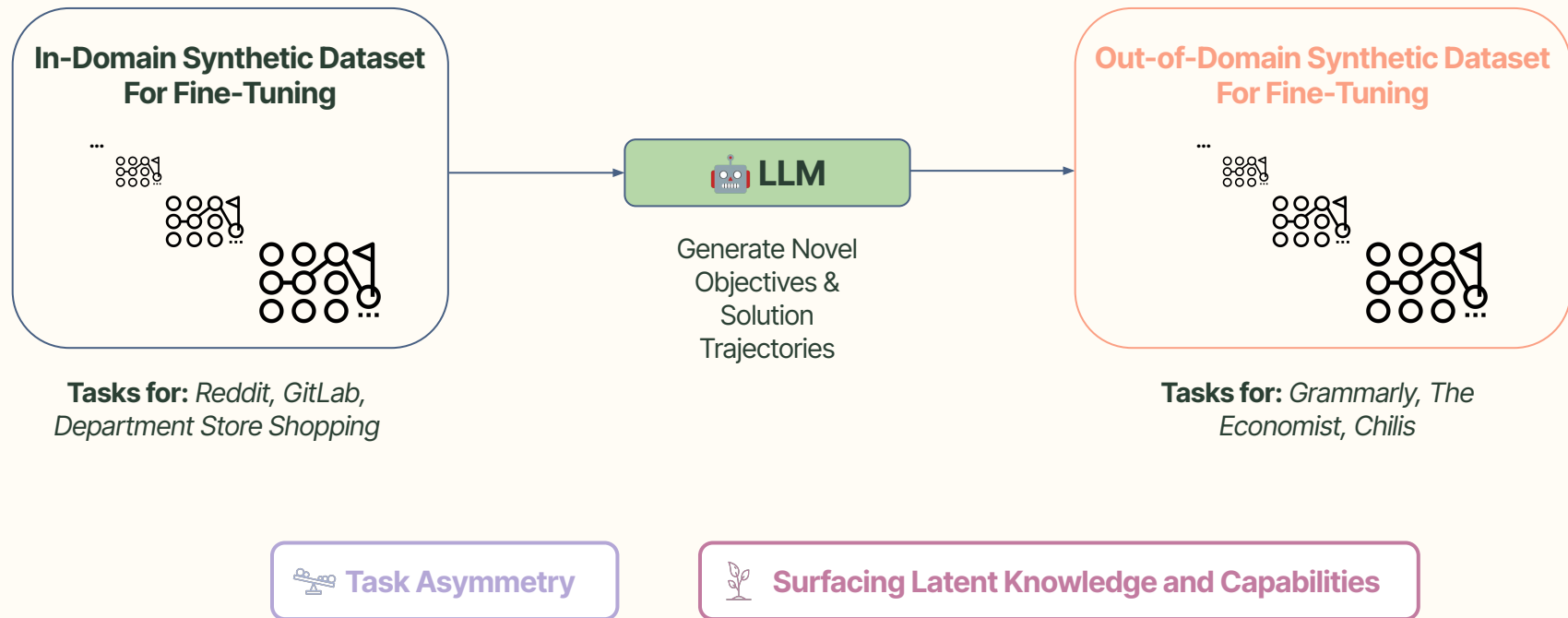


(Patel et al., 2024)

Procedure



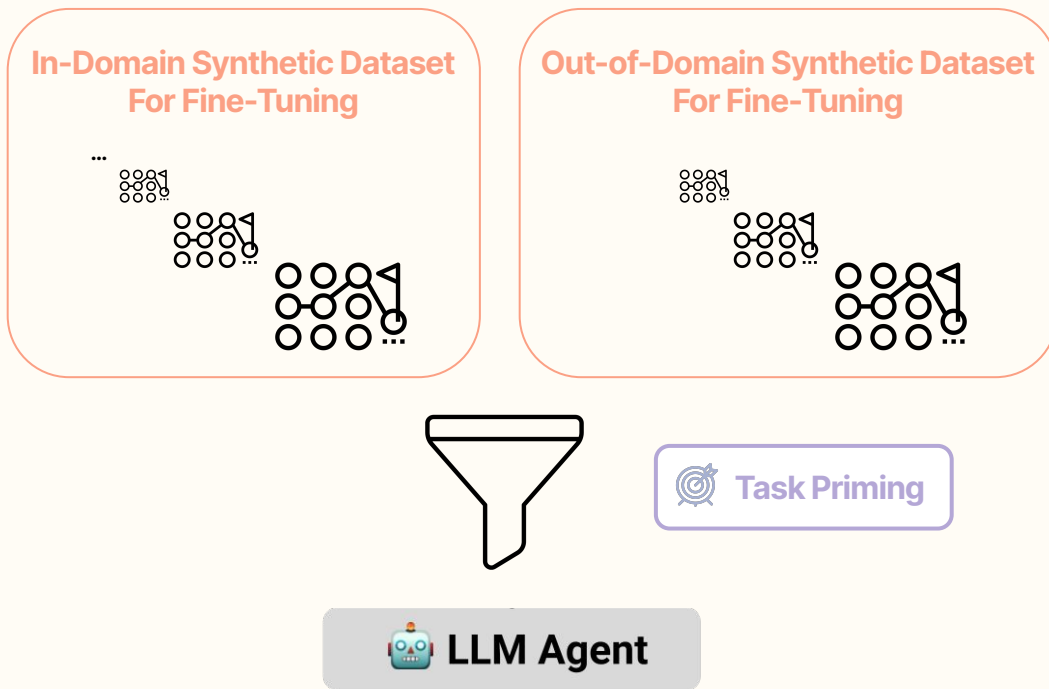
Procedure



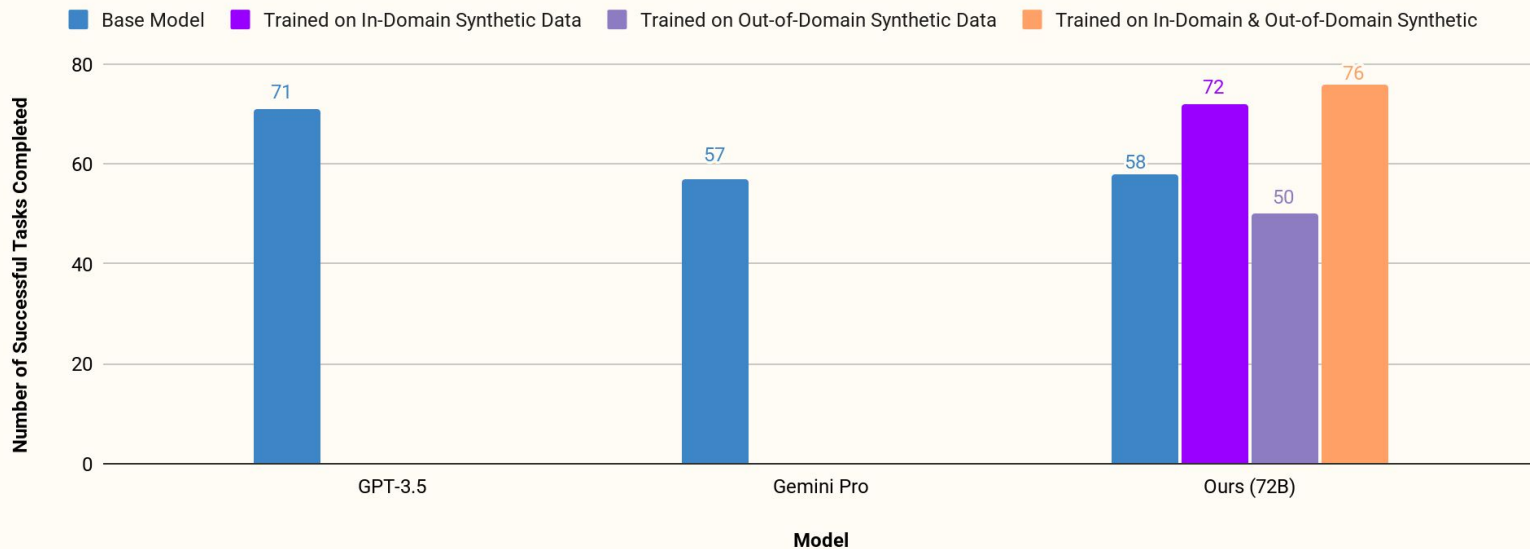
(Patel et al., 2024)

Procedure

$$L_{\text{FT}}(\theta) = -\mathbb{E}_{[(\text{task}, \text{obs}_t, \text{action}_{t-1}), \text{action}_t] \sim \mathcal{D}_{\text{dataset}}} [\log P_{\theta}(\text{action}_t \mid (\text{task}, \text{obs}_t, \text{action}_{t-1}))]$$



Results

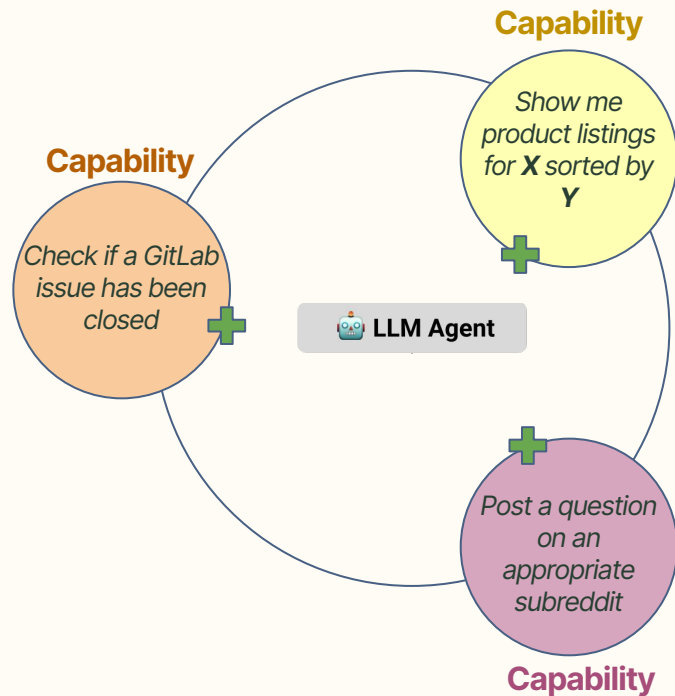
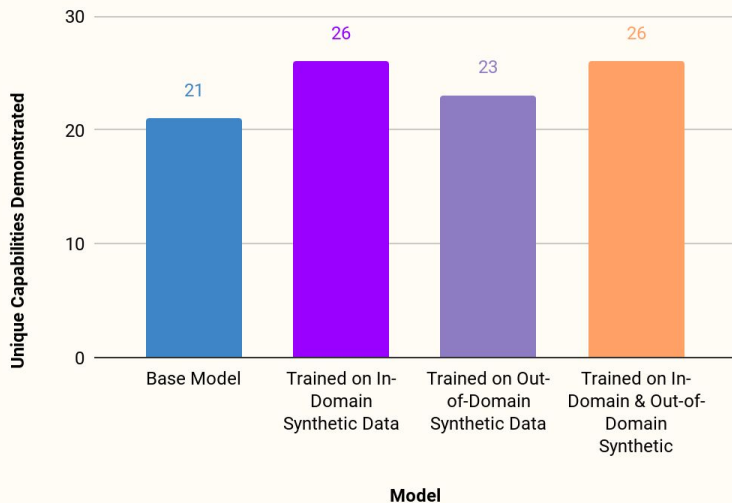


on 812 tasks in the WebArena benchmark

(Patel et al., 2024)

Results

After training on synthetic data, the model acquires **new capabilities**.



(Patel et al., 2024)

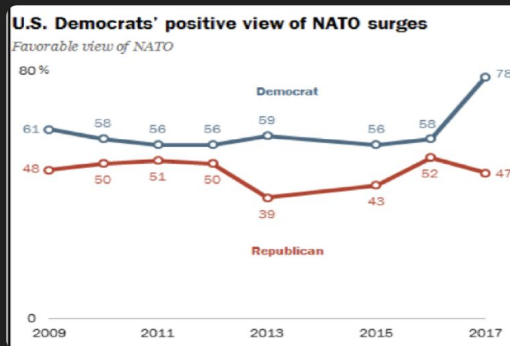
Enabling New Model Capabilities by Fine-Tuning on Synthetic Data

Example 3/3

Enabling Vision Language Models to
Understand Text-Rich Figures

Motivation

Current vision-language models (VLMs) struggle to understand **text-rich** images.



According to this chart what year was Republican support of NATO the most positive?



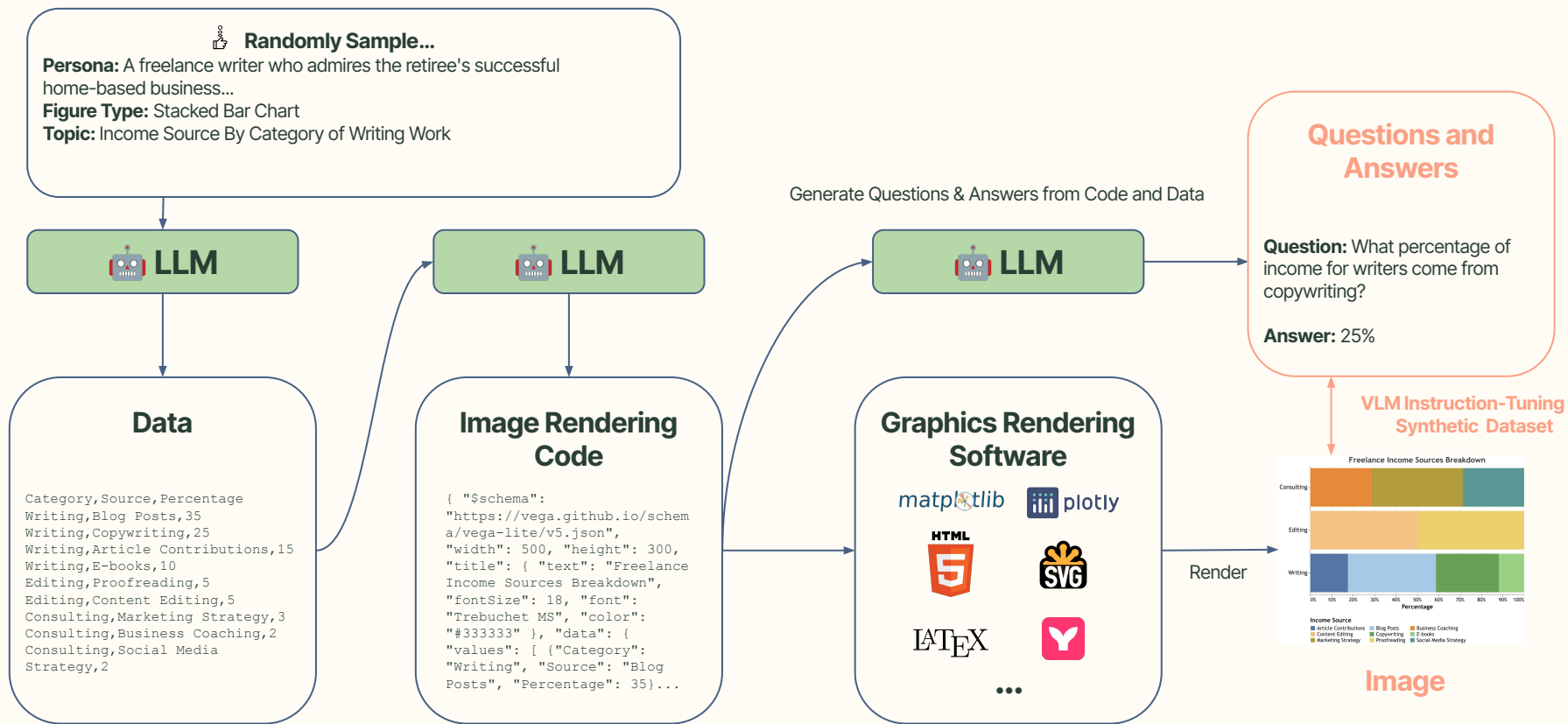
According to the chart, Republican support for NATO was most positive in 2011, where it reached 51%.



← Wrong Answer ❌

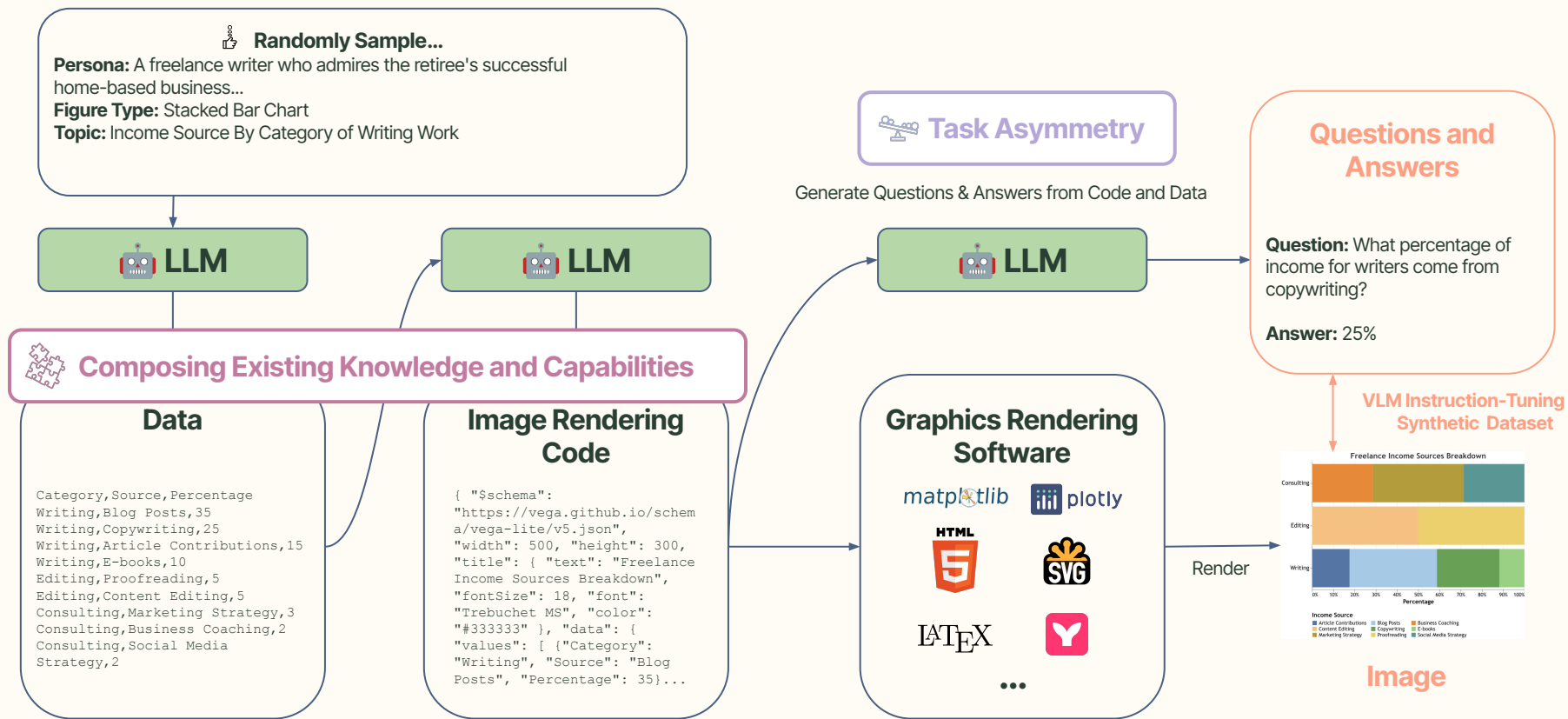
(Yang et al., 2024)

Procedure



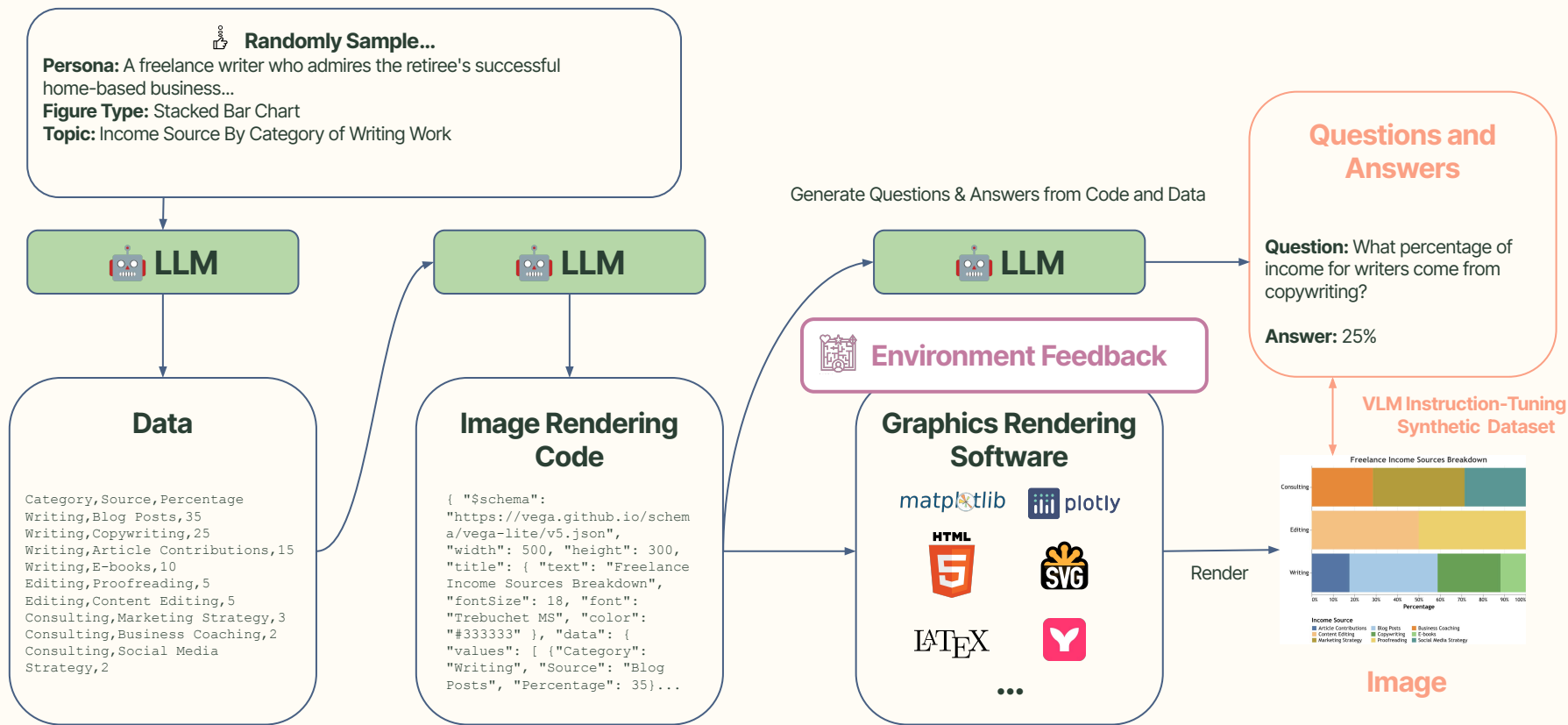
(Yang et al., 2024)

Procedure



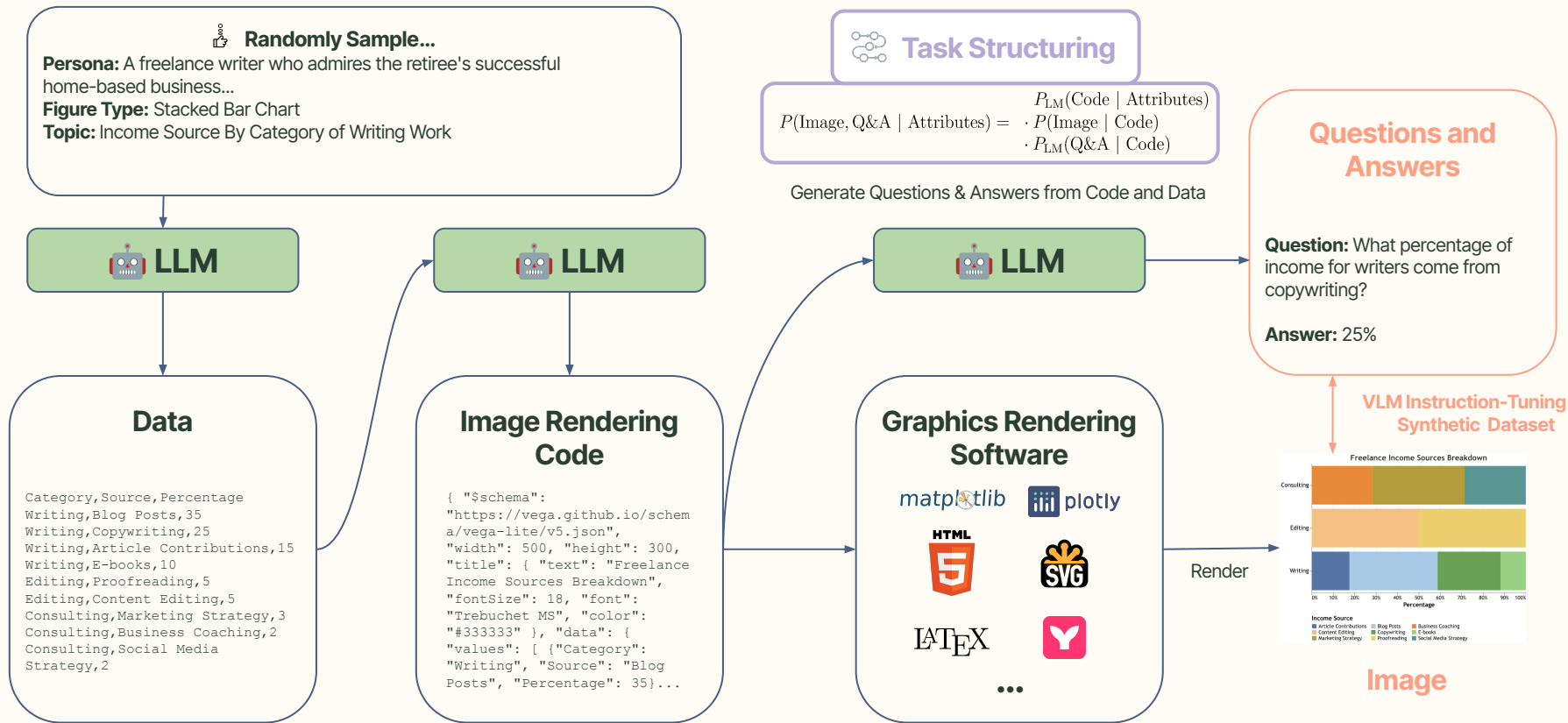
(Yang et al., 2024)

Procedure



(Yang et al., 2024)

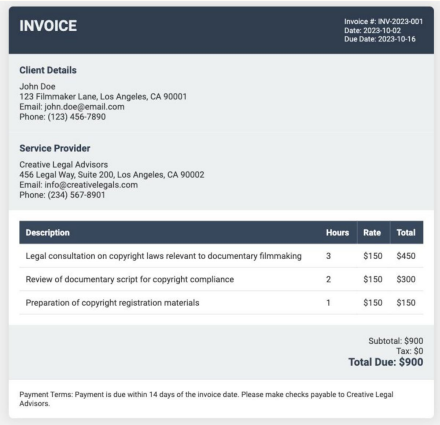
Procedure



(Yang et al., 2024)

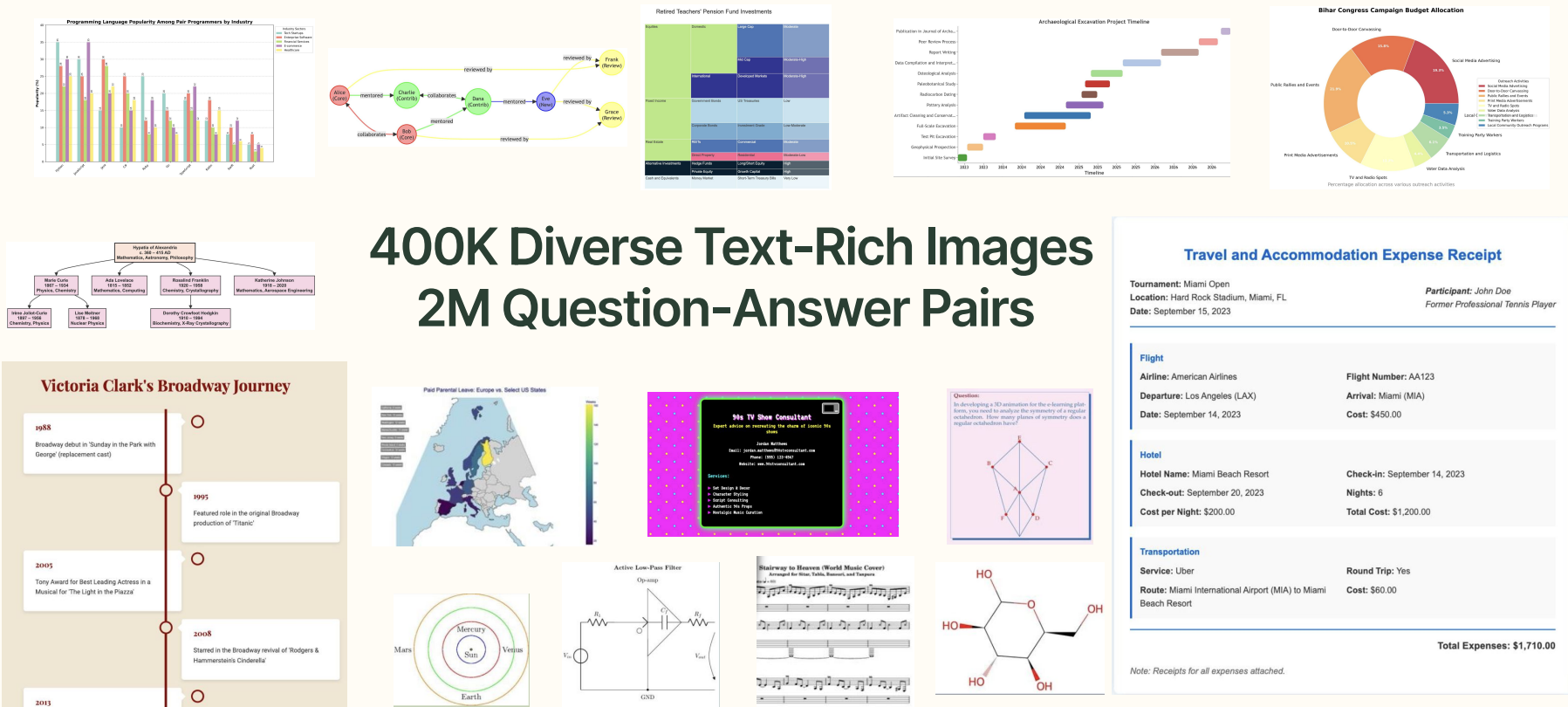
Synthetic Vision-QA Dataset

Each synthetically generated image is paired with high-quality questions and answers for vision instruction-tuning.

Image	Question	Answer
	What is the invoice number?	INV-2023-001
	What is the rate per hour for the legal consultation?	\$150
	What payment terms are mentioned?	Payment is due within 14 days of the invoice date
	...	...

(Yang et al., 2024)

Synthetic Vision-QA Dataset



400K Diverse Text-Rich Images 2M Question-Answer Pairs

Travel and Accommodation Expense Receipt

Tournament: Miami Open
Location: Hard Rock Stadium, Miami, FL
Date: September 15, 2023

Participant: John Doe
Former Professional Tennis Player

Flight
Airline: American Airlines
Departure: Los Angeles (LAX)
Date: September 14, 2023

Flight Number: AA123
Arrival: Miami (MIA)
Cost: \$450.00

Hotel
Hotel Name: Miami Beach Resort
Check-out: September 20, 2023
Cost per Night: \$200.00

Check-in: September 14, 2023
Nights: 6
Total Cost: \$1,200.00

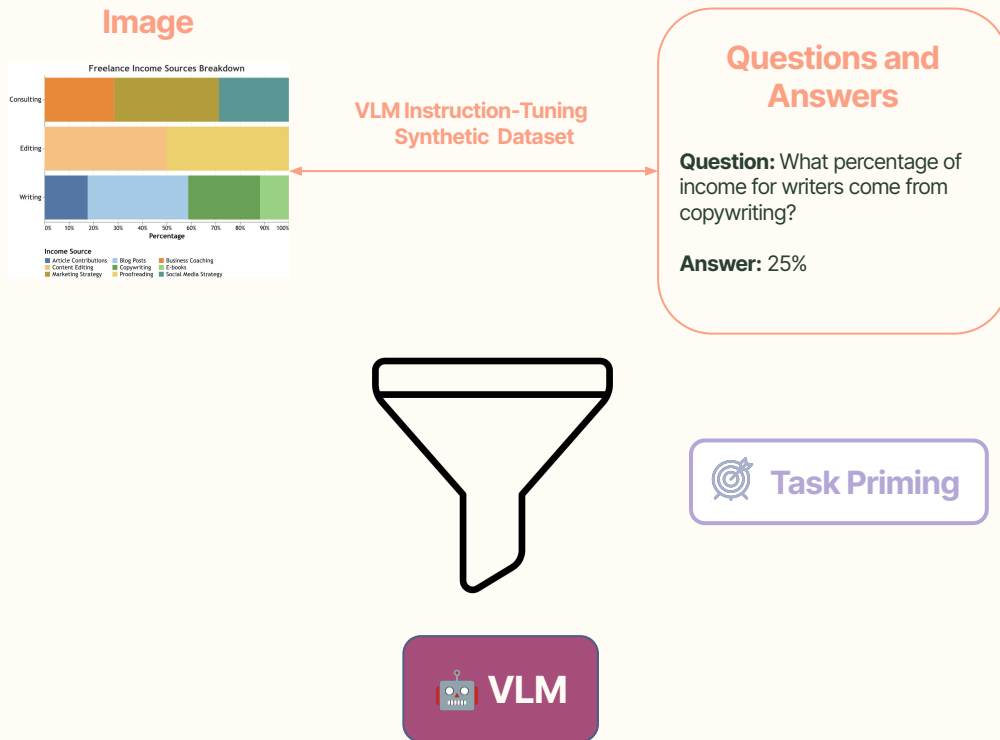
Transportation
Service: Uber
Route: Miami International Airport (MIA) to Miami Beach Resort

Round Trip: Yes
Cost: \$60.00

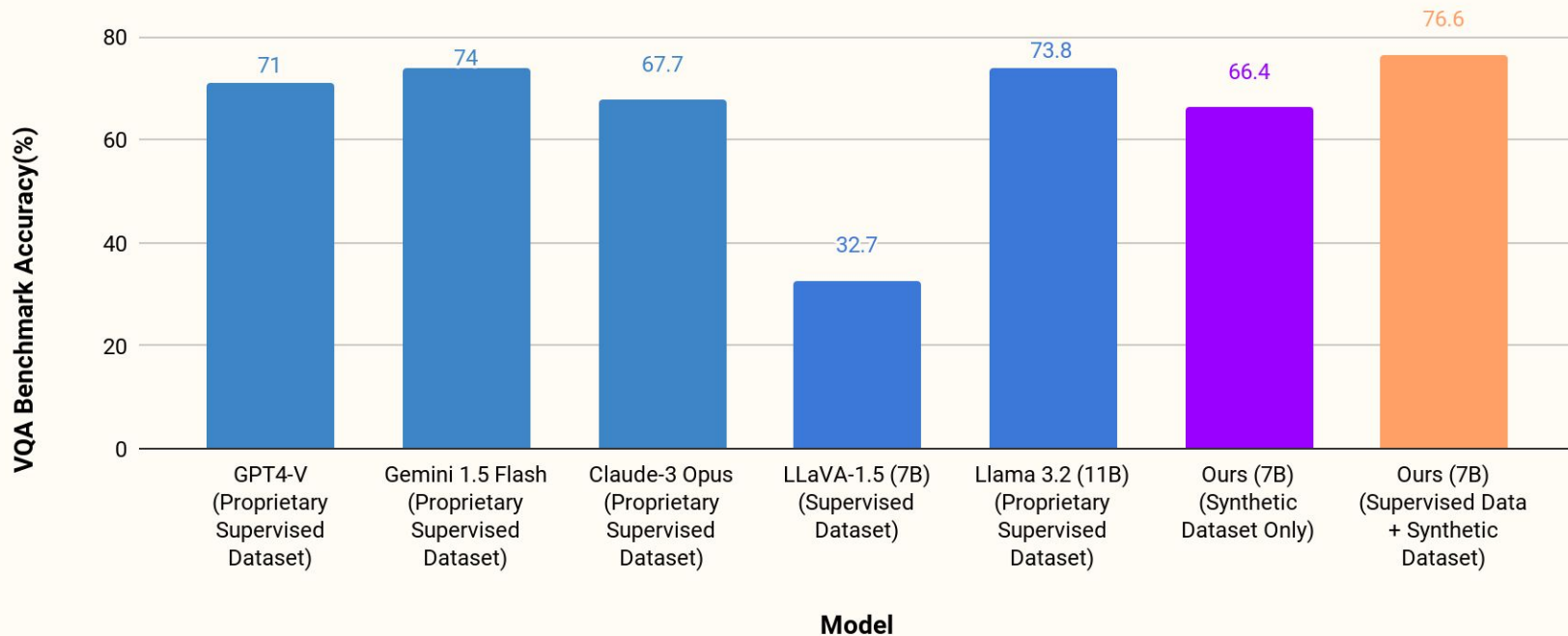
Total Expenses: \$1,710.00

Note: Receipts for all expenses attached.

Vision-Language Instruction-Tuning



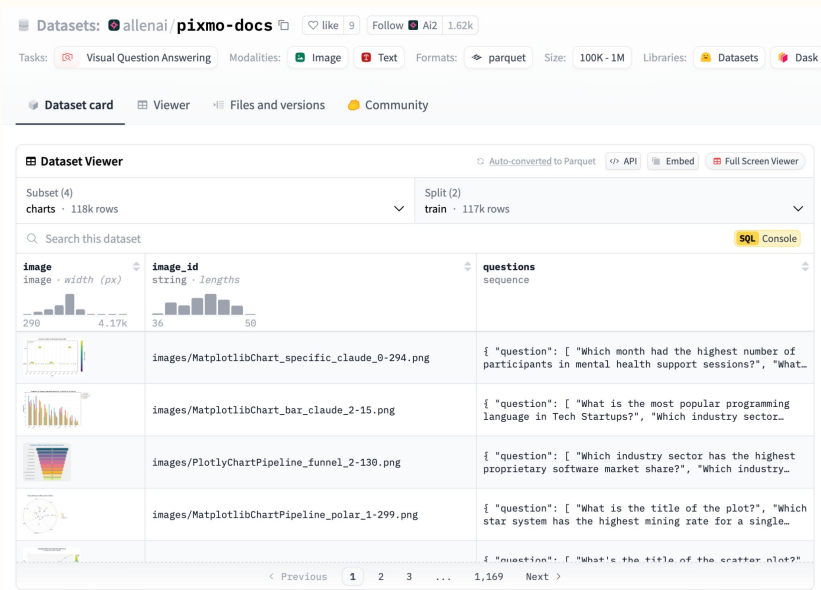
Results



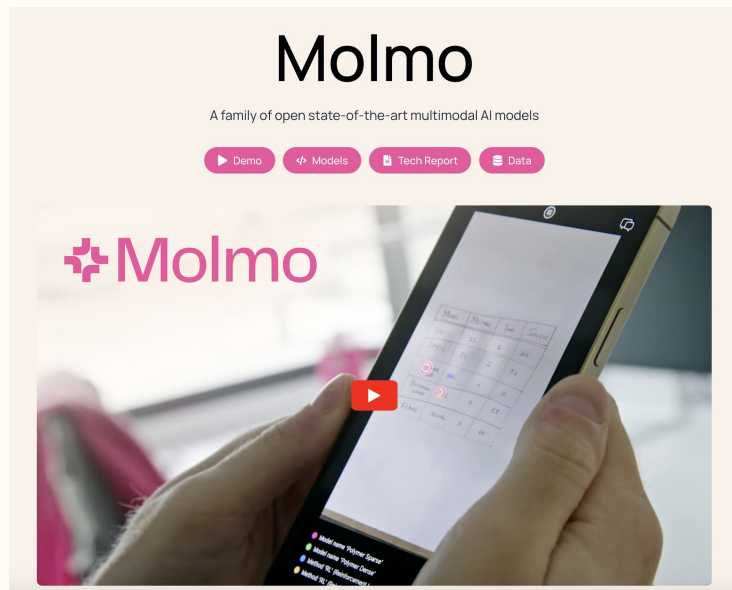
Average over: ChartQA, DocVQA, InfoVQA, TableVQA, AI2D, TextVQA, ScreenQA, MathVista

(Yang et al., 2024)

Results



PixMo-Docs Dataset

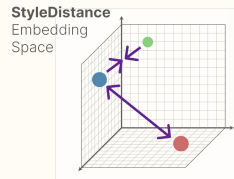


Molmo Multimodal Model

(Deitke et al., 2024)

Summary

Embedding Models



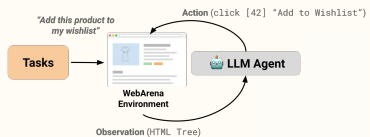
Inductive Biases

Task Rules and Heuristics

Sources of Knowledge

Surfacing Latent Knowledge and Capabilities

Agent Models



Task Priming

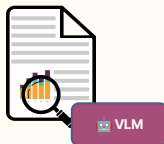
Task Asymmetry

Surfacing Latent Knowledge and Capabilities

Composing Existing Knowledge and Capabilities

Environment Feedback

Vision-Language Models



Task Priming

Task Asymmetry

Task Structuring

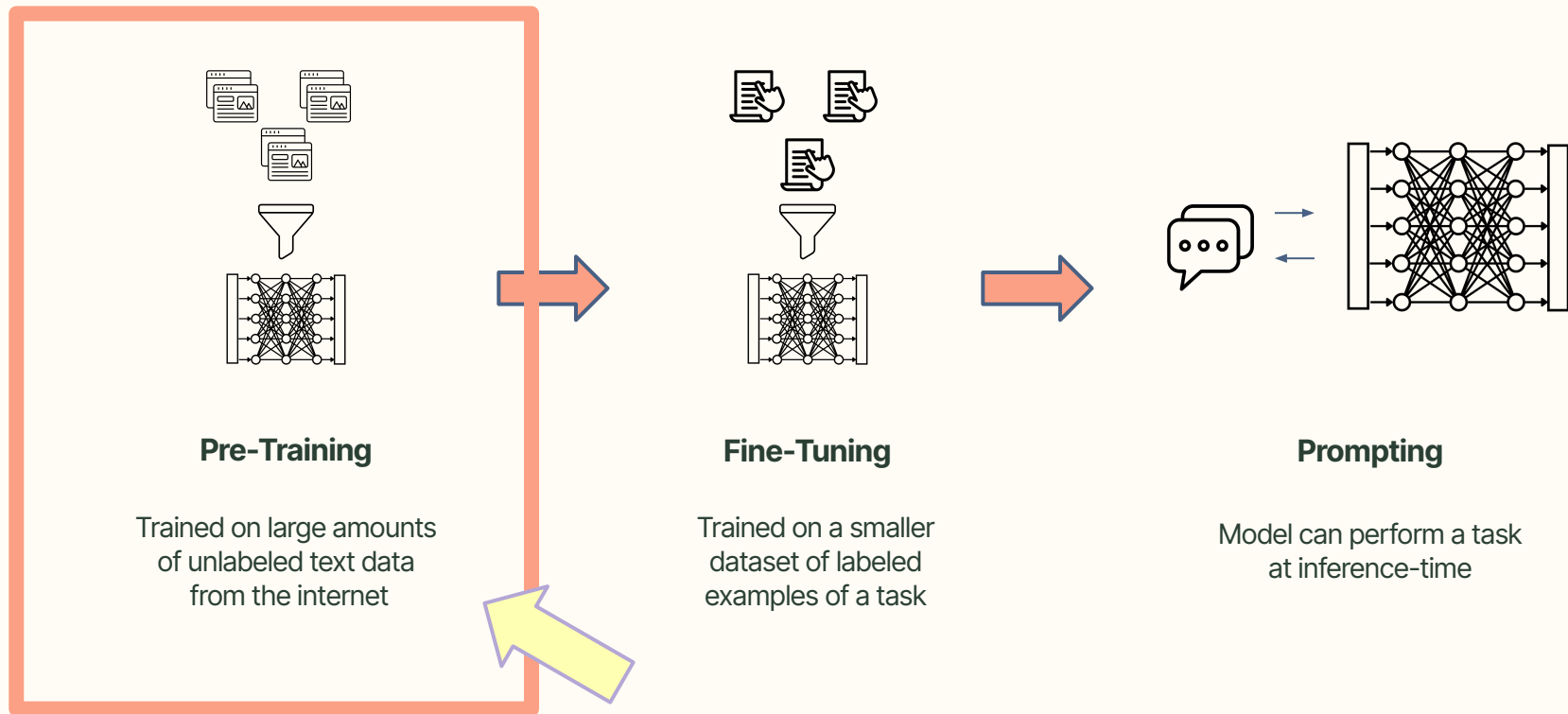
Composing Existing Knowledge and Capabilities

Environment Feedback

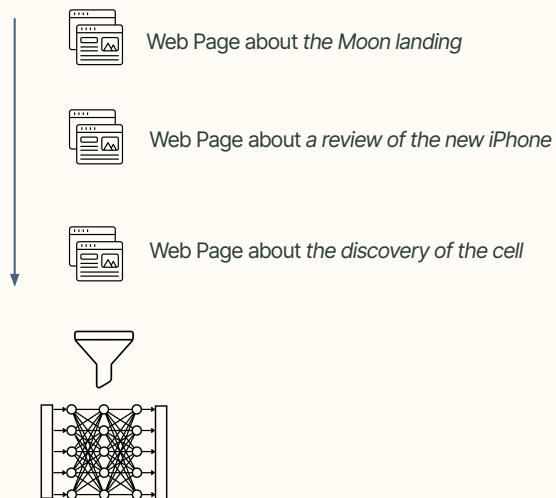
Advancing Large Language Model Pre-Training with Synthetic Data

- Motivation
- Procedure & Experimental Setup
- Results
- Future Work
- Summary

Motivation



Motivation

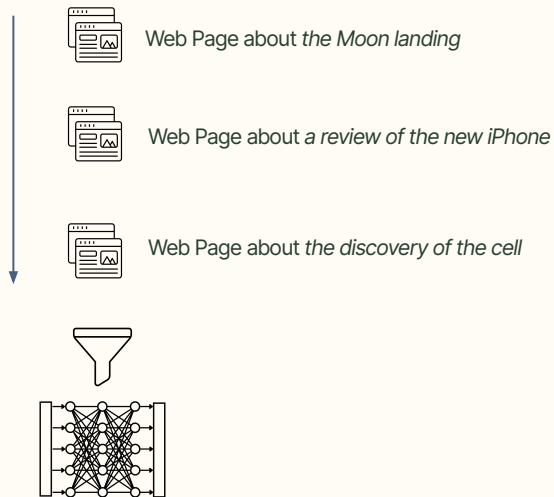


Pre-Training

Trained on large amounts
of unlabeled text data
from the internet

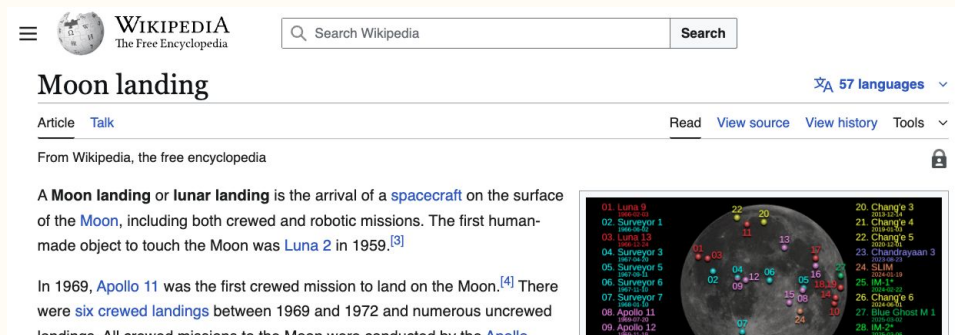
(Radford et al., 2019; Firth 1957)

Motivation

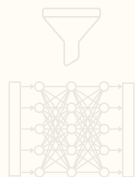
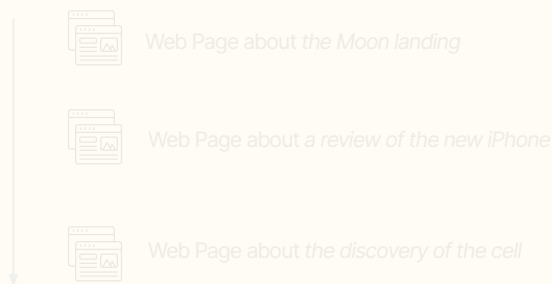


Pre-Training

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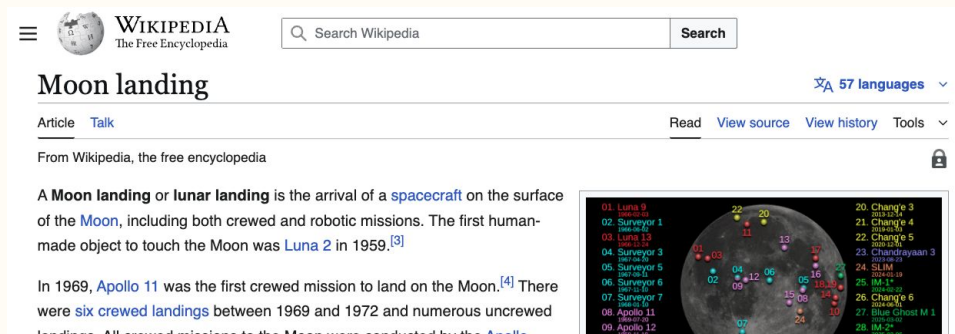


Motivation



Pre-Training

Trained on large amounts
of unlabeled text data
from the internet



“Predict the Next Word” Unsupervised Training Data

Input

Output

Neil Armstrong became the
first person to step foot on
the....

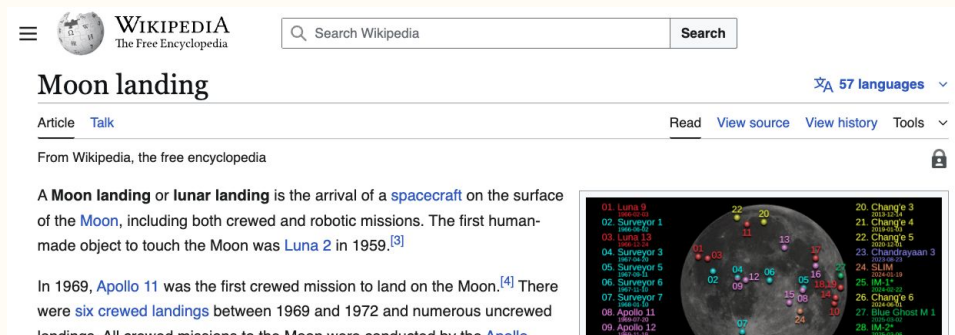
Moon

Apollo 11 was launched
from...

Florida

(Radford et al., 2019; Firth 1957)

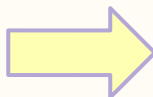
Motivation



"Predict the Next Word" Unsupervised Training Data

To predict the next word accurately:

- The model must **understand** the sentence
- It must build some **knowledge** about the real world



Input

Output

Neil Armstrong became the first person to step foot on the....

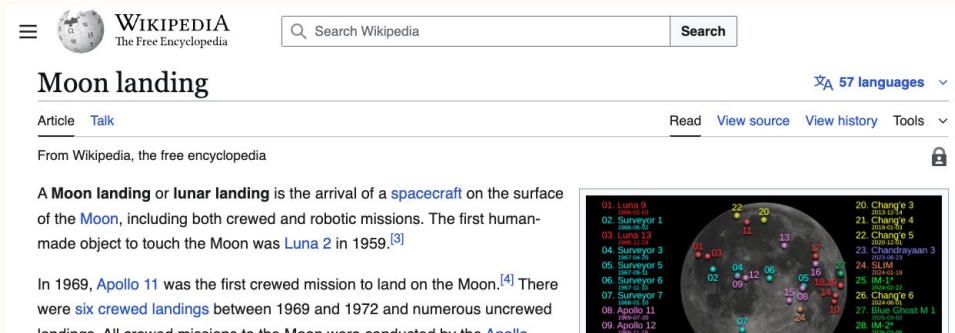
Moon

Apollo 11 was launched from...

Florida

(Radford et al., 2019; Firth 1957)

Motivation

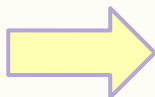


"Predict the Next Word" Unsupervised Training Data

Input

Output

We can create **billions** of these kinds of training examples with ease with the text from **all of the pages on the internet...**



Neil Armstrong became the first person to step foot on the....

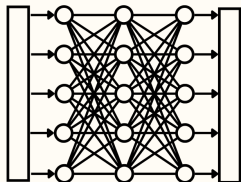
Moon

Apollo 11 was launched from...

Florida

(Radford et al., 2019; Firth 1957)

Motivation



Neil Armstrong became the first person to step foot on the **Moon**.



"Predict the Next Word" Model

Can you give me a recipe for roasted red pepper pesto?

Here's a clean, concise recipe for roasted red pepper pesto—bright, smoky, and great on pasta, sandwiches, or grilled vegetables.

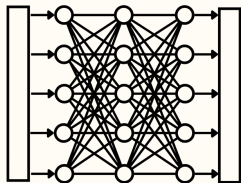
Roasted Red Pepper Pesto

Ingredients

- 2 large roasted red peppers (jarred or homemade), drained
- 1 cup fresh basil leaves, lightly packed
- 2–3 cloves garlic
- 1/2 cup toasted almonds, walnuts, or pine nuts

ChatGPT

Motivation



Neil Armstrong became the first person to step foot on the **Moon**.



"Predict the Next Word" Model



Continue
Training On....

"Task & Answer" Instruction-Tuning Training Data

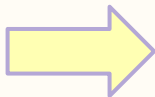
Input	Output
Give me a recipe for...	Here's a recipe for...
Who invented the...?	In 1837, Samuel Morse invented...
Write an email requesting a refund...	Sure! Here's an email formatted...

(Ouyang et al., 2022)

Motivation



We may only have **thousands** of these training examples ...



"Task & Answer" Instruction-Tuning Training Data

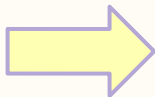
Input	Output
Give me a recipe for...	Here's a recipe for...
Who invented the...?	In 1837, Samuel Morse invented...
Write an email requesting a refund...	Sure! Here's an email formatted...

(Ouyang et al., 2022)

Motivation



We may only have **thousands** of these training examples ...



CBS NEWS

60 Minutes - Newsmakers

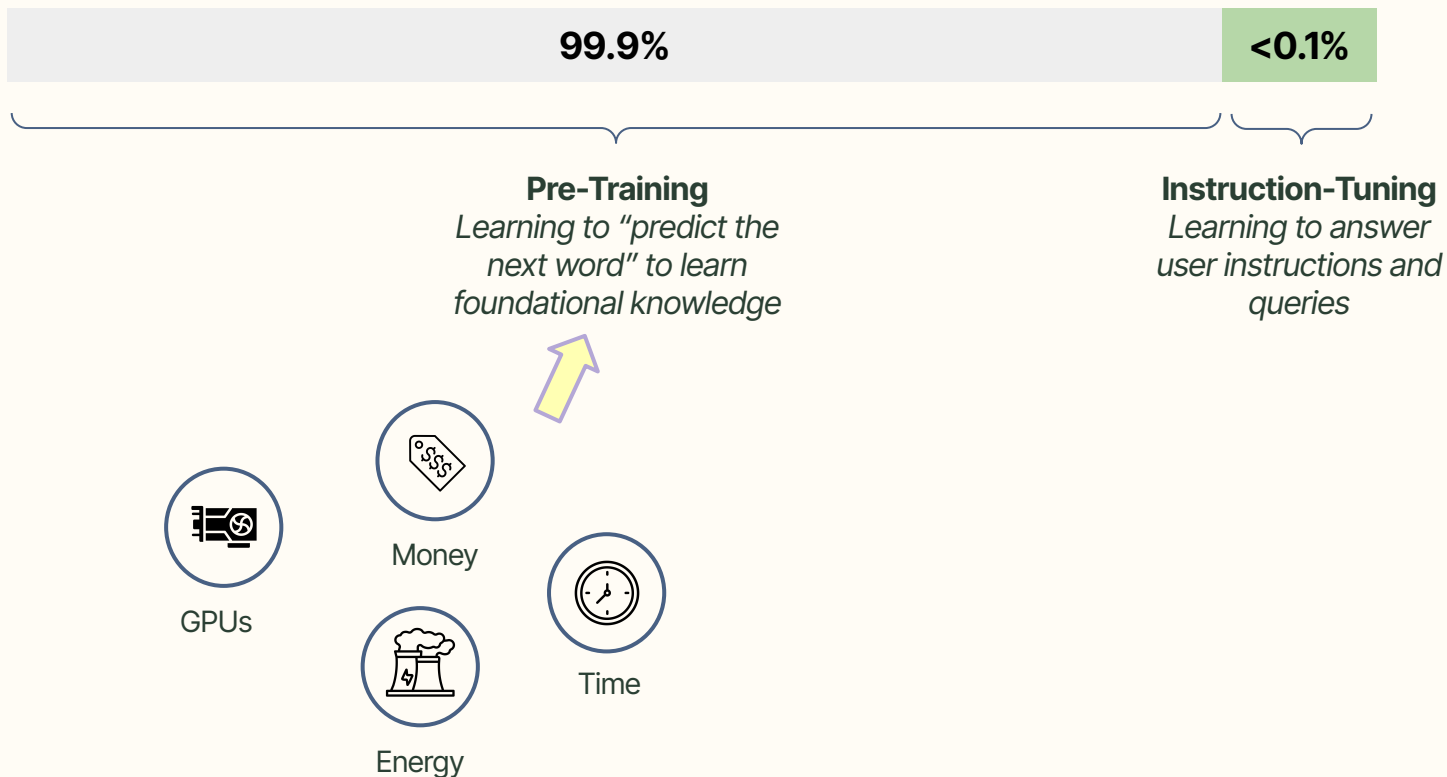
Labelers training AI say they're overworked, underpaid and exploited by big American tech companies

"Task & Answer" Instruction-Tuning Training Data

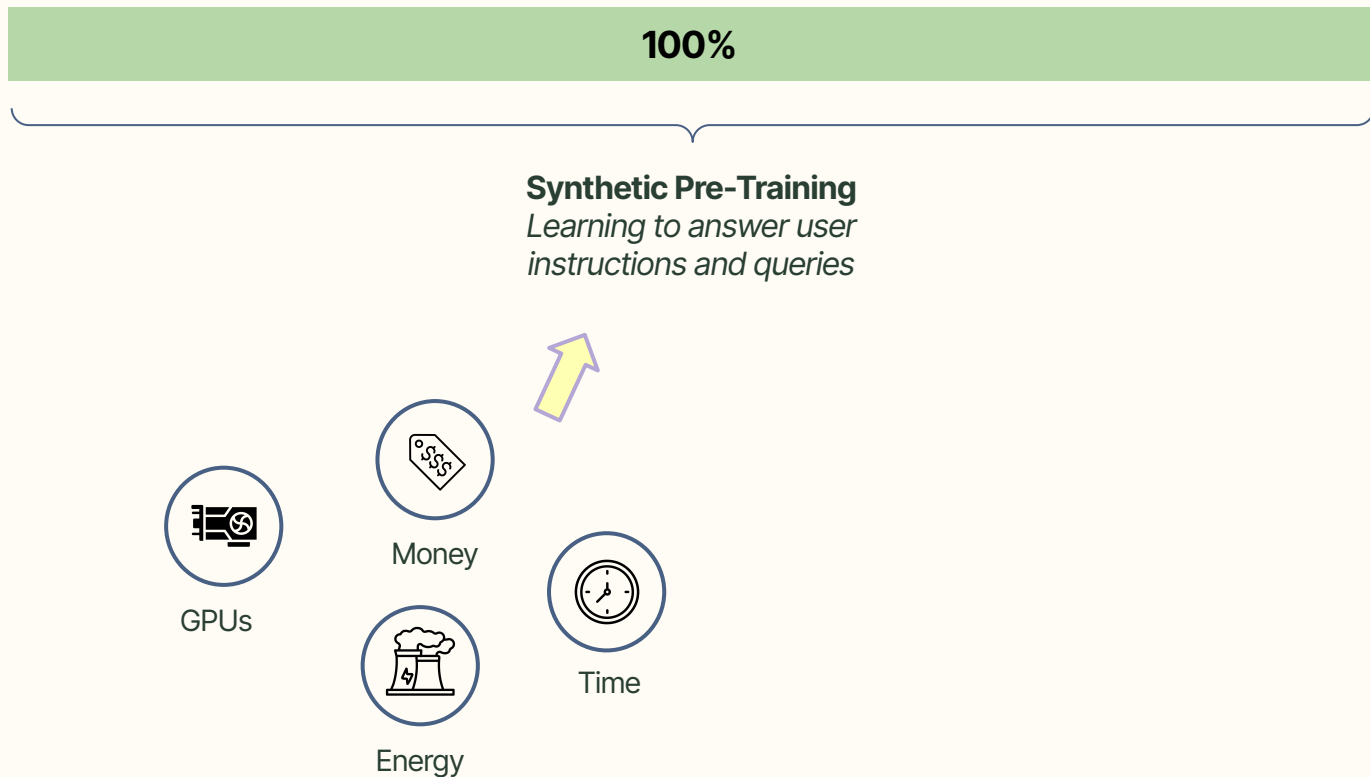
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Motivation



Motivation



Motivation

DISHING out HEALTH RECIPES ▾ 30 MINUTES ABOUT CONTACT 🔍

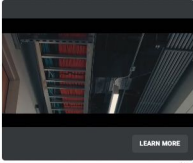
Skinny Chicken Alfredo Stuffed... 30 Minute Creamy Tomato Gnocchi... 30 Minute Zucchini Pesto Pasta Perfect Sautéed Broccoli 30 Minute Chicken Florentine Pasta Shaved Brussels and Pear Salad... Pesto-Roasted Whole Chicken

Step-by-Step Instructions:

The beauty of this sauce is that it all comes together in the food processor. No slicing or dicing necessary! Here's how it's done:

Step 1: Combine all pesto ingredients in a food processor. This includes the red pepper, sun-dried tomatoes, basil, olives, garlic, red pepper flakes, salt, and pepper.

Step 2: Blend until smooth. While blending, gradually stream in the olive oil until the mixture is cohesive.



Restructure

Input:

Give me a recipe for red pepper pesto sauce.

Output:

Here's a recipe for red pepper pesto:

Step 1: Combine all pesto ingredients in a food processor. This includes the red pepper...

Motivation

DISHING out HEALTH RECIPES ▾ 30 MINUTES ABOUT CONTACT 🔍

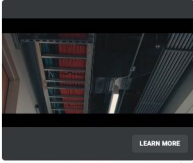
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Restructure

Input:

Give me a recipe for red pepper pesto sauce.

Output  **Task Structuring**

Here's a recipe for red pepper pesto:

Step 1: Combine all pesto ingredients in a food processor. This includes the red pepper...

Procedure

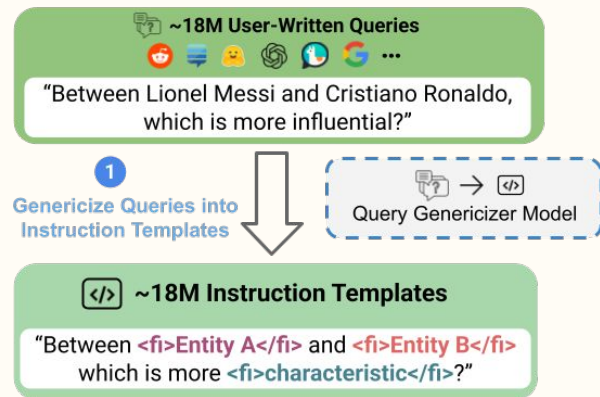
 ~18M User-Written Queries



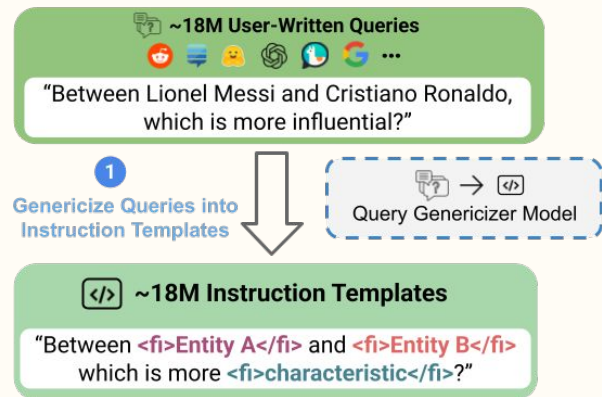
"Between Lionel Messi and Cristiano Ronaldo,
which is more influential?"

(Patel et al., 2025)

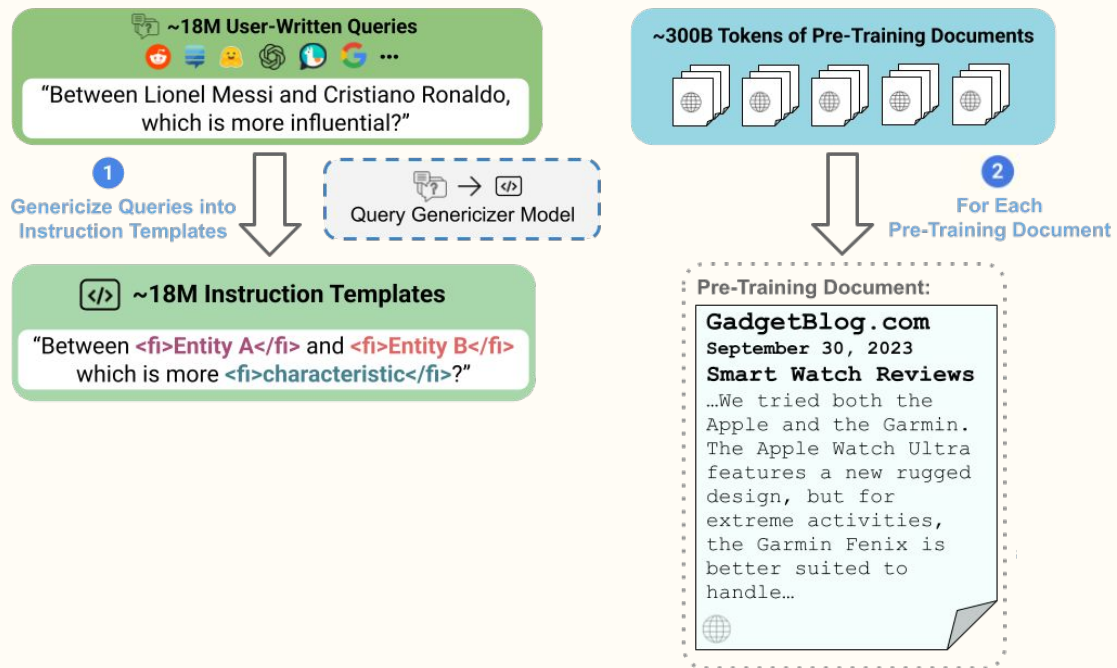
Procedure



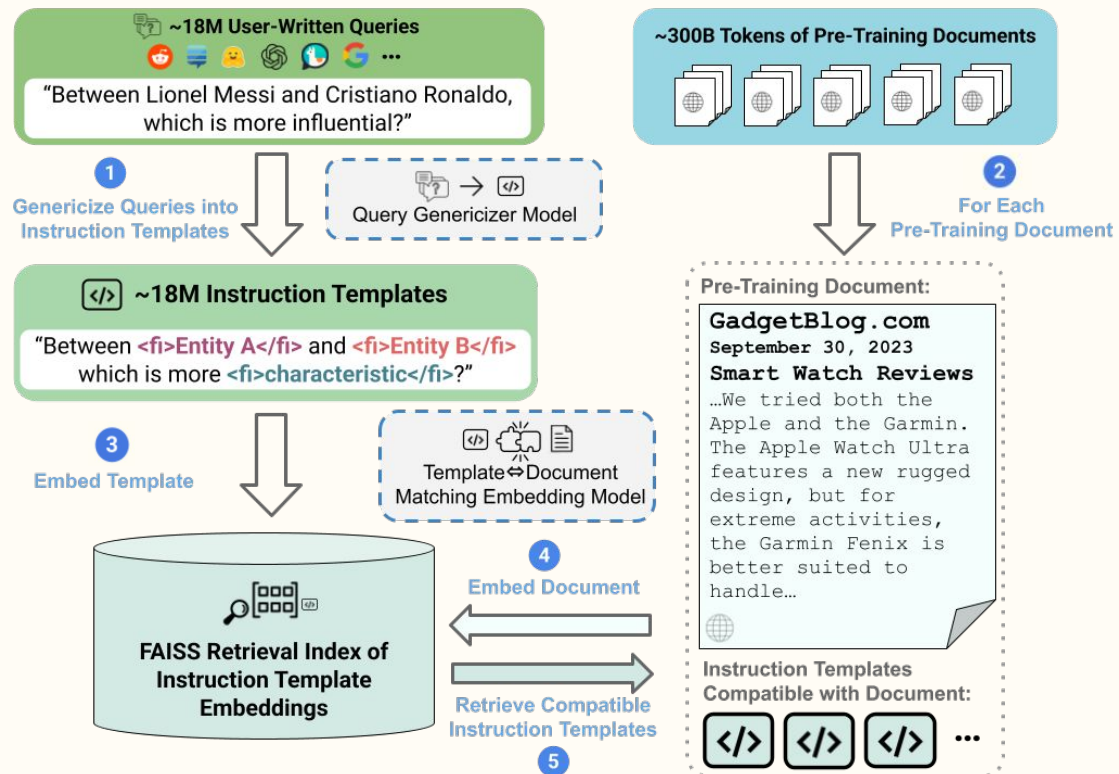
Procedure



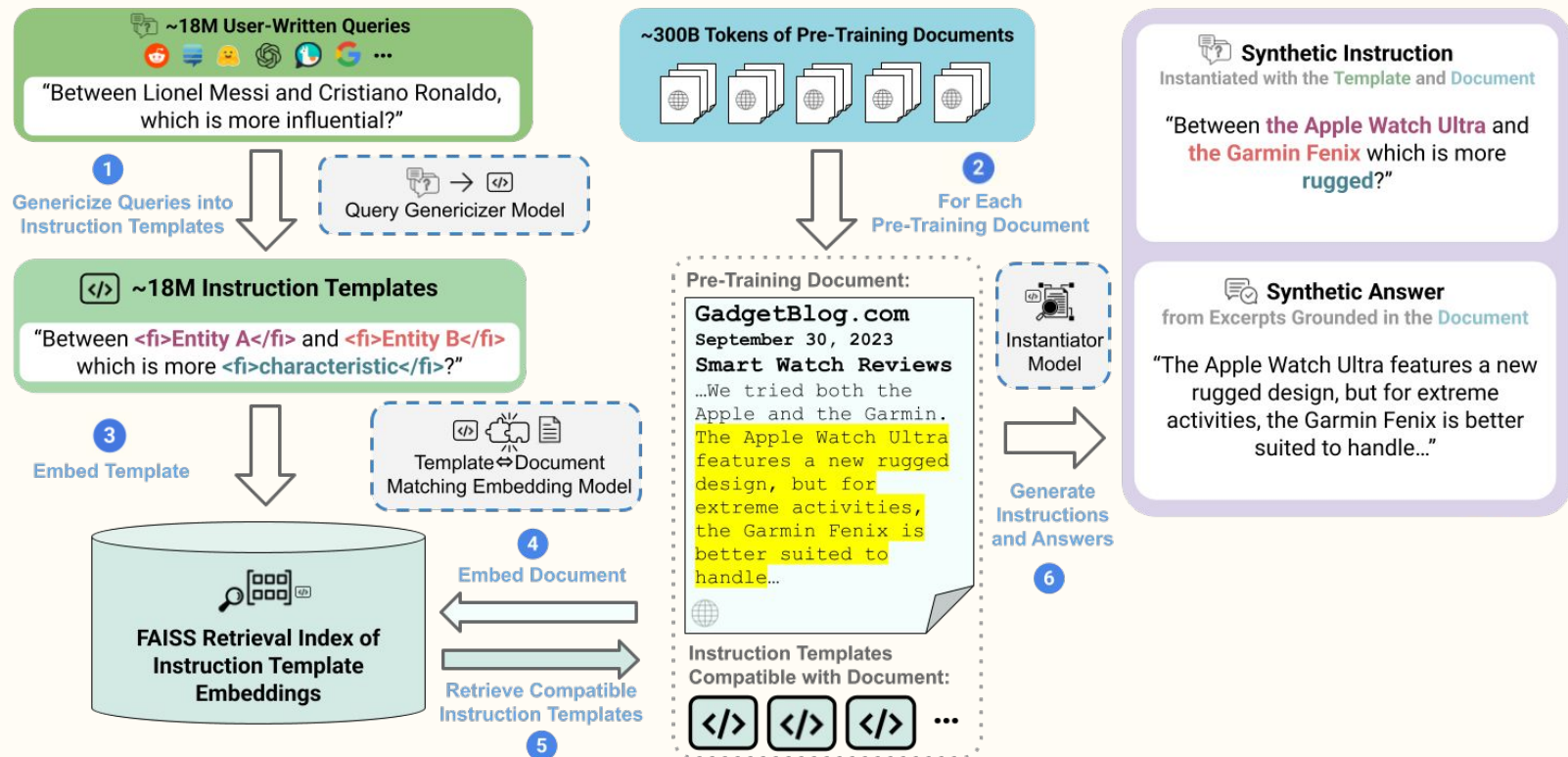
Procedure



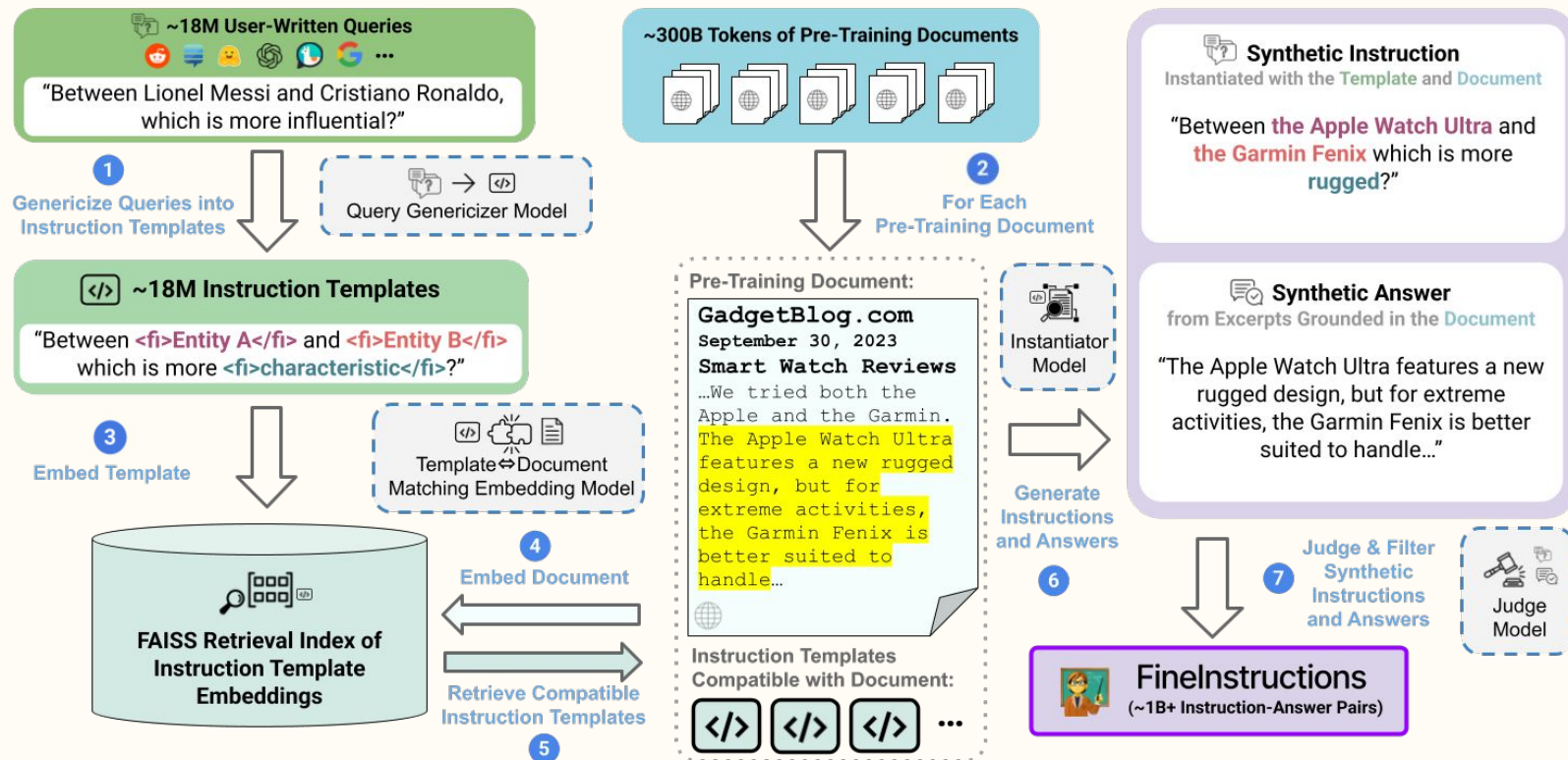
Procedure



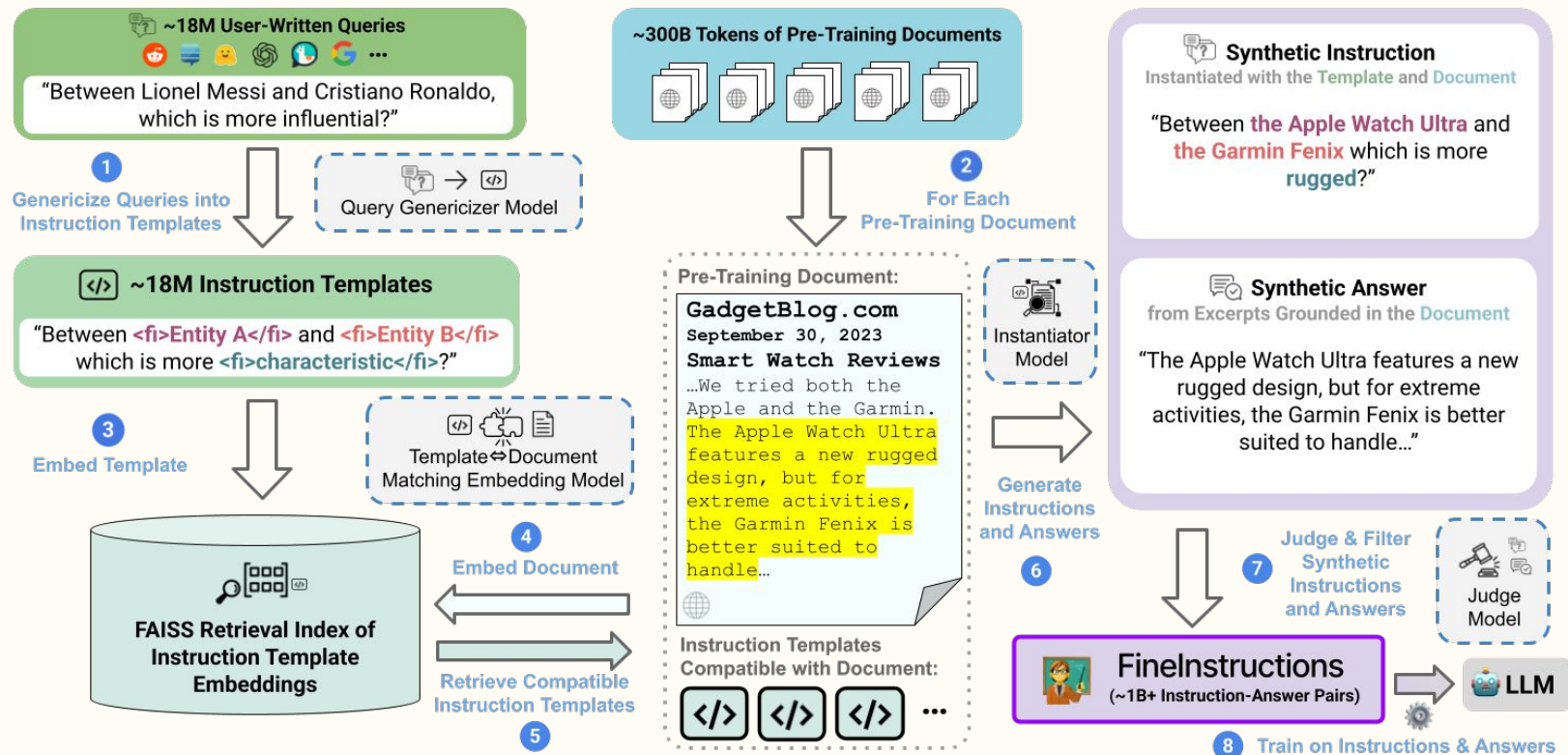
Procedure



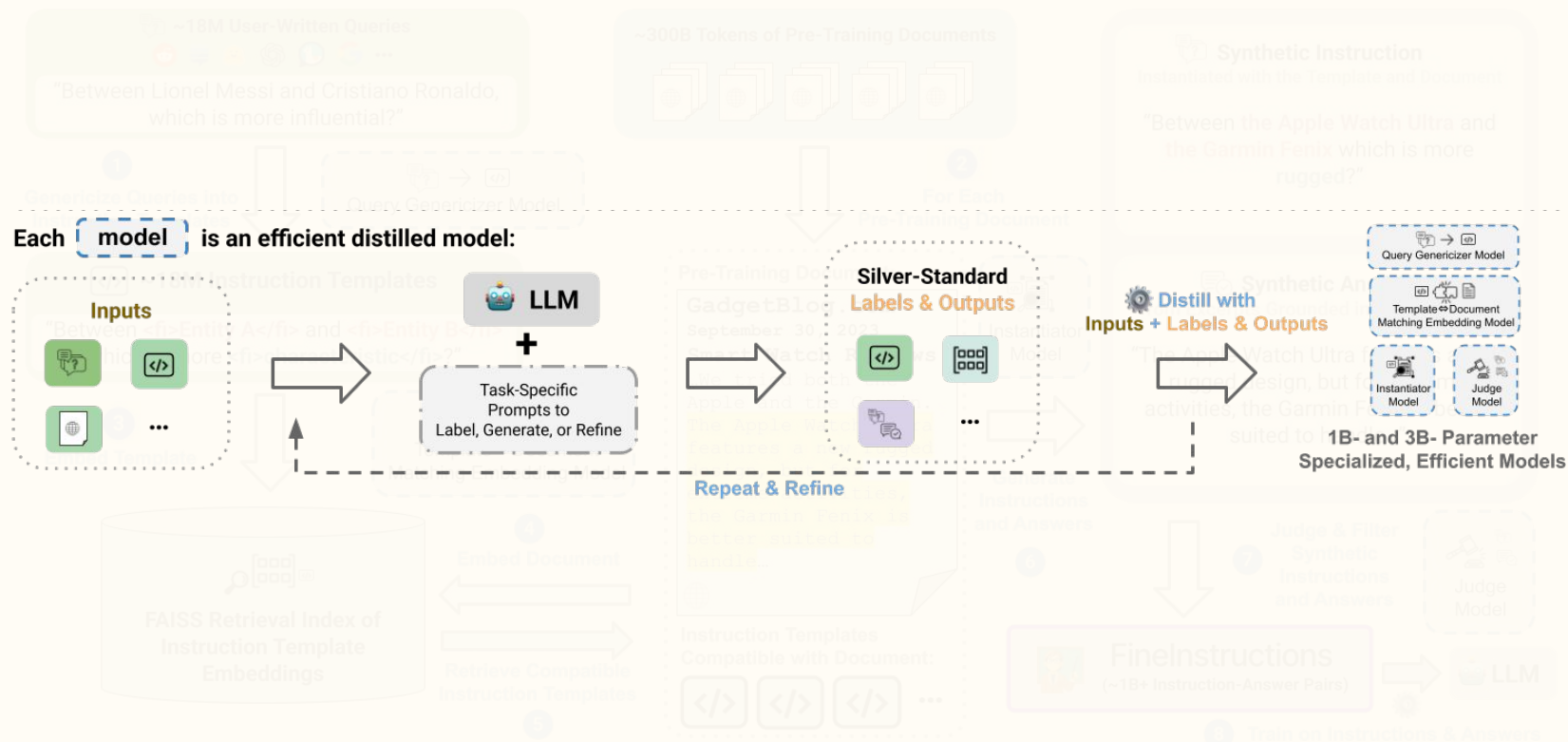
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Procedure

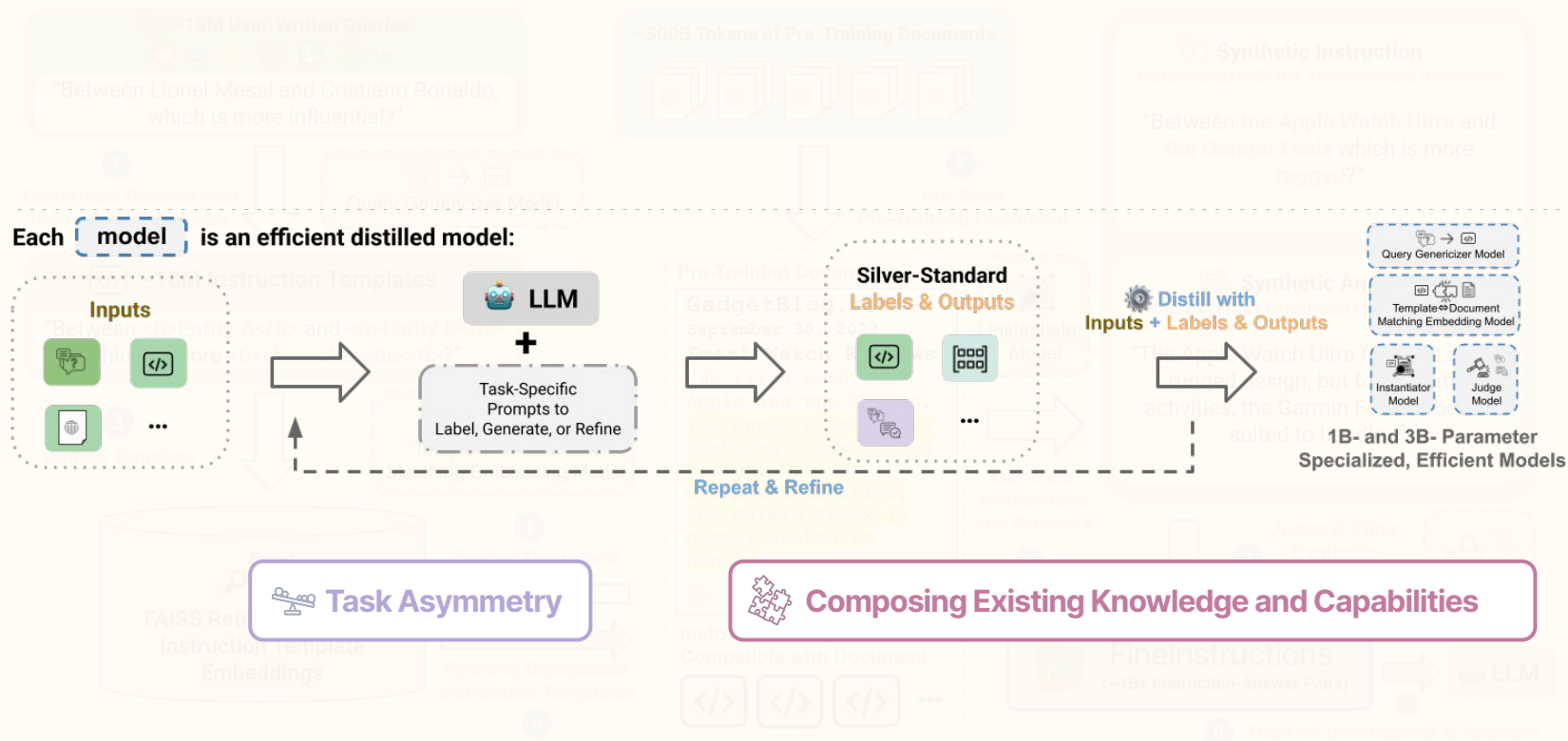


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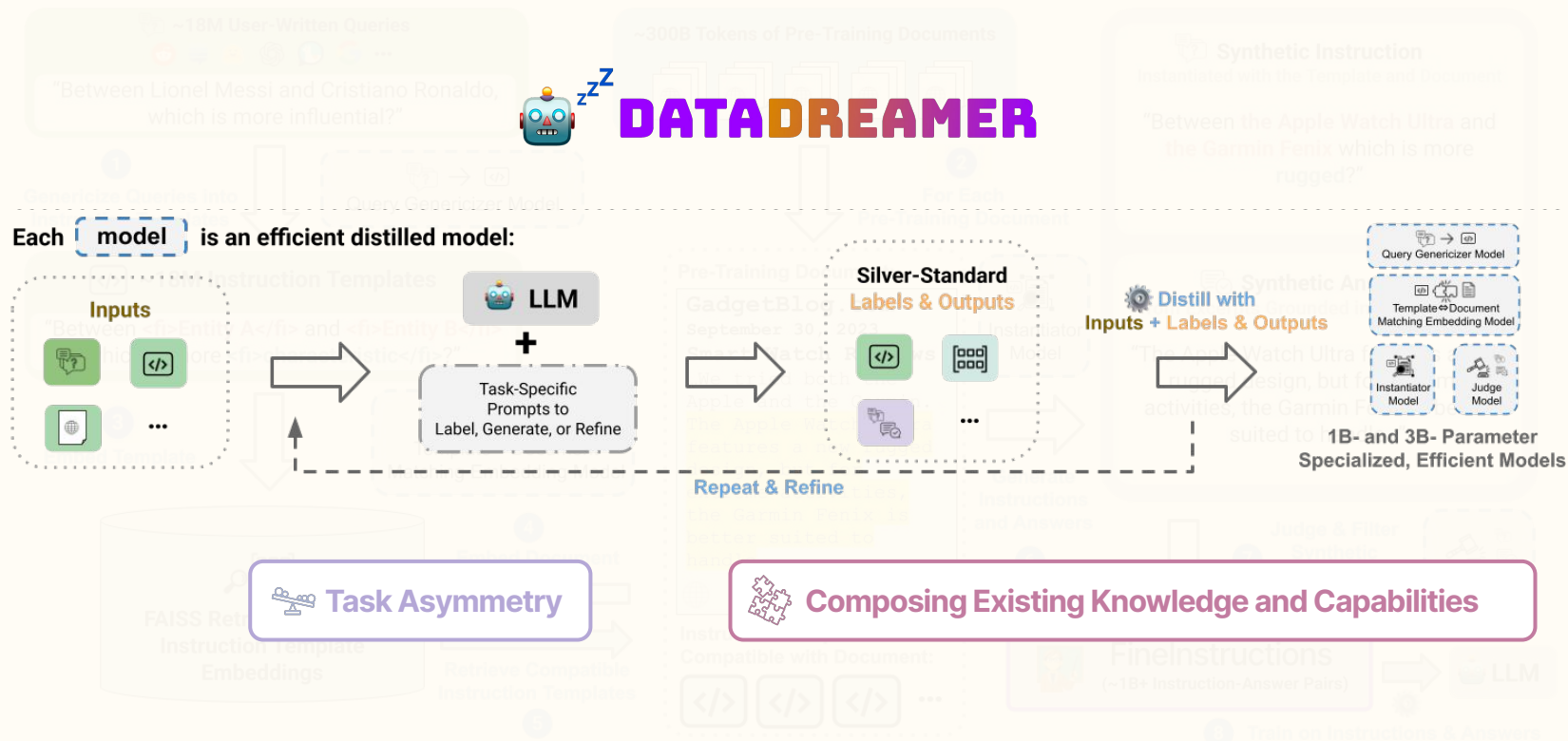


(Penedo et al., 2024)

Procedure

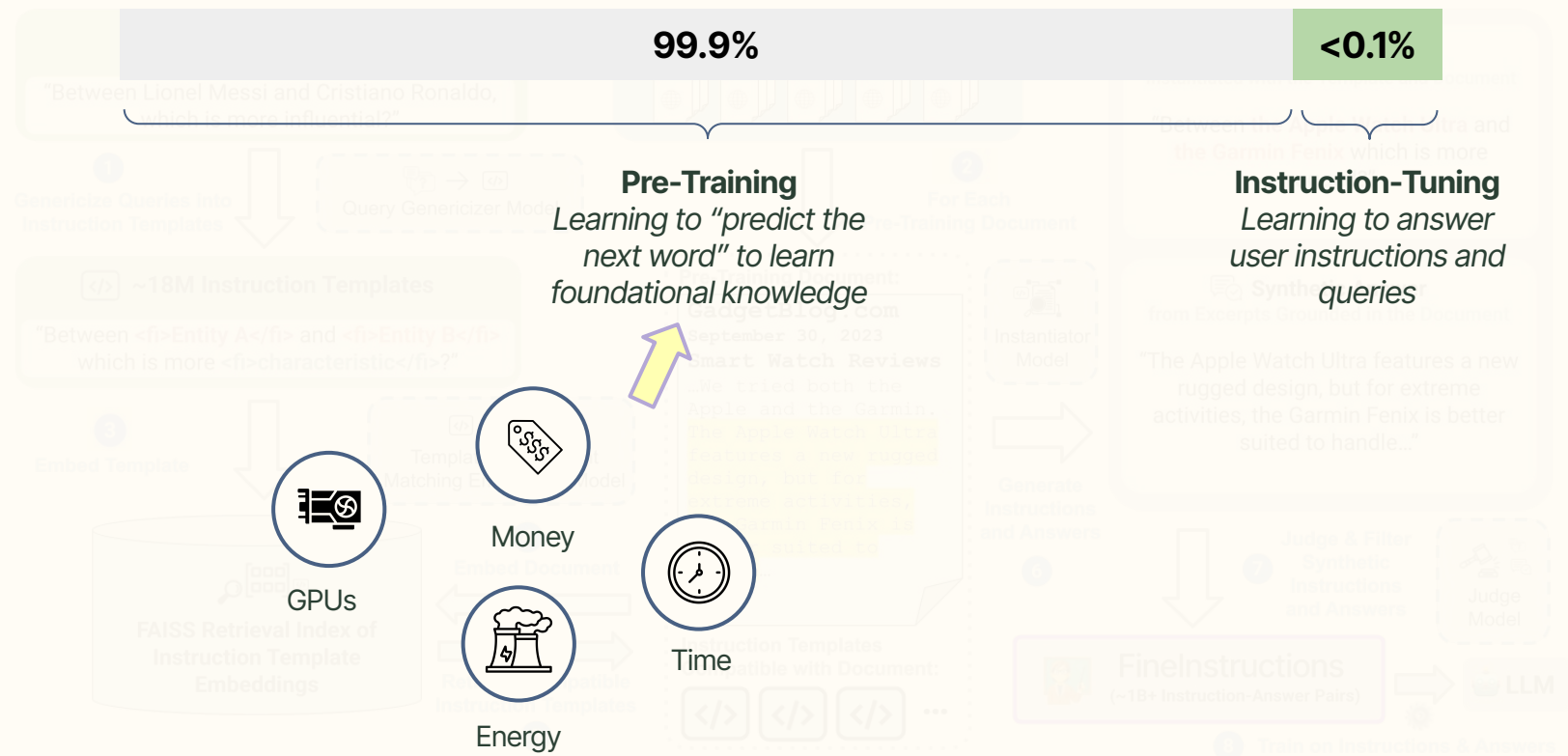


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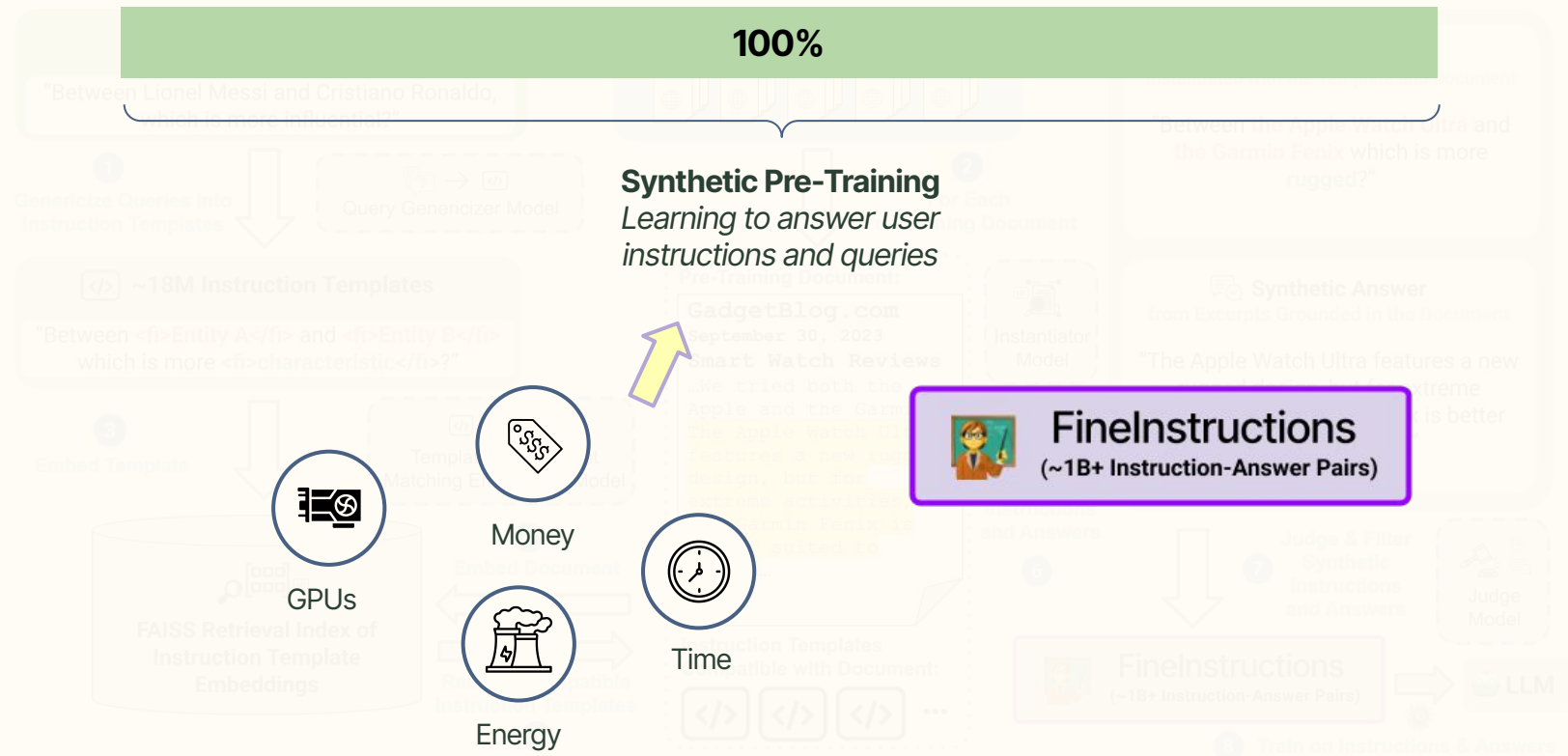


(Penedo et al., 2024)

FineInstructions



FineInstructions



FineInstructions

Pre-Training Document

Structure and Synchronicity for Better Charting

Two key characteristics will help you to ensure that your notes communicate not only what you did, but also what you were thinking.

Fam Pract Manag. 2011 Jul-Aug;18(4):15-17.

How many times have you read a medical note that does not make the selection of the diagnosis or treatment clear? Have you ever read your own notes after receiving notice of a malpractice suit and winced at the inconsistencies? Poorly constructed medical notes are a widespread problem. I've seen it while reviewing the charts of medical students and residents to ensure that they met the standard of care and avoided malpractice risk. While most physicians document history and physical data with the required number of CPT elements, few clearly convey a line of reasoning that reveals their clinical thought process. A note that documents a detailed history and moderately complex decision making does not necessarily illuminate why certain decisions were reached or why a particular treatment was justified.

I wanted to find out where in the educational process students learned key documentation concepts, so I did some research that included informally querying medical students from multiple medical schools who were rotating at our residency. None of them could describe learning a formal structure for completing assessments and plans, which is consistent with a study that found only 4 percent of standardized encounters were accurately charted by medical students. I also discovered that the 2010 United States Medical Licensing Examination clinical skills exam guide states that students are expected to present a list of differential diagnoses in order of likelihood along with desired evaluations, but no requirements existed for discussing the clinical rationale. As a result of these and other findings, I developed a formal framework, described in this article, to teach residents and students an appropriate way to construct their notes. The initial feedback has been highly positive.

Documenting a confirmed diagnosis

To begin, let's review what should be included when documenting a confirmed diagnosis. Generally, six elements are needed...

Compatible Instruction Template

I'm looking for guidelines for how to write a `<fi>type of document, e.g. technical report, research paper, etc.</fi>` .

The `<fi>type of document</fi>` is `<fi>context or field of study</fi>-related and intended for a <fi>role or position</fi> to review the work of the <fi>task or investigation</fi> of a <fi>type of issue or problem</fi> related to <fi>field or industry</fi> .`

Synthetic Instruction

I'm looking for guidelines for how to write a medical note.

The medical note is healthcare-related and intended for a physician to review the work of the clinical investigation of a medical issue related to patient care.

Synthetic Answer

To write a medical note, consider the following guidelines. `<excerpt>`To begin, let's...`</excerpt>`

FineInstructions

Pre-Training Document

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Synthetic Instruction

I'm looking for guidelines for how to write a medical note.

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Using excerpts:

- Ensures the answer is grounded in the document.
- Allows compute-efficient generation.

Synthetic Answer

To write a medical note, consider the following guidelines. `<excerpt>`To begin, let's...`</excerpt>`

FineInstructions

Pre-Training Document

"Between Lionel Messi and Cristiano Ronaldo,
History of Civilization which is more influential?"

The history of civilization is, at its core, the story of how human communities learned to live together in increasingly complex ways. While the earliest members of our species survived in small, mobile groups, it was the gradual shift toward settled life, including farming, permanent dwellings, and shared labor, that set the stage for what we now call civilization.

... ~18M Instruction Templates

By the third millennium BCE, these settled societies had transformed into some of the earliest urban civilizations. Mesopotamia produced cities such as Uruk and Ur, complete with monumental temples and administrative records preserved on clay tablets. In Egypt, centralized rule under the pharaohs supported grand architectural projects and a highly organized bureaucracy. Farther east, the Indus Valley cities of Harappa and Mohenjo-daro developed sophisticated urban planning, including grid layouts and drainage systems.

...

Modern civilization, shaped by industrialization, global communication, and rapid technological change, is in many ways an extension of earlier patterns. Larger populations and more intricate systems of knowledge continue to define daily life, and networks of exchange now span the entire planet. Yet these same forces introduce new challenges. Environmental pressures, social inequality, and the strain placed on political institutions all raise questions about how societies can sustain stability and fairness in a rapidly changing world. Despite these uncertainties, the central thread remains. Humans continue to adapt, organize, and imagine new forms of collective life.

~300B Tokens of Pre-Training Documents



For Each

Pre-Training Document

Pre-Training Document:

GadgetBlog.com
September 30, 2023
Smart Watch Reviews
We tried both the
Apple and the Garmin.
The Apple Watch Ultra
features a new rugged
design, but for
extreme activities,
the Garmin Fenix is
better suited to
handle...



Generate
Instructions
and Answers

Instruction Templates
Compatible with Document:



Synthetic Instruction

Instantiated with the Template and Document

"Between the Apple Watch Ultra and
the Garmin Fenix which is more
rugged?"



Synthetic Answer

from Excerpts Grounded in the Document

"The Apple Watch Ultra features a new
rugged design, but for extreme
activities, the Garmin Fenix is better
suited to handle..."

Judge & Filter
Synthetic
Instructions
and Answers



FineInstructions
(~1B+ Instruction-Answer Pairs)

Train on Instructions & Answers

FineInstructions

Pre-Training Document

History of Civilization

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Embedding of Section 1/5



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Pre-Training Document

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Instruction Templates
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Generate
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Synthetic Instruction

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Judge & Filter
Synthetic
Instructions
and Answers

Judges
Model

FineInstructions
(~1B+ Instruction-Answer Pairs)

LLM

Train on Instructions & Answers

FineInstructions

Pre-Training Document

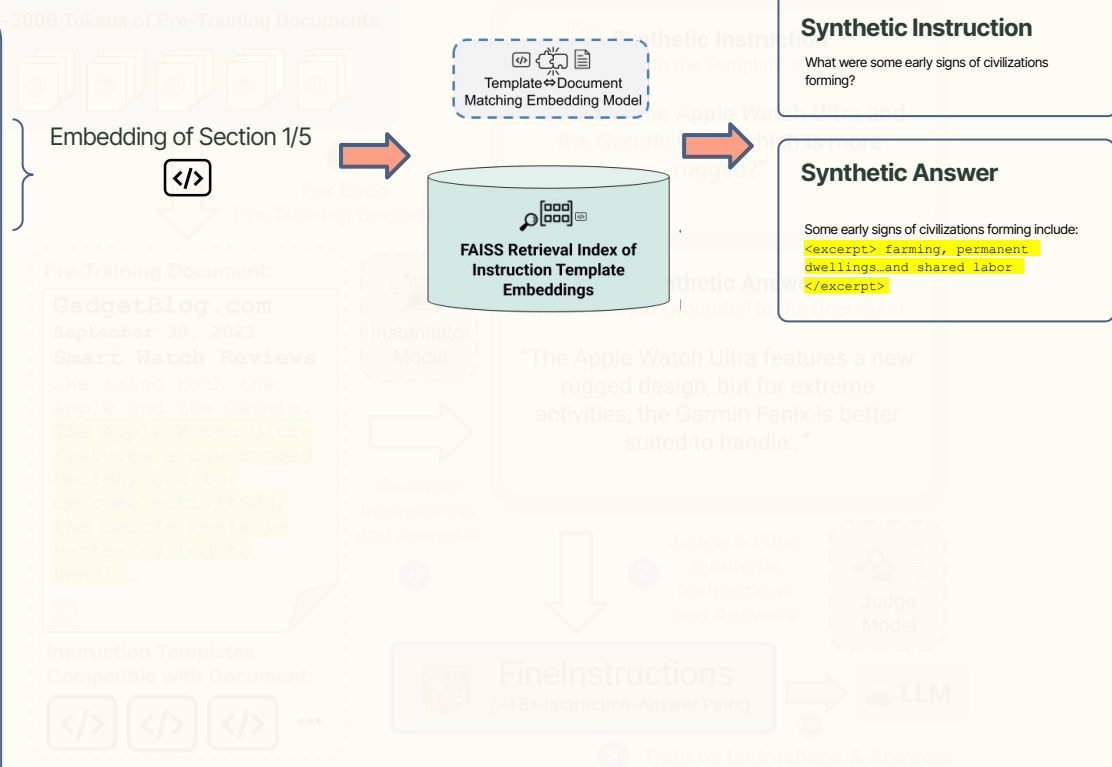
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(Patel et al., 2025)

FineInstructions

Pre-Training Document

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300B Tokens of Pre-Training Documents



For Each
Pre-Training Document

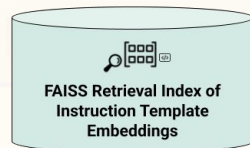
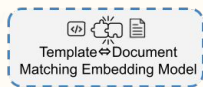
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Generate
Instructions
and Answers

Embedding of Section 5/5



Synthetic Instruction
Instantiated with the Template and Document

"Between the Apple Watch Ultra and
the Garmin Fenix which is more
rugged?"

Synthetic Answer
from Excerpts Grounded in the Document

"The Apple Watch Ultra features a new
rugged design, but for extreme
activities, the Garmin Fenix is better
suited to handle."

Synthetic Instruction

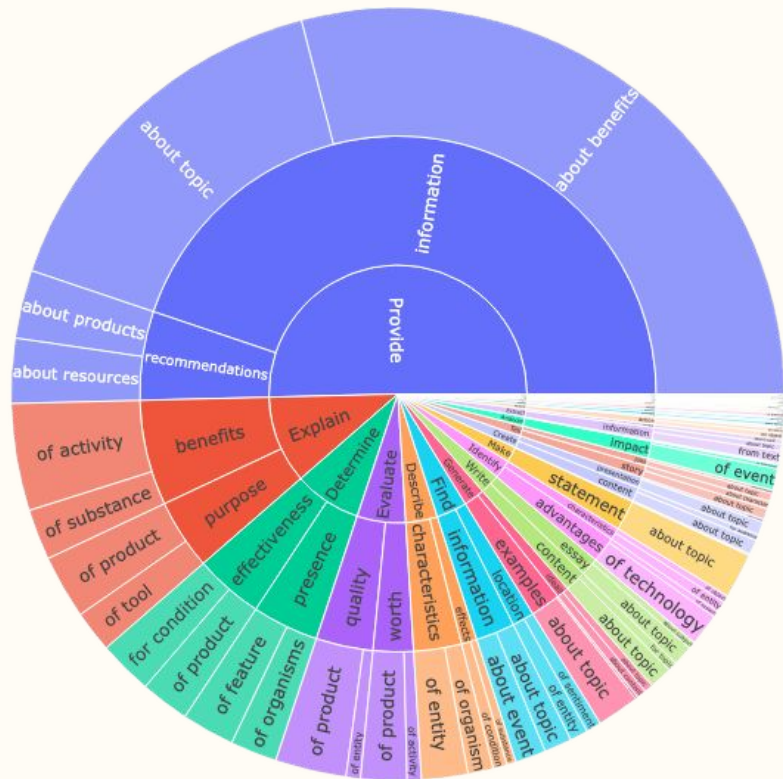
What kinds of challenges do civilizations face as
populations increase?

Synthetic Answer

Civilizations face a number of challenges with
scale. **<excerpt> Environmental
pressures, social rapidly changing
world.</excerpt>**

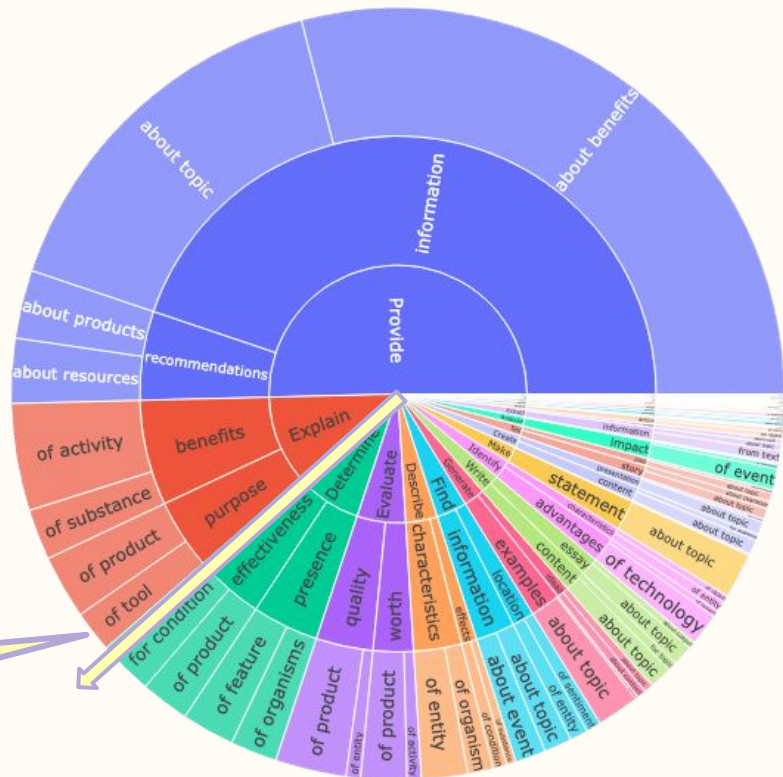
(Patel et al., 2025)

FineInstructions



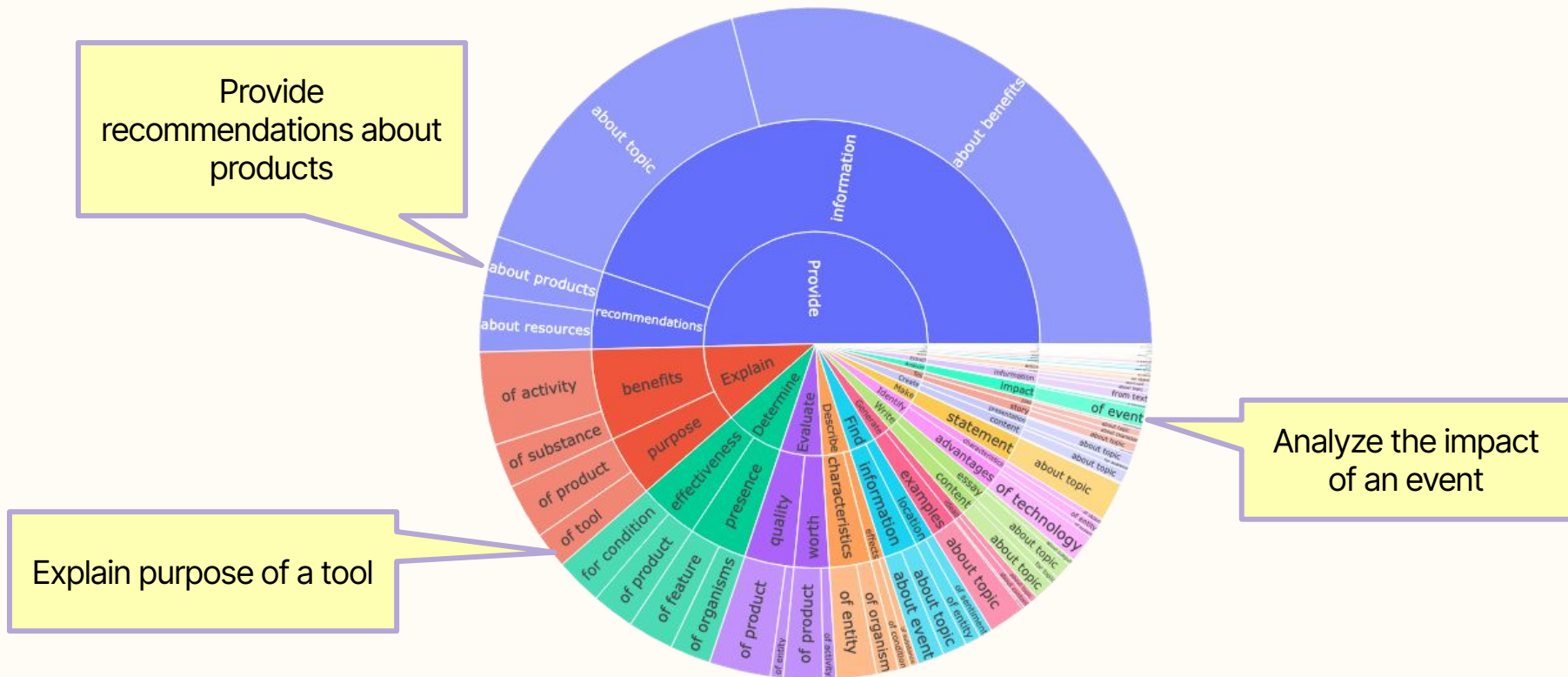
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FineInstructions

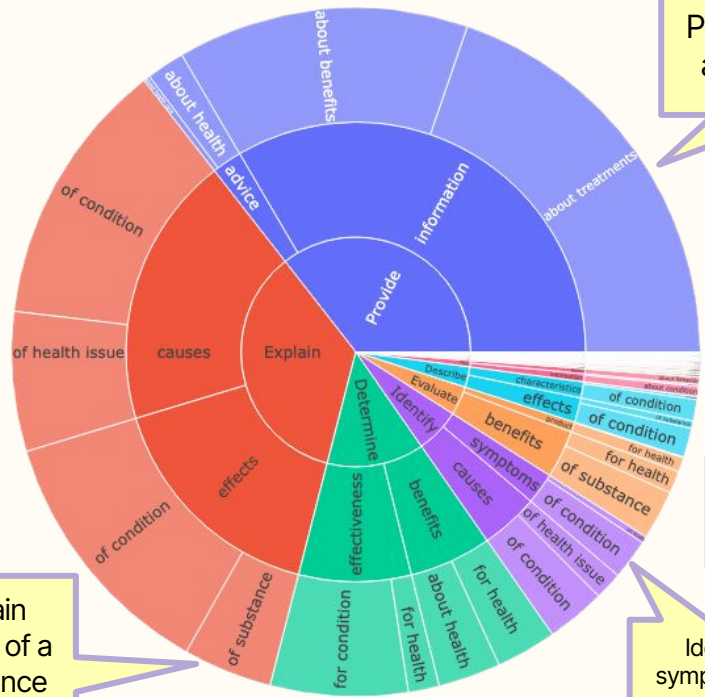


Explain purpose of a tool

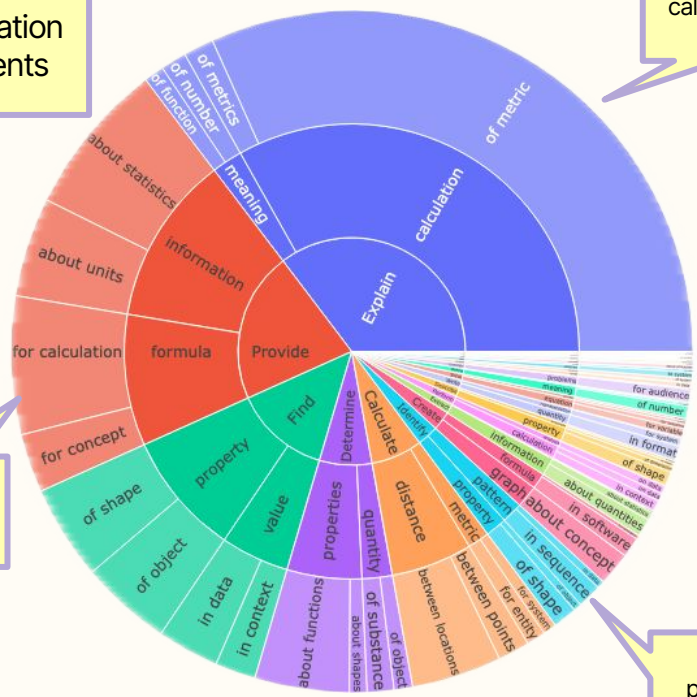
FineInstructions



FineInstructions



Medicine-Related FineInstructions



Math-Related FineInstructions

Baselines

We compare against standard pre-training as well as other proposed synthetic approaches.

FinelInstructions



FinelInstructions

(~1B+ Instruction-Answer Pairs)



Synthetic Instruction

Instantiated with the **Template** and **Document**

"Between the **Apple Watch Ultra** and
the **Garmin Fenix** which is more
rugged?"

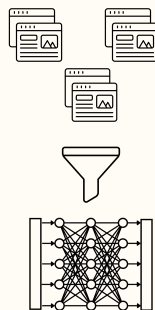


Synthetic Answer

from Excerpts Grounded in the **Document**

"The Apple Watch Ultra features a new
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Standard Pre-Training



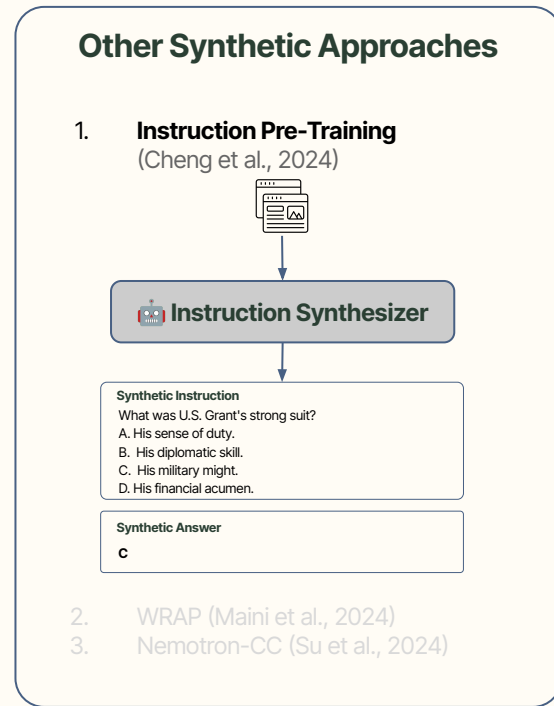
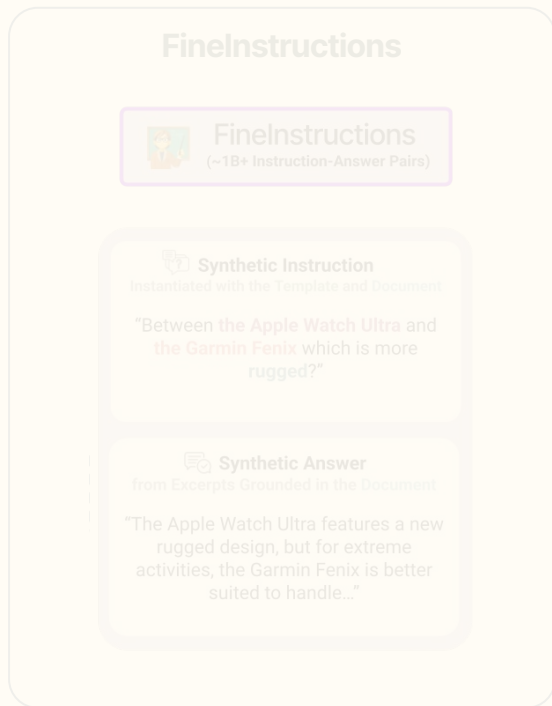
Neil Armstrong became the first
person to step foot on the **Moon**.

Other Synthetic Approaches

1. Instruction Pre-Training
(Cheng et al., 2024)
2. WRAP (Maini et al., 2024)
3. Nemotron-CC (Su et al., 2024)

Baselines

We compare against standard pre-training as well as other proposed synthetic approaches.



Baselines

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FineInstructions



FineInstructions
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Synthetic Instruction

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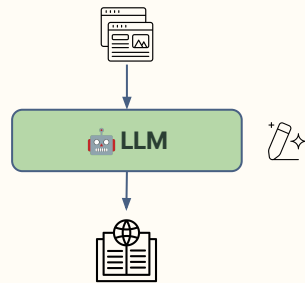
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FineInstructions



FineInstructions
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Synthetic Instruction

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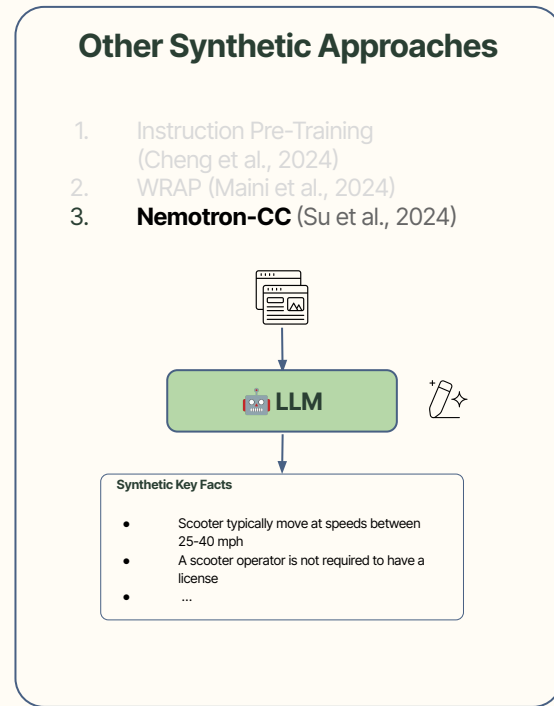
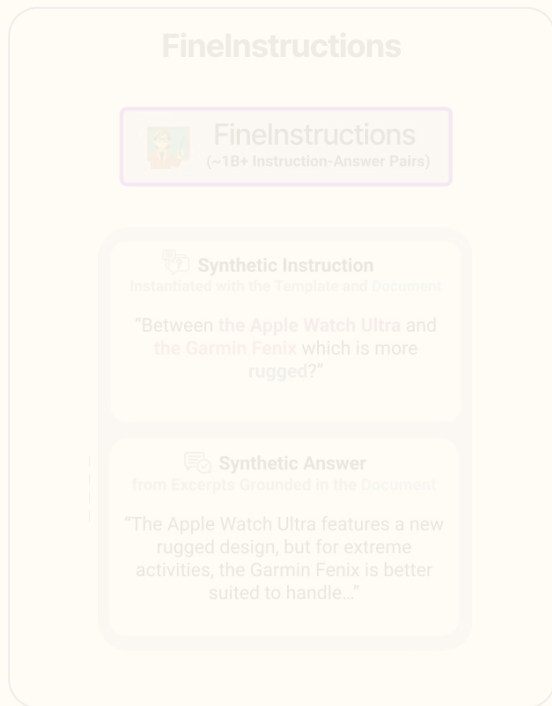


Synthetic Instruction
What is the actual top speed of most scooters on
the road today?

Synthetic Answer
25 to 40 mph.

Baselines

We compare against standard pre-training as well as other proposed synthetic approaches.



Baselines

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FinelInstructions

- ✓ Diverse Tasks
- ✓ Long Form Answers
- ✓ Instructions & Questions
- ✓ In-Distribution With Real User Queries

Standard Pre-Training



Neil Armstrong became the first person to step foot on the Moon.

Other Synthetic Approaches

- ✗ Limited Diversity
- ✗ Multiple Choice and Short-Form Answers
- ✗ Questions Only
- ✗ Optimized for Academic LLM Benchmarks

Baselines

We compare against standard pre-training as well as other proposed synthetic approaches.

FinelInstructions



FinelInstructions
(~1B+ Instruction-Answer Pairs)



Synthetic Instruction

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"Between the Apple Watch Ultra and the Garmin Fenix which is more rugged?"

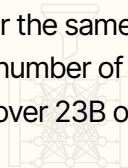


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Standard Pre-Training



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We pre-train LLMs from scratch with data from *each* method:

- Controlled for the same set of documents
- Controlled for the number of tokens used to pre-train
- 1.8B Parameters over 23B or 300B Token Datasets

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FinelInstructions



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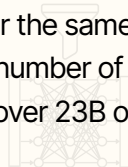


Synthetic Answer

from Excerpts Grounded in the Document

"The Apple Watch Ultra features a new
rugged design, but for extreme
activities, the Garmin Fenix is better
suited to handle..."

Standard Pre-Training



Neil Armstrong became the first
person to step foot on the Moon.



We pre-train LLMs from scratch with data from *each* method:

- Controlled for the same set of documents
- Controlled for the number of tokens used to pre-train
- 1.8B Parameters over 23B or 300B Token Datasets

Other Synthetic Approaches

1. Instruction Pre-Training
(Cheng et al., 2024)
2. WRAP (Maini et al., 2024)
3. Emotron-CC (Su et al., 2024)

**Acknowledgements For
Compute & Storage Resources:**



Hugging Face

+ Colin for facilitating

Evaluation Benchmarks

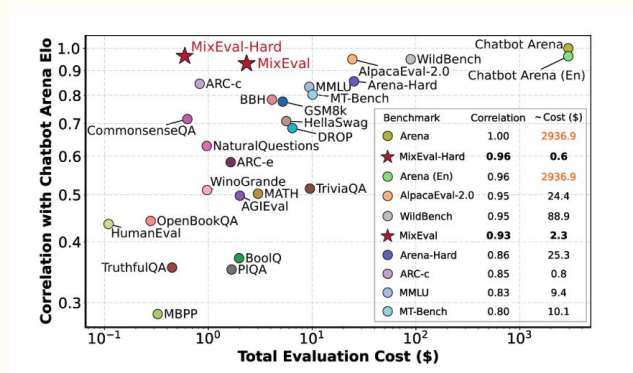
Knowledge & Factual-Recall Focused:

- **MixEval (Accuracy %):** MMLU, HellaSwag, TriviaQA, CSQA, GSM8K, BBH, AGIEval, etc.

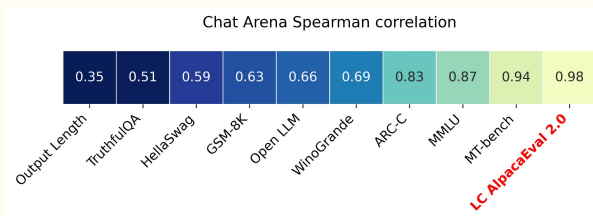
Realistic User LLM Queries:

- **MT-Bench-101 (Likert Score):** Measured via Rubric
- **AlpacaEval (Win Rate %):** Pairwise Model Response Comparison (Model A v. Model B)

Evaluation Benchmarks



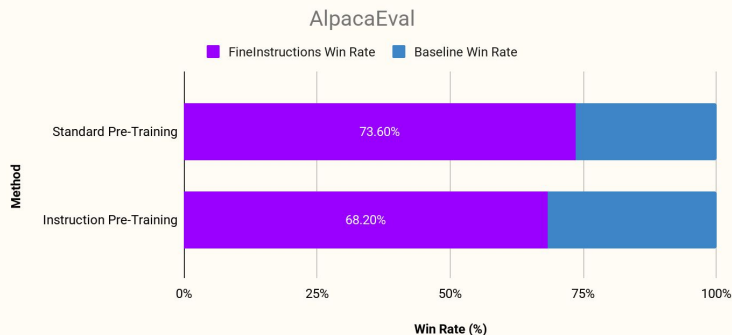
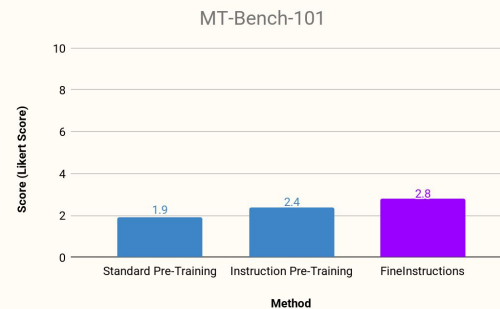
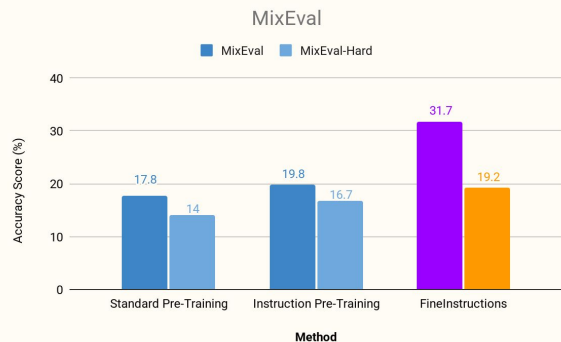
Evaluation Method	Agreement	Δ
Human Experts	80%	0%
MT-Bench-101	87%	+7%
w/o scoring guidelines	77%	-3%
w/o minimum values metrics	75%	-5%



(Ni et al., 2024; Bai et al., 2024; Li et al., 2023)

Results

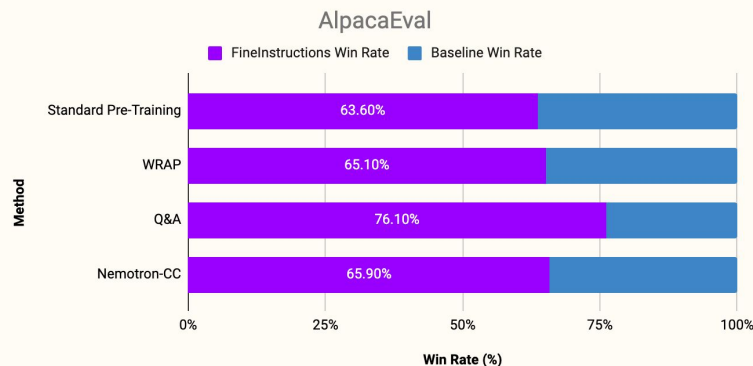
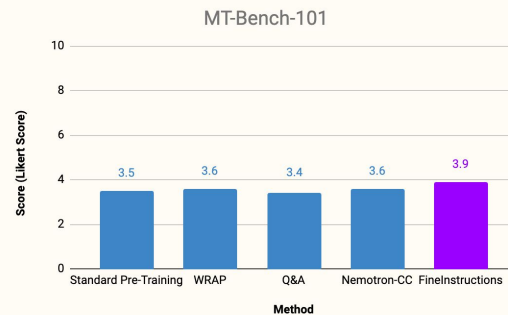
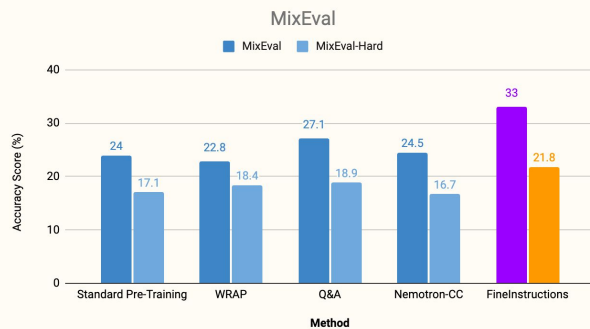
Pre-training on FinelInstructions improves performance on **both knowledge and factual-recall benchmarks and on realistic user LLM query benchmarks.**



(Patel et al., 2025)

Results

Pre-training on FinelInstructions improves performance on **both knowledge and factual-recall benchmarks and on realistic user LLM query benchmarks.**



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Future Work

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Future Work

- **Scale of Distilled Models:** Larger models would help with more complex templates and processing longer documents.
- **Tuning the Distribution and Mix of Queries/Templates:** Math, Science, Factual Q&A, “Task-y”, Writing Generation, etc.
- **Evaluation:** Evaluations for long-tail tasks and knowledge and evaluation beyond log-probability based evaluation.

(Wang et al., 2024; Chandak et al., 2024)

Summary

- We develop a method for synthetically generating instructions and answers at pre-training scale to introduce an inductive bias and improve the foundational learning process of LLMs.
- We evaluate the effectiveness of training on FinelInstructions and demonstrate improvements in human-correlated LLM benchmarks.

Inductive Biases



Task Asymmetry



Task Structuring

Sources of Knowledge

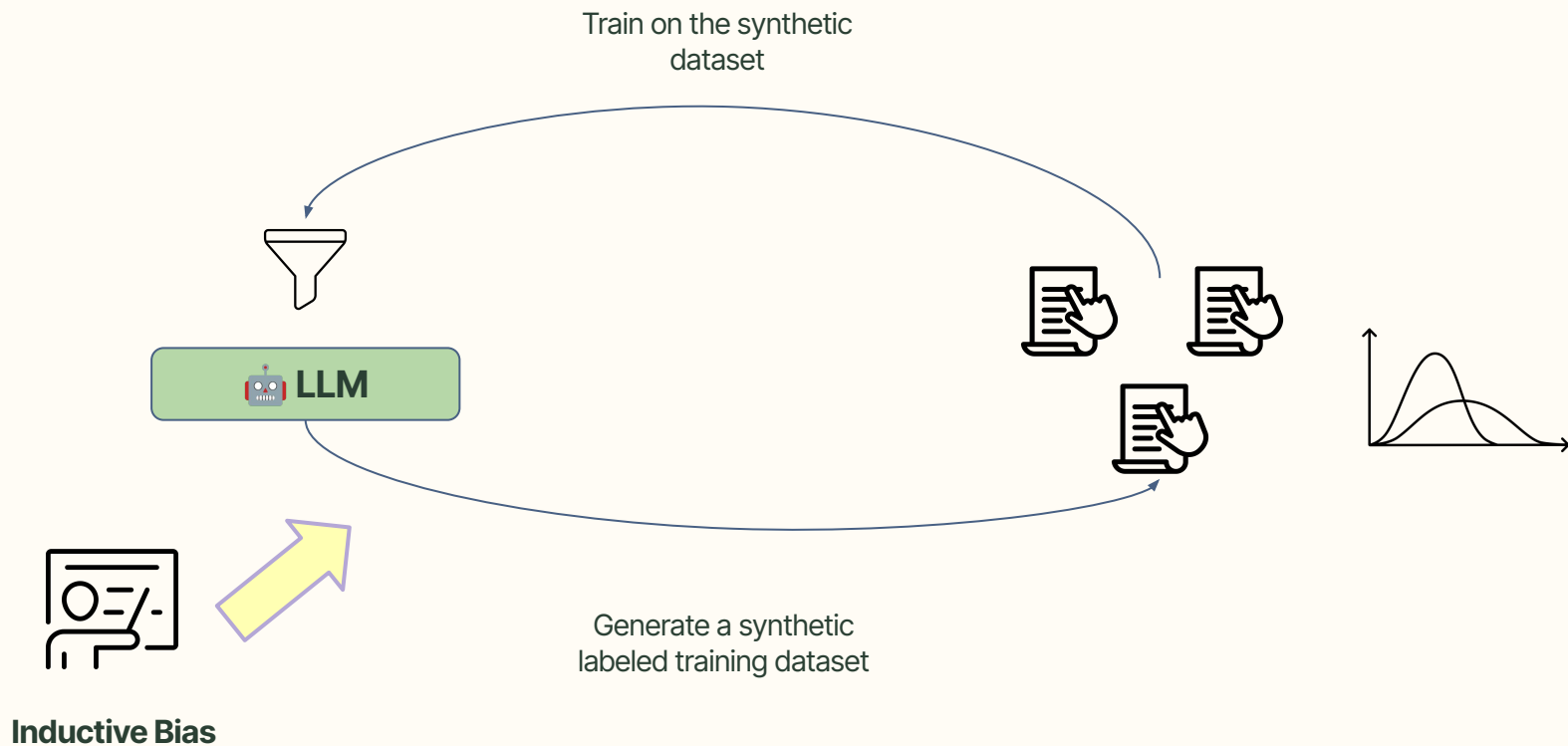


Composing Existing Knowledge and Capabilities

Conclusion

- Synthetic Data as an Inductive Bias
- Summary
- Future Directions
- Acknowledgements

Synthetic Data as an Inductive Bias



Inductive Biases and Sources of Knowledge

Inductive Biases



Task Priming



Task Asymmetry



Task Rules and Heuristics



Task Structuring

Sources of Knowledge



Surfacing Latent Knowledge and Capabilities

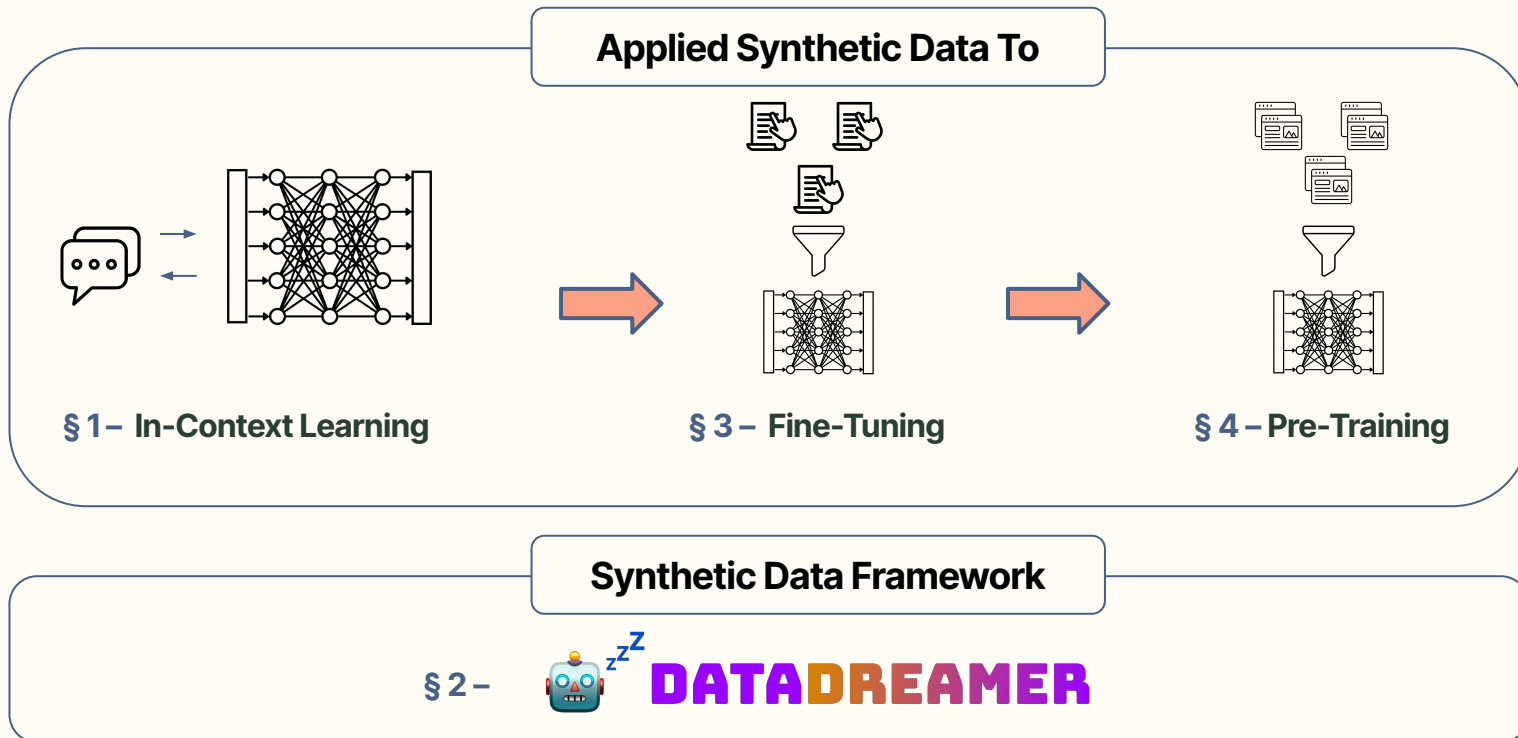


Composing Existing Knowledge and Capabilities



Environment Feedback

Summary



Summary

Leveraging LLMs to generate **synthetic datasets** allows practitioners to scale their prior knowledge and **solve tasks that were previously out of reach.**

Synthetic Data for Applied Tasks

Text Style Transfer

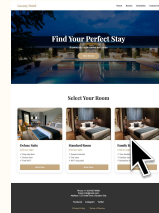
"It's keeping me up at night, I have to know what it is"

Tiny Styler

"What is it? It is keeping me up at night."

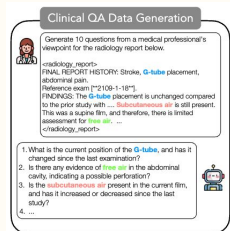
(Horvitz et al., 2024; Patel et al., 2022)

Multimodal Understanding for Agents



(Yang et al., 2024)

Clinical QA



(Bai et al., 2024)

Information Retrieval

Answer the following question:
{query}
Give the rationale before answering

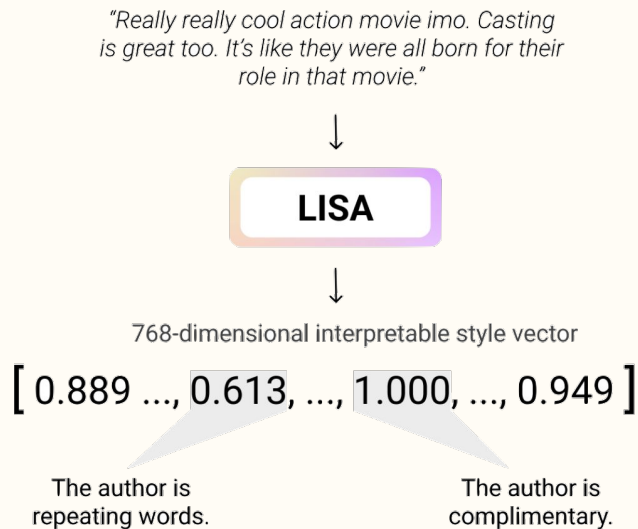
Large Language Model (LLM)

Concat({query}, {model output})

Retrieval System (BM25)

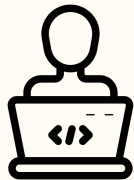
(Jagerman et al., 2023)

Synthetic Data for Interpretability



Text	Style Attribute
Subscribed. Interesting idea. I would like to see some advanced stats on hitting percentages to different locations on the court. For example, having the court broken up into maybe 12 zones and then hitting percentages from each position to those zones. I remember seeing an article that did this years ago and I have never been able to find anything online. I said recently on this sub that the deep angle shot from the left side or right side was the highest percentage shot in volleyball, but I was not able to back up my claim with any sources or anything. Anyways, I am a VB nerd, no doubt. Interested to see what direction you take this. Cheers!	The author is being polite.
Yeah I also work in QA, and seeing this kind of stuff get released is maddening. About a year ago working on a new platform we were seeing bugs in the hundreds each week, we pushed back the release of the product 3 months because basically it didn't work. If it was up to the devs, they'd have released it on time, because the stuff they'd written code for worked. Thorough doesn't even cover the work we go through every 3 months, and Niantic's approach seems completely amateur from this side. They're putting bandaids on problems and hiding things like the 3 step problem behind curtains without seemingly fixing anything, although I do have to say their balance tweaks to battling have been a big step in the right direction.	The author is using a personal anecdote to illustrate their point.
Thank you. I'd be interested in reading more about your experiences, in addition to the "American Wedding" story. Are you watching the stream? I wish there was a way to find out how many people in the world are watching it. The music is lovely, huh? God damn. He's got his bunny Fair Isle sweater on, drinking Dunkin' Donuts coffee. I would have thought him a Starbucks man. :-)	The author is using an emoji.

Automating Synthetic Data Procedures



```
from datadreamer import DataDreamer
from datadreamer.steps import DataFromPrompt, ProcessWithPrompt
from datadreamer.llms import OpenAI
from datadreamer.trainers import TrainHFFineTune
from peft import LoraConfig

with DataDreamer("./output"):

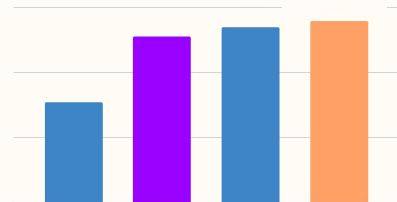
    documents = load_dataset("fineweb-edu")["train"]["text"]

    teacher_llm = OpenAI(model_name="gpt-5-large")

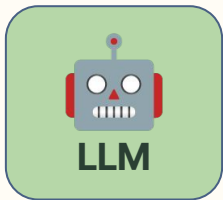
    prompts = ProcessWithPrompt(
        "Generate Prompts from Documents",
        inputs={"docs": documents},
        args={
            "llm": teacher_llm,
            "instruction": (
                "Given the document, create a useful user-style prompt..."
            ),
            "temperature": 1.0,
        },
        outputs={"generations": "prompts"},
    )

    ...

    trainer = TrainHFFineTune(
        "Distill into a small model",
        model_name="meta-llama/Llama-3.3-3B-Instruct",
        peft_config=LoraConfig(r=16, lora_alpha=32),
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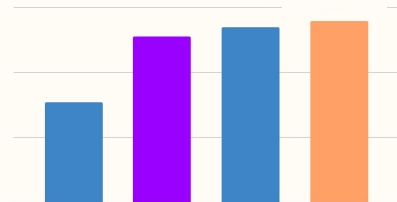
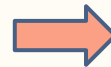
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(Si et al., 2024; Starace et al., 2025; Trirat et al., 2024; Patel et al., 2024)